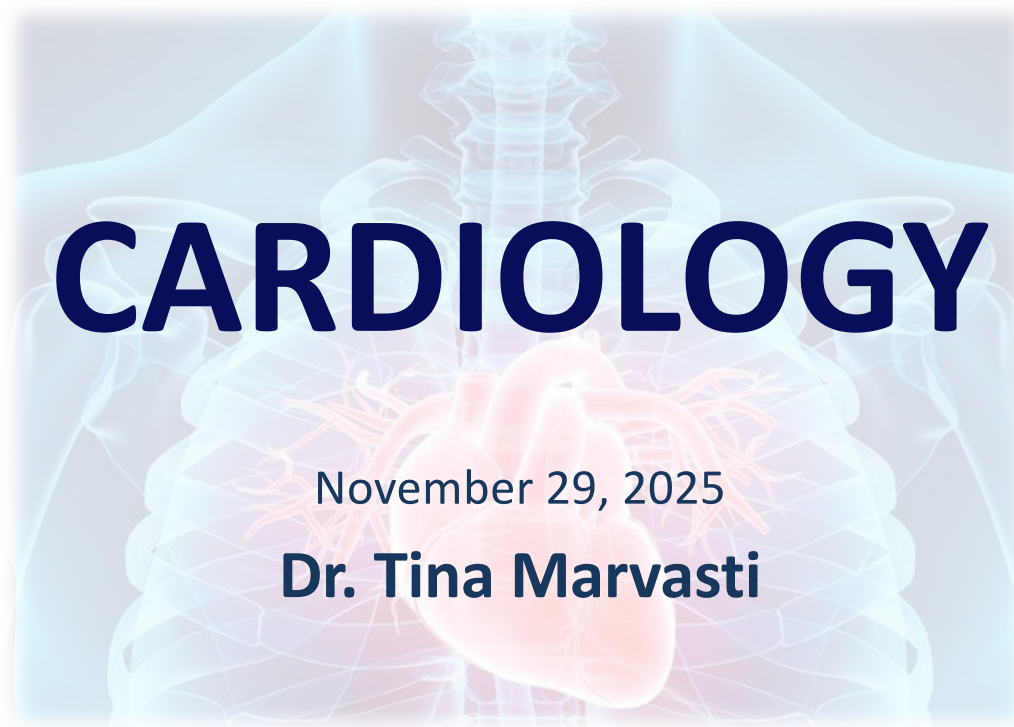




November 29, 2025

Dr. Tina Marvasti



Guiding Principles

Broad strokes:

- Help symptomatic patients feel better
- Help patients live longer
- Identify and treat risk factors
- Practice evidence-based medicine

RC tips:

- Use **Canadian Guidelines (CCS)** when possible
- Do not worry about understanding advanced cardiovascular techniques (ablation, angiography, etc.)
- Be safe!

Outline

1. Coronary artery disease
2. Heart failure / cardiomyopathy
3. Valvular heart disease
4. Aortopathy
5. Pericardial disease
6. Arrhythmias and implantable cardiac devices
7. Peripheral arterial disease

NB. ECGs and Cardiac Physical Exam will be self study online

ACS	Acute Coronary Syndrome
CRT (D) vs. (P)	Cardiac Resynchronization Therapy (with Defibrillator) vs. (Pacing only)
CCTA	Coronary CT angiography
DAPT	Dual Antiplatelet Therapy (ASA + clopidogrel or prasugrel or ticagrelor)
EST	Exercise Stress Test
METs	Metabolic Equivalents
DOACs	Direct Oral Anticoagulants (ex apixaban, dabigatran, rivaroxaban)
NSTEACS	Non ST Elevation Acute Coronary Syndrome
OAC	Oral Anticoagulants (Includes VKA, DOAC)
OMT	Optimal Medical Therapy
POBA	Plain Old Balloon Angioplasty
PCI	Percutaneous Coronary Intervention
VKA	Vitamin K Antagonist (warfarin)
SAPT	Single Antiplatelet Therapy (ASA OR Clopidogrel)
SPECT	Single photon emission computed tomography

Coronary Artery Disease

Key resources:

- CCS 2014 Guidelines – Diagnosis and management of stable ischemic heart disease
- ACC/AHA 2023 Guidelines – Chronic CAD



Coronary Artery Disease (CAD)

Two major presentations:

Chronic stable CAD

2014 CCS guidelines on stable ischemic heart disease

2023 ACC/AHA Guideline on Management of Patients with Chronic Coronary Disease

Acute coronary syndromes

Various CCS/ACC/AHA guidelines on ACS, antiplatelets

How Do Patients Present?

Ischemic cascade – with increasing ischemic time:

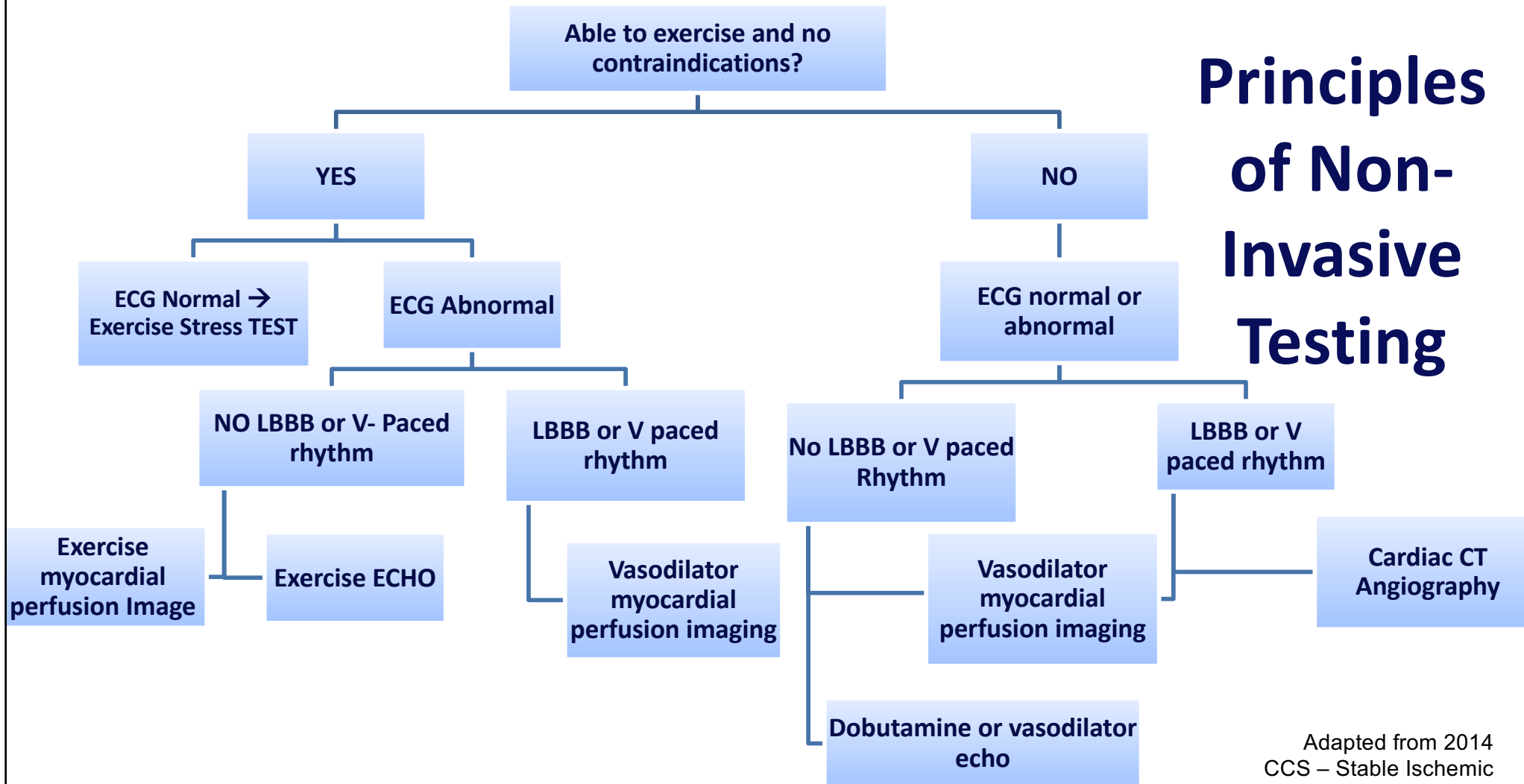
1. Blood flow changes (can be seen on myocardial perfusion)
2. Diastolic, then systolic dysfunction (wall motion abnormalities)
3. ECG changes
4. Symptoms
5. Myocardial necrosis

Diagnostic Tests for CAD

- Functional
 - Non-invasive stress tests:
 - *STRESS* = exercise, drugs (inotropes, vasodilators)
 - *TEST* = ECG, ECG+echo, ECG+nuclear
- Structural
 - Coronary angiography
 - CT coronary angiography
- *STRESS*: pick exercise whenever possible!
 - Provides prognostic info: e.g. duration of exercise, METs
 - Not possible if physical limitations or contraindications (e.g. critical aortic stenosis)
- *TEST*: consider functional imaging (e.g. nuclear) if:
 - Cannot accurately assess for ischemia on ECG
 - LBBB, paced rhythm, preexcitation, significant ST changes at rest → the ECG is **not interpretable** in these cases
 - RBBB interpretable generally
 - Need specific anatomic correlation (e.g. prior revascularization)

All patients with stable CAD should have resting LVEF imaging (eg TTE) – CCS 2014

Principles of Non-Invasive Testing



Adapted from 2014
CCS – Stable Ischemic
Heart Disease

Absolute Contraindications to EST

- Acute MI (within 2 days)
- Ongoing unstable angina
- Uncontrolled hemodynamically-significant arrhythmia
- Active endocarditis
- Symptomatic severe AS
- Decompensated heart failure
- Acute PE, pulmonary infarction, DVT
- Acute myocarditis, pericarditis
- Acute aortic dissection
- Physical limitations

Mnemonic:

I – Inflammation

D – Dissection

O – Ongoing angina

N – No consent

O – Ongoing MI

T – Thrombosis

S – Severe AS

T – Technical issues

R – Rhythm

E – Endocarditis

S – Systolic dysfunction

S – Slow



EST Results

- Positive
- Negative
- Equivocal
- Uninterpretable

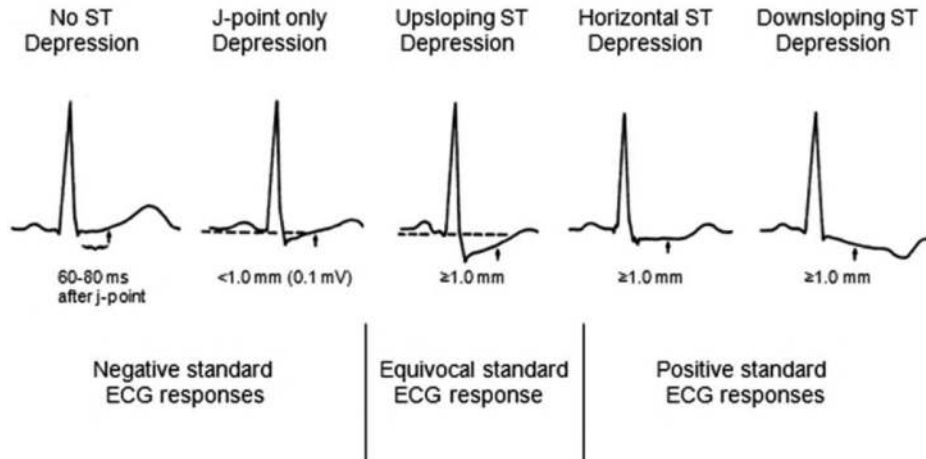
- Maximal vs. submaximal test
 - Patient should reach **85%** of age predicted maximum heart rate to **rule out** ischemia
 - ($Max\ HR = 220 - age$)

EST Results

Positive test

- $\geq 1\text{mm}$ STE
- $\geq 1\text{ mm}$ STD (horizontal or downsloping)

ST SEGMENT DEPRESSION DURING EXERCISE



High risk features*

- Duke Treadmill Score ≤ -11 or less
- <5 METs achieved
- Low threshold angina / ischemia
- STE
- Severe STD $\geq 2\text{mm}$
- Ischemia on ≥ 5 leads
- Ischemia ≥ 3 mins into recovery
- Abnormal BP response [failure to achieve SBP >120 , drop in BP >10 , drop below baseline]
- Ventricular arrhythmia

Myocardial Perfusion Imaging

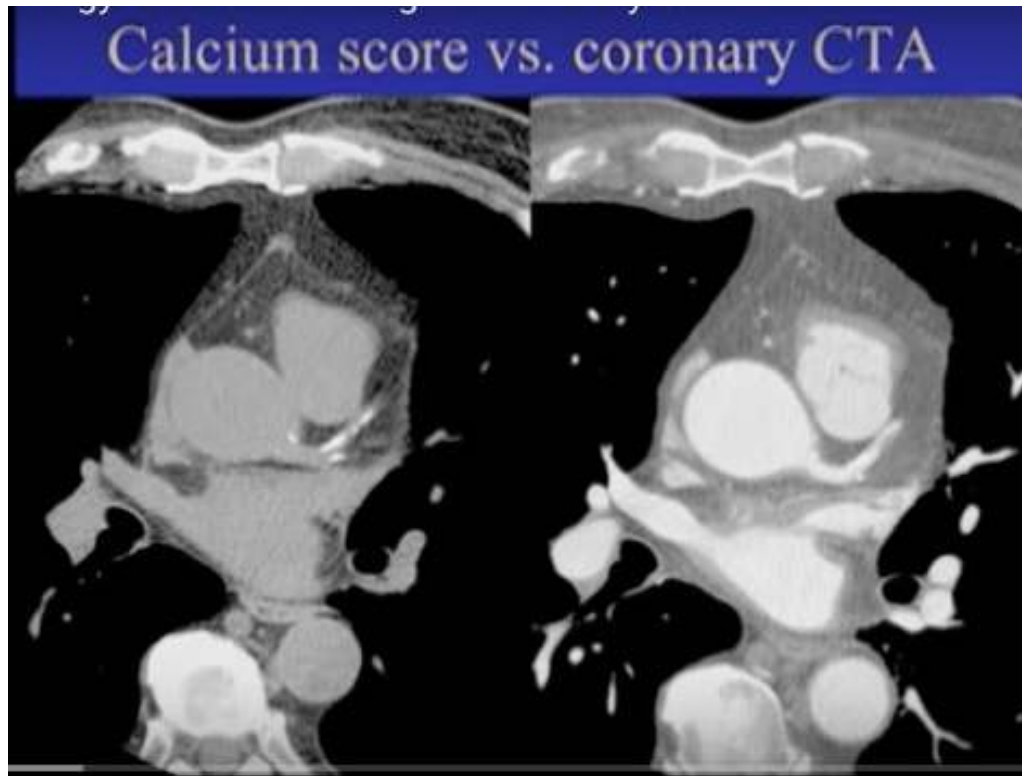
- **Radioactive tracer** (e.g. ^{99m}Tc) distributes in myocardium proportional to blood flow; **SPECT** detects tracer decay
- **Stress options:** exercise or pharmacologic (dipyridamole >> adenosine and regadenoson*; dobutamine used in stress echo)
- Pharmacologic Vasodilator 101:
 - **Mechanism:** vasodilation → as diseased vessels are already maximally dilated → perfusion mismatch
 - **Use:** option if patient cannot do exercise stress test
 - **False negatives:**
 - Caffeine/theophylline (hold before test)
 - Severe triple-vessel or left main disease (“balanced” ischemia)
 - **Contraindications:** high grade AV block, active/severe asthma or COPD (risk of bronchospasm)
 - **Reversal agent:** aminophylline

*Regadenoson = selective adenosine reuptake inhibitor (vs nonselective dipyridamole.) Same precautions, also reversed by aminophylline. Easier to administer but more costly so not universal in Canada.

Coronary CT Angiography (CCTA)

- Procedure specifics:
 - Low dose CT with beta blockade +/- IV nitro given (HR target <60), breath hold
 - 2-4mSv (~background annual radiation dose)
 - Less accurate (sensitivity/specificity) if the patient has a lot of coronary artery calcium or previous stent
- Indications:
 - Diagnosis of CAD for **low to intermediate pre-test prob** patients
 - Risk stratification in patients with stable CAD
- Contraindications:
 - ACS
 - Severe structural heart disease (AS or HCM)
 - Usual CT precautions: Contrast Allergy, Renal Failure, Pregnancy

Coronary CT Angiography (CCTA) vs CAC

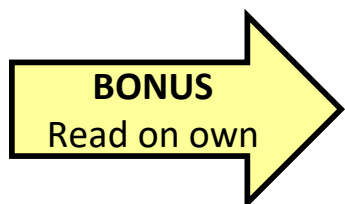


CCTA $\neq$ Coronary Artery Calcium (CAC) score.

CAC scoring: Recommended for **intermediate-risk** (FRS 10–19%), asymptomatic patients >40 who **aren't otherwise statin candidates**.

Can also be considered for **low-risk patients** with **family history of premature CVD** or **genetic dyslipidemia**.

CAC >100: Statin indicated regardless of FRS; start therapy.



Risk Stratify and Follow patients

Table 4. Summary estimates of pooled sensitivity and specificity (with 95% confidence intervals) for noninvasive cardiac tests for the diagnosis of coronary artery disease

Technology	Sensitivity	Specificity
Exercise treadmill	0.68 (0.23-1.0)	0.77 (0.17-1.0)
Attenuation-corrected SPECT	0.86 (0.81-0.91)	0.82 (0.75-0.89)
Gated SPECT	0.84 (0.79-0.88)	0.78 (0.71-0.85)
Traditional SPECT	0.86 (0.84-0.88)	0.71 (0.67-0.76)
Contrast stress echocardiography (wall motion)	0.84 (0.79-0.90)	0.80 (0.73-0.87)
Exercise or pharmacologic stress echocardiography	0.79 (0.77-0.82)	0.84 (0.82-0.86)
Cardiac computed tomographic angiography	0.96 (0.94-0.98)	0.82 (0.73-0.90)
Positron emission tomography	0.90 (0.88-0.92)	0.88 (0.85-0.91)
Cardiac MRI (perfusion)	0.91 (0.88-0.94)	0.81 (0.75-0.87)

MRI, magnetic resonance imaging; SPECT, single photon emission computed tomography.

Data from Gianrossi et al.,³² Medical Advisory Secretariat,³³ and McArdle et al.³⁴

- None of the noninvasive cardiac tests are perfect
 - Do not memorize table!
- Clinical history, pretest probability (based upon gender, age etc) vitally important in interpreting stress results
- All patients with stable CAD should have resting LVEF imaging (eg TTE) – CCS 2014

Treatment of Chronic Stable CAD



General Principles

- Educate on symptom management, lifestyle changes, medication adherence, target **#SDOH** risk factors, multidisciplinary team based care [ACC/AHA 2023]
- Treat symptoms with medical therapy first
- Consider revascularization if refractory symptoms, high risk structural disease (e.g. LM disease), LV dysfunction, severe MR

Optimal Medical Therapy (OMT) is non-inferior to revascularization (PCI/CABG) for patients with **Stable CAD**

- *COURAGE 2007, ORBITA 2017, ISCHEMIA 2019*
- **ISCHEMIA**: Median 3.2 years follow-up; no difference in primary outcome (MACE) invasive strategy vs. conservative strategy
 - Extended 7-year follow-up data showed no difference in all-cause mortality was no different with invasive strategy. However, CV death was lower (offset by non-CV death for reasons that are unclear). **Bottom line is all-cause was not reduced by an invasive strategy.**

Treatment of Chronic Stable CAD



- **ALL patients with CAD:**
 - Commence medical treatment for CAD
 - **Aspirin + statin** for evidence of coronary atherosclerosis regardless of the modality of diagnosis
 - Clopidogrel can be used as SAPT if ASA intolerant
 - CAPRIE trial 1996 showed reduced MACE with clopidogrel over ASA in patients with CAD/stroke/PAD
 - HOST-EXAM study (Circulation, 2023) showed lower rates of cardiovascular death, MI, stroke and bleeding with long term clopidogrel compared to ASA after 1 year of DAPT post PCI (12.8% vs 16.9%)
 - For now ASA monotherapy is the long term antiplatelet of choice for chronic stable CAD though this may change in the next few years

Treatment of Chronic Stable CAD



- **Adjunctive therapies:**

- Smoking cessation (see RESP + Med Onc lecture)
 - Counselling + smoking cessation meds (bupropion, varenicline, NRT) more effective than either alone
- Cardiac rehabilitation
 - After recent MI/PCI/CABG – I-A
 - Stable CAD or post heart transplant – I-B
 - Recommended to reduce risk of CV complications with sexual activity (4 METs) (AHA/ACC 2023)
- Weight management counselling
- Physical activity (150 minutes/wk aerobic, 2 days/wk weight training if no contraindications)
- Alcohol – reduce $\leq 1/d$ women, $\leq 2/d$ men
- Vaccines – annual flu (I-C), COVID + boosters (I-C), pneumococcal (II-a)

Stable CAD “Don’t”:

- Non-prescription dietary supplements NOT recommended! NO in $\downarrow$ MACE/Death (Omega 3 Fatty acids, Vit C, D, E, Ca, beta-carotene)
- Do not recommend alcohol for purpose of CV protection.
- Do not use chronic NSAID = harm (CV/bleeding)

- **Antianginal (symptomatic benefit):**

- Beta blockers: reduce HR/contractility, indicated for most patients*
- CCBs: reduce HR/contractility (non-dihydropyridine, eg. Verapamil, beware if LVEF < 40%), reduce preload (dihydropyridine eg amlodipine)
- Nitrates: venodilate, reduce LVEDP
- Others: ranolazine (calcium modulator that reduces angina, just so you’re aware of but use in Canada is rare)

- **Disease modifying therapies:**

- ACE inhibitors: HTN, T2DM, LVEF < 40%, CKD, can be considered for all for vascular protection
- Beta blockers: LVEF < 40%
 - *If no previous MI and LVEF > 50 = **use of BB therapy does not $\downarrow$ MACE**, in absence of other indication for BB (eg for control of HTN or rapid afib) – ACC/AHA 2023
- CAD + DM: SGLT2i or GLP1RA
- Hypertension, dyslipidemia, diabetes management

Treatment of Chronic Stable CAD



To be aware of:

- COMPASS trial results:
 - Low dose ASA + rivaroxaban 2.5 mg BID
 - “Another option for secondary prevention in patients with chronic stable CAD” (buried in CCS antiplatelet 2018 text)
 - Reasonable alternative to ASA alone in patients with CAD + AF at low risk of stroke (CHADS65 = 0) (CCS 2020 AF guidelines)
- POST-PCI trial results:
 - Investigated patients post-PCI for high risk CAD (left main, multiple lesions, bifurcating/long lesions, diabetes) undergoing routine stress testing at 1 year vs. usual care (symptom driven)
 - No differences in all cause death, MI or hospitalization for angina with surveillance strategy on routine stress testing → **Don't need to stress patients routinely unless they have a change in symptoms!**

Eikelboom JW et al. Rivaroxaban with or without aspirin in stable cardiovascular disease. NEJM. 2017 Oct 5;377(14):1319-30.

Park DW et al. Routine Functional Testing or Standard Care in High-Risk patients after PCI. NEJM 2022 Sept 8;387(10):905-15.

Revascularization - PCI vs. CABG

In general, invasive angiography should be considered for:

- High risk features on non-invasive testing
- Medically refractory symptoms

For MCQs:

- Conflicting stroke data
- **Less repeat revascularization with CABG**
- ↑ Survival with CABG in highly select scenarios
- “Team-based decision” beyond scope of GIM

Revascularization with Chronic CAD (ACC/AHA 2023)

(multivessel = 2 or more)

5.2. Revascularization: PCI Versus CABG

Recommendations for Revascularization: PCI Versus CABG Referenced studies that support the recommendations are summarized in the Online Data Supplement.		
COR	LOE	Recommendations
Patients With CCD		
1	B-R	1. In patients with CCD who require revascularization for significant left main involvement associated with high-complexity CAD, CABG is recommended in preference to PCI to improve survival. ^{1,2}
2a	B-R	2. In patients with CCD who require revascularization for multivessel CAD with complex and diffuse CAD (eg, SYNTAX score >33), it is reasonable to choose CABG over PCI to improve survival. ^{1,3-6}
Patients With CCD at High Surgical Risk		
2a	B-NR	3. In patients with CCD who are appropriate for revascularization but poor candidates for surgery, it is reasonable to choose PCI over CABG to improve symptoms and reduce MACE. ⁷⁻⁹
Patients With CCD and Diabetes		
1	A	4. In patients with CCD, diabetes, and multivessel CAD with involvement of the left anterior descending artery who are appropriate candidates for CABG, CABG (with a left internal mammary artery to the left anterior descending artery) is recommended in preference to PCI to reduce mortality and repeat revascularizations. ^{4,10-17}
2b	B-R	5. In patients with CCD and diabetes who have left main stenosis and low- or intermediate-complexity CAD (eg, SYNTAX score ≤33), PCI may be considered as an alternative to CABG to reduce MACE. ^{10,18}

*Modified from the 2021 ACC/AHA/SCAI Guideline for Coronary Artery Revascularization.¹⁹

Scenario	Survival benefit
Left Main >50% stenosis	↑survival over GDMT alone
Multivessel dz w LVEF ≤ 35%	
Left Main w high complexity CAD	↑survival over PCI
Multivessel dz with high complexity CAD	
Multivessel dz in DIABETES w LAD involvement amenable to LIMA	↑survival & ↓revascularization over PCI

“High complexity CAD” = High SYNTAX score (beyond GIM scope!)

- PCI if:
 - Poor surgical candidate
 - Single vessel disease
 - Diabetes with LM and low/intermediate complexity CAD consider as alternative to CABG

Acute Coronary Syndromes

Key resources:

- CCS 2019 Guidelines – Management of STEMI: focused update on regionalization and reperfusion
- CCS 2018 Guidelines – Use of antiplatelets
- CCS 2023 Focused Update on the Use of Antiplatelets
- CCS 2024 MINOCA Clinical Practice Update
- AHA/ACC 2013 Guidelines – Management of STEMI
- AHA/ACC 2014 Guidelines – NSTEMI
- AHA/ACC 2024 Perioperative Guidelines for non-cardiac surgery

MCQ 1 2026

A 65F with DM2 and CCS3 angina presents with LAD and RCA disease on angiogram. Her surgeon advises that she go for CABG but she wants stents (PTCA). With regards to future risk of revascularization, What do you advise:

- a) CABG
- b) PTCA
- c) Medical treatment
- d) No difference

Acute Coronary Syndromes

The spectrum of acute coronary syndromes:

	Pathophysiology	Symptoms	ECG	Biomarkers
NSTE-ACS (NSTEMI/ UA)	Subtotal occlusion with subendocardial ischemia	Angina or equivalent	Normal/ST-segment depression/T wave inversion	(+) in “NSTEMI” (-) in “UA”
STEMI	Total occlusion with transmural infarction	Angina or equivalent	ST-segment elevation/Hyper acute T waves	(+) <i>but can be initially (-) in early presenters</i>

ACS = plaque rupture with thrombus formation in epicardial coronary artery

Management of ACS

1. Immediate Medical Management
2. Reperfusion strategies
3. Chronic Medical Management / risk factor optimization
4. Driving Restrictions

Immediate Medical Management

The ACS “cocktail”

- **Antiplatelet:** ASA + 2nd agent (P2Y₁₂ inhibitor: **ticagrelor**, clopidogrel, prasugrel)
 - Loading doses → ASA (160-325 mg CHEWED), Ticagrelor (180 mg), Prasugrel (60mg)
Clopidogrel (300-600 mg)
 - Ticagrelor C/I if previous intracranial hemorrhage, prasugrel C/I if ANY prior TIA/Stroke
 - *If **thrombolysis** → ASA + Clopidogrel 300 mg (NO TICAGRELOR OR PRASUGREL)*
 - *AHA 2025 suggests clopidogrel 75mg only if patient >75y and receiving thrombolysis*
 - Maintenance doses → ASA (81 mg OD), Ticagrelor (90 mg q12h), Clopidogrel (75 mg OD),
- **Anticoagulation:** UFH or LMWH (enoxaparin) or Fondaparinux or Bivalirudin
 - Continued for 48 hours until discharge or 8 days, stop if revascularized
- **Antianginals:** beta-blocker, cautious administration, within 24 hours for patients without contraindications* to reduce risk of reinfarction and ventricular arrhythmias
 - *usual contraindications: heart failure, unstable, heart block, bronchospasm
 - PRN: nitrates, opioids (*use sparingly*)



Reperfusion Therapy – STEMI



- Principle: achieve coronary patency in ALL patients presenting <24h from symptom onset, or with ongoing symptoms
- Options:
 - Primary PCI
 - Fibrinolysis/pharmaco-invasive (“drip & ship”)

Key Guideline: 2019 CCS
Acute Management of STEMI
Can J Cardiol 35 (2019) 107-132.

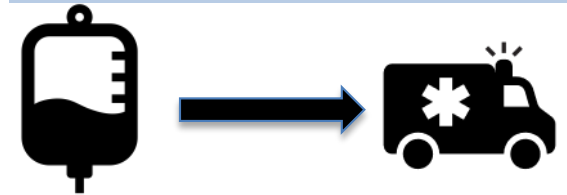
GIM takeaways:

- Avoid routine IV opioid administration for pain during STEMI
- Suggest avoidance of **routine prehospital O2 admin** if SpO₂ ≥ 90%
- No P2Y_{12i} in ambulance; administer in ED

Reperfusion Therapy – STEMI

- Primary PCI > fibrinolysis:
 - *if timely*
 - PCI capable hospital → FMC-to-balloon time <90 min
 - Non-PCI capable hospital → FMC-to-balloon time <120 min
 - *FMC = First Medical Contact → ER triage if walk-in, or EMS arrival if 911*
 - *if later presentation (12-24h of symptom onset)*
 - *if cardiogenic shock*
- Fibrinolysis indicated if PCI is not readily available (>120min from FMC)
 - **Drip** (give lysis) **then ship** (send immediately to PCI centre for angiogram/PCI within 3-24 hours)
 - If Fibrinolysis → give within 30 minutes of FMC
 - If Fibrinolysis → PCI should occur within 24 hr.
 - Timing of fibrinolysis → the earlier, the better, but can be given *up to 12h* after onset of chest pain w STE

No quick (<120min) access to PCI?
DRIP (TNK)Then SHIP!



Contraindications to fibrinolysis for STEMI

Absolute: “HABITS”

- Hemorrhage (intracranial, ever)
- Aortic dissection
- Bleeding (diathesis or active) and BP (>180 / >110 despite therapy)
- Intracranial (lesion, AVM, malignancy)
- Trauma (significant closed head w/in 3 mos)
- Stroke (ischemic within 3month)

Reperfusion Therapy – STEMI

- Reperfusion if the patient presents after 12h-24h hours of symptom onset?
 - Nuanced topic, *“the benefit of fibrinolytic therapy in patients who present >12 hours after symptom onset has not been established”*
 - Transfer for PCI if 12-24h post symptom onset, ask interventionalist if they want fibrinolytics given
 - If they received lysis within 24 hours, PCI can be considered after 24 hours (ideally try to get within 3-24 hours)
 - If the vessel is completely occluded after 24 hours from symptom onset
 - PCI can cause harm (reperfusion injury) without benefit especially if pt stable (muscle has infarcted/scarred) → discuss with interventional cardiologist

The 2nd Antiplatelet... CCS 2023

Key points

Ticagrelor is generally preferred (conditional) vs clopidogrel due to greater efficacy

If patient is High Bleeding Risk:

- DAPT for 1-3 months non-inferior to longer durations
- “practice point” – if stepdown to SAPT choose P2Y₁₂inhibitor over ASA
- Step-down to SAPT should incorporate cardiologist recommendations – (eg) avoid if complex PCI or high risk of stent thrombosis

Consideration for P2Y₁₂ Inhibitors Before Coronary Angiography

ASA before angiography for possible PCI

PRACTICAL TIP

SIHD

NSTEACS

STEMI

Routine pretreatment with a P2Y₁₂ inhibitor **not** recommended

Coronary angiography < 24 hours:
Pretreatment with P2Y₁₂ **not** recommended

Coronary angiography ≥ 24 hours:
Pretreatment with P2Y₁₂ is recommended

Routine pretreatment with a P2Y₁₂ inhibitor recommended
Going for 1^o PCI, no fibrinolysis

If ad hoc PCI performed:

Clopidogrel 600 mg loading dose in the cath lab

PRACTICAL TIP

If PCI performed,

Ticagrelor 180 mg

OR

Prasugrel 60 mg loading dose preferred over

Clopidogrel 600 mg loading dose in the cath lab

PRACTICAL TIP

Ticagrelor 180 mg preferred over

Clopidogrel 600 mg*

PRACTICAL TIP

Ticagrelor 180 mg

OR

Prasugrel 60 mg loading dose preferred over

Clopidogrel 600 mg loading dose**

PRACTICAL TIP

SIHD = stable ischemic heart disease

Weak recommendation

The 2nd Antiplatelet... Nuances

- Ticagrelor is contraindicated if history of:
 - intracranial hemorrhage
 - active pathological bleeding
 - moderate-severe hepatic impairment
 - combinations with CYP3A4 inhibitors (ketoconazole, clarithromycin, ritonavir)
 - Should consider avoiding in patients with evidence of heart block or bradycardia.
 - Dose: 180mg load then 90 mg bid x 12 months then 60mg bid thereafter
- Prasugrel is contraindicated if:
 - active bleeding
 - any prior TIA/stroke [even ISCHEMIC stroke]
 - hypersensitivity reaction
 - Dose: 60mg load then 10mg daily (reduce to 5mg if <60kg)
-  **Ticagrelor & Prasugrel have NOT been adequately evaluated in the setting of fibrinolysis in STEMI** → Load with clopidogrel when lysing the patient. Post lysis (and PCI as part of a “drip and ship” strategy, clopidogrel can be changed to ticagrelor or prasugrel)

Reperfusion Therapy - NSTEMI-ACS

Timing & risk stratification are the major differences vs. STEMI

1. Risk stratify using clinical judgment + risk scores (TIMI, GRACE)
 - Do not memorize these, but understand what factors involved
2. High risk patients: early invasive strategy (angiogram within 24hr – AHA, 48 CCS)
 - Early invasive reduces risk of rehospitalization for ACS but **NO mortality benefit**
3. Intermediate risk patients: routine invasive (angiogram before discharge, <72h)
4. Low risk patients and/or unclear diagnosis: non-invasive testing risk stratification during hospitalization or recurrent symptoms OR coronary angiography can be considered

Reperfusion Therapy:

PCI vs CABG for STEMI and NSTEMI-ACS with multivessel disease

STEMI:

- PCI of culprit vessel acutely

THEN

- Elective CABG for complex MVD involving LAD/LM to reduce risk of CV events

or

- PCI of significantly stenosed non culprit, non complex arteries to ↓ risk of death/MI, ↑ angina related QOL (can be at time of primary PCI)

Unless: Cardiogenic shock – no PCI to non culprit lesions.

NSTEMI-ACS

- Mode of revascularization is based upon disease complexity / comorbidities
- CABG should be considered with:
 - Significant LM stenosis with complex MVD
 - MVD with diffuse or complex CAD
 - Diabetes and involvement of LAD
 - MVD or complex left main CAD with severe LV dysfunction
- If having multivessel PCI,

Multivessel disease: stenosis $\geq 50\%$ in ≥ 2 epicardial arteries
Complex MVD = beyond scope of GIM exam
“significantly stenosed” : $\geq 70\%$, unless Left main ($\geq 50\%$)

PCI

- **Plain-Old-Balloon-Angioplasty (POBA)**
 - Rare
- **Bare metal stent (BMS)** – rarely used
 - Endothelialize quickly = lower risk of stent thrombosis after 4+ weeks
 - But...higher risk of re-stenosis
- **Drug eluting stent (DES)** – standard of care
 - Elute anti-proliferative agents
 - Lower rates of restenosis vs. BMS = can be used in smaller vessels, CABG grafts
 - But...take longer to endothelialize
- **Drug coated balloon (DCB)**
 - Expand a blood vessel and deliver antiproliferative agents (e.g. Paclitaxel) without delivering a stent
 - Useful for in-stent restenosis (ISR), bifurcating/branch lesions, buying time for definitive treatment

Stents and DAPT Durations

CCS Antiplatelet 2018,2023

AHA ACS guideline 2025

POST ACS (STEMI or NSTEMI/UA): Aim for 12 months of DAPT, whether PCI or medically managed

- ACS DAPT= ASA + Ticagrelor 90 BID or Prasugrel 10 OD (preferred over ASA+Clopidogrel)
- If tolerating ASA+Ticagrelor DAPT post PCI, transition to Ticagrelor monotherapy ≥1 month post PCI can ↓ bleeding without ↑ in MACE (AHA 2025)
- High risk of bleeding:
 - SAPT with P2Yi after 1-3 months of DAPT – preferring ticagrelor monotherapy (I-A, AHA 2025, CCS 2023)
 - Descalating to a less potent P2Yi (eg. ASA+ticagrelor → ASA + clopidogrel (II-B, AHA 2025, CCS 2023)
- ASA monotherapy within first 12 months increases thrombotic risk in patients who have had PCI
- PPI recommended in patients at high risk of GI bleeding on DAPT (AHA 2025)
- Indication for OAC (Atrial Fibrillation, thrombosis, mechanical valve etc)
 - Triple therapy (ASA+P2Yi+OAC) for 1-30 days (CCS / 1-4weeks AHA 2025) then Dual pathway (clopidogrel + OAC)

(More in AFIB section on durations with OAC)

Stents and DAPT Durations (**without AF**)

CCS Antiplatelet 2018,2023

AHA ACS guideline 2025

Reassess bleeding at 1 year (CCS 2023)

If HIGH RISK bleed: SAPT ASA 81 or Clopidogrel 75

If LOW RISK bleed: Continue DAPT -
Good evidence for up to 3 years
(DAPT trial)

DAPT after 12 months: Suggest ASA + one of:

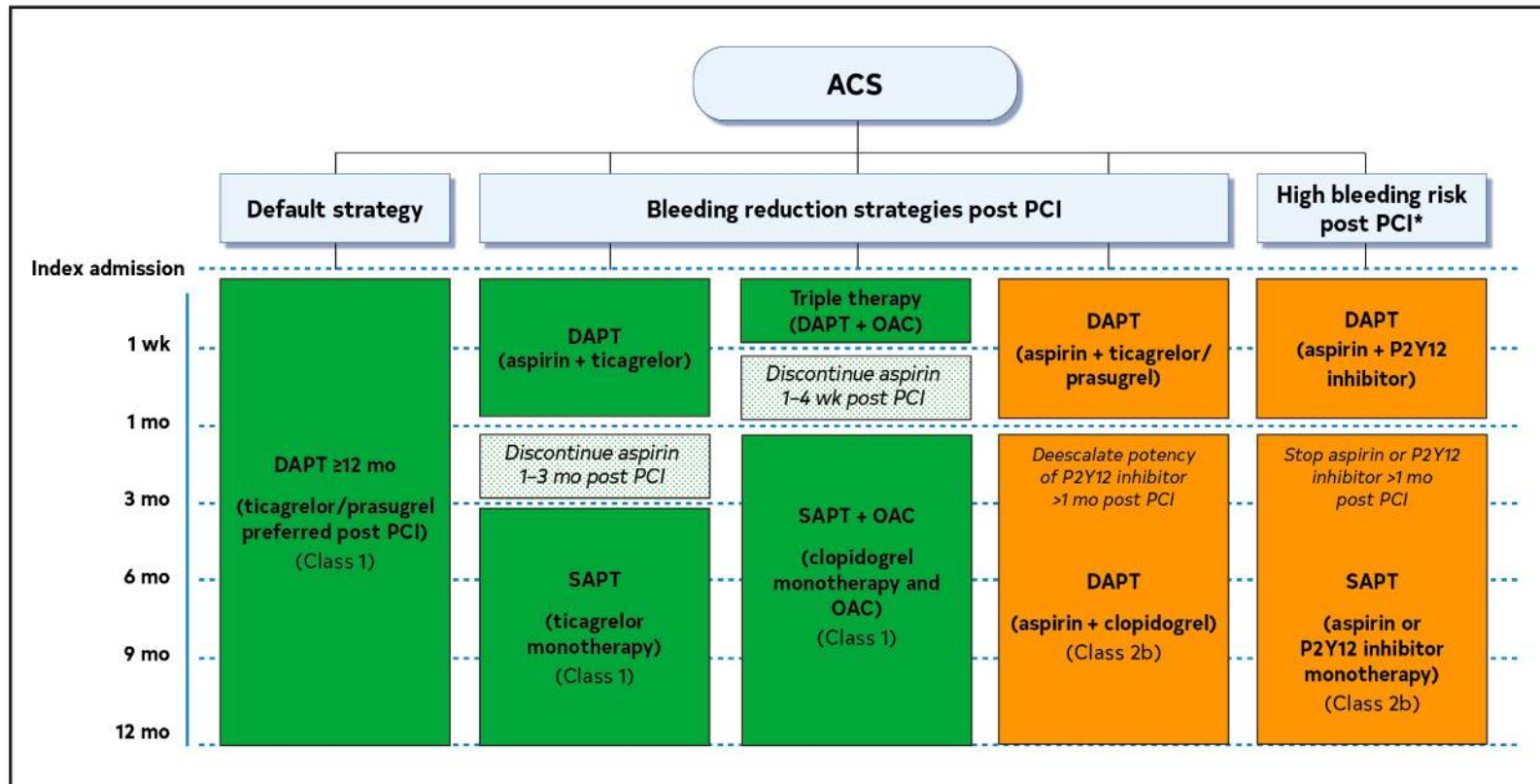
- Ticagrelor (**60 mg po bid**) (reduced dose)
- Clopidogrel (75 mg po daily)
- Prasugrel (10mg po daily) (weaker recommendation than others for extended therapy)

Non-ACS situations (ELECTIVE PCI)

- High Risk of Bleeding: Elective PCI DAPT = ASA 81+ Clopidogrel 75
 - BMS = DAPT for **1 month** then SAPT with ASA 81 or Clopidogrel 75 indefinitely
 - DES = DAPT for **3 months** then SAPT with ASA 81 or Clopidogrel 75 indefinitely
- Not at high risk of bleeding: DAPT for **6 months**, then reassess
 - If High Risk thrombotic events: extend DAPT up to 3 yrs
 - If not at high risk of thrombosis *or if now at high risk bleeding*: SAPT (ASA or Clopidogrel)

Stents and DAPT Durations post ACS

AHA 2025



High bleeding risk patient

If 1 major or 2 minor consider a shorter duration or de-escalated regimen of DAPT

Need 1 major

1. Advanced CKD (eGFR<30)
2. Liver cirrhosis with portal HTN
3. Active malignancy (except melanoma) in the last year, undergoing treatment
4. Major bleeding (transfusion or needing hospitalization) in last 6 months or chronic bleeding diathesis
5. Hb<110, thrombocytopenia PLT<100
6. Prior spontaneous/traumatic ICH <12mo
7. Brain AVM/stroke<6mo
8. Requires long-term anticoagulation (excluding COMPASS riva)
9. Non-deferable major surgery on DAPT
10. Recent major surgery/trauma <30d before PCI

Or 2 minor:

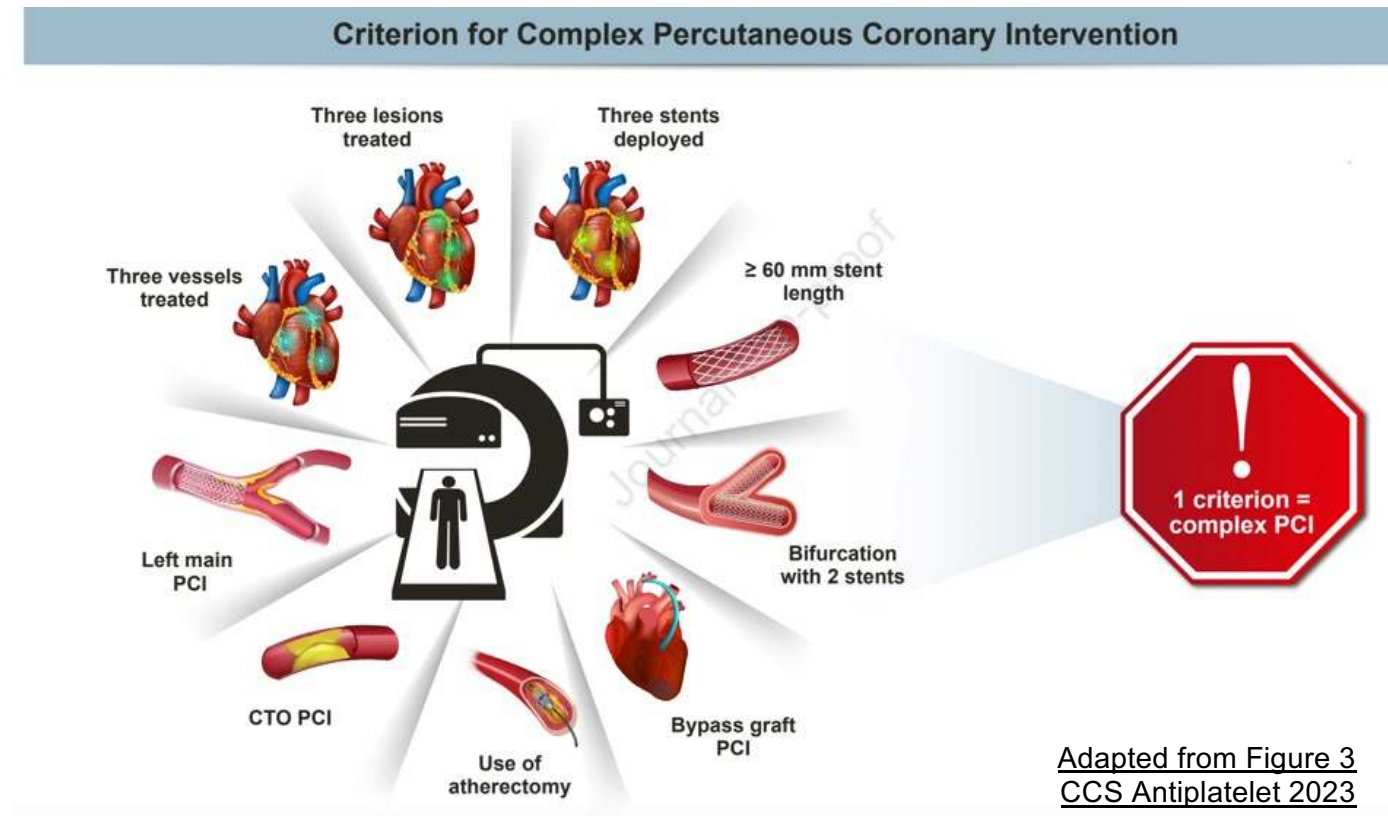
1. Moderate CKD (eGFR 30-59)
2. Spontaneous bleeding with hospitalization or transfusion <12mo not meeting major criterion
3. Hb 110-129 men/110-119 women
4. Any ischemic stroke not meeting major criterion
5. Long term NSAID/steroids
6. Age>75

Complex PCI :FYI only

Complex PCI = extend DAPT where safe. Always talk with the interventionalist about durations!

Just need 1:

- Left main
- 3 vessels
- 3 lesions
- 3 stents
- >60mm stent
- Bifurcation stents
- Bypass graft PCI
- Atherectomy
- Chronic total occlusion procedure



Adapted from Figure 3
CCS Antiplatelet 2023

CCS 2023 Antiplatelet Guideline Updates : 1 Slide “All you need to know”

Scenario	Preferred Regimen	DAPT duration	Alternative regimen	Notes
1° Prevention	No antiplatelet! (HTN Canada, CDA, CCS, Stroke all agree)			
Stable CAD, elective PCI	ASA + clopidogrel 600mg load in cath lab	HBR: 1-3 mos Standard: 6 mos ASA+ clopidogrel Extend: up to 3yrs ASA+ clopidogrel	Other P2Yi not studied extensively in this setting	HBR = high bleeding risk Individualize duration based upon pt factors
ACS, no PCI (medical management)	ASA + Ticagrelor	“shared care decision” HBR: 1-3 mos DAPT then SAPT SAPT = P2Yi favouring ticagrelor (AHA 2025) Standard bleed risk: 1 yr DAPT ASA+ Ticagrelor 90 BID or ASA+ Prasugrel 10 OD [*do not pick if no PCI, prefer ticagrelor here] Extend : Up to 3yrs ASA+Ticagrelor 60 BID (dose reduced) or ASA + clopidogrel 75mg OD or ASA + prasugrel 10mg OD	ASA + clopidogrel preferred over ASA + prasugrel (=↑ bleed)	Type I MI, not applicable to “demand ischemia” / type 2 MI
NSTEMI + PCI	IN ER: Load with ASA + ticagrelor 180mg <i>Going to PCI in <24h*?</i> Can instead load P2Yi in cath lab (180mg ticagrelor or 60mg prasugrel)		ASA + clopidogrel	*“Pretreatment” (eg out of hospital or in ED) with P2Yi does not change outcomes compared to giving load of P2Yi in cath lab <i>if having angiogram within 24h.</i>
STEMI + PCI	ASA + P2Yi prior to cath If TNK: Clopidogrel If no TNK: dealer’s choice if 1° PCI			Post PCI: load 180mg ticagrelor or 60mg prasugrel load

BONUS
Read on own

What to do with P2Yi if anticipating CABG

- Ideally, P2Yi load would be given in cath lab at time of treatment decision (PCI vs CABG) as high risk patients ideally should have cath within 24h
- In Canadian Reality: if you strongly suspect CABG will be required, can withhold P2Yi for NSTEMI-ACS prior to Angiogram even if expected delay >24h post ACS [CCS 2023, on exam: discuss with cardiology!]
- Stopping P2Yi prior to CABG
 - Limited evidence... “multidisciplinary team” to decide, hold 2-7d pre-op
 - Weigh factors such as coronary anatomy, hemodynamic stability, bleeding risk, surgical team expertise
 - Suggest hold Ticagrelor 2-3 days rather than 5-7d [weak, conditional CCS 2023]
- Postop Antiplatelet Regimen: [weak recommendation, CCS 2023]
 - If ACS: Favour DAPT w ticagrelor > prasugrel >> clopidogrel up to 12 months postop
 - If **NO** ACS: Favour DAPT with ASA+Clopidogrel up to 12 months postop
 - DAPT improves graft patency; if “off-pump” CABG, DAPT improves MACE
 - AFIB? – consider OAC monotherapy post CABG

We think this is beyond scope of GIM –
Usually our exam questions pertain to NON CARDIAC SURGERY!

Periop Mgmt - Stents and DAPT

Minimum delay Non-Cardiac Surgery

- BMS – Delay surgery **for Elective** for at least **1 month** post PCI
- DES – Delay surgery for at least **3 months** post PCI

Minimum delay for Semi-Urgent Non-Cardiac Surgery

- BMS – Delay surgery for at least **1 month** post PCI*
 - DES – Delay surgery for at least **1 month** post PCI*
- *weak recommendation, low quality evidence – individualize as semi-urgent sx usually can't be delayed 1 month

Pre-Op instructions:

- Hold Clopidogrel and Ticagrelor 5-7d preoperatively
 - 5-7 days if neuraxial anesthesia or very high risk bleeding, speak to anesthetist and surgeon
- Hold prasugrel 7-10d preoperatively
- Continue ASA perioperatively “whenever possible”

CCS Antiplatelet 2018

NO CHANGES in CCS 2023**
See perioperative medicine lecture

Post-Op

- Restart DAPT post-op as soon as deemed safe by surgical team

***NEW* ASSURE-DES trial** (ESC Meeting 2024) showed that in patients who had PCI with a DES more than 1 year ago, stopping ASA for non-cardiac surgery showed no differences in ischemic events (0.6% vs 0.9%) and major bleeding (5.2% vs. 6.5%) but did show a difference in minor bleeding (10.1% vs. 14.9%)

***NEW* AHA/ACC 2024 Perioperative Guidelines for non-cardiac surgery** are consistent with the above recommendations with some differences in wording (e.g. post ACS with PCI “ideally” delay 1 year; interrupting DAPT after 3 months “may be considered” for urgent surgery if risk of delaying surgery outweighs risk of MACE).

BONUS

Read on own

Post-MI Complications

- Heart failure
- Arrhythmias
 - Tachy: Atrial, ventricular
 - Brady: Heart block (esp. inferior MI)
- Mechanical complications
 - Papillary muscle dysfunction and acute MR
 - Ventricular septal rupture
 - Free wall rupture
 - RV infarction (esp. inferior)
- Pericarditis
 - Post MI pericarditis = Early (5d) vs. delayed [*Dressler syndrome*] (2-8wks)
 - Fever, pleuritic chest pain, pericardial rub and/or pleural effusion; Rx is high dose ASA + colchicine

Chronic Management/Risk factors

Initiate **BEFORE** Discharge

- High potency statin, reassess lipids 4-8 weeks post discharge (AHA 2025)
- ACEi if concurrent hypertension, diabetes
- Beta blockers? - Still controversy, go case by case
 - Start within 24 hours for patients without contraindications* to reduce risk of reinfarction and ventricular arrhythmias (AHA 2025)
 - Contraindications: usual, e.g. heart block, cardiogenic shock, uncontrolled asthma, etc.
 - Duration?
 - **Controversy in evidence – conflicting studies about duration of beta blockers and benefit once stable, with preserved LVEF**
 - **What to do?** If stable CAD+no MI can stop beta blocker (ACC/AHA Stable CAD Guidelines 2023); if MI / post ACS then shared-decision making once stable. If LVEF reduced <40% continue if no contraindications. On MCQ, if the patient is admitted post-ACS, start BB if no contraindications per AHA.
- Identify +/- optimize diabetes
 - SGLT2 inhibitors and GLP1RAs noted to reduce MACE in patients with diabetes in general. SGLT2i studied post ACS and not found to reduce MACE immediately post ACS (no harm though); it is reasonable to incorporate to improve diabetes control
- Influenza vaccine (I-A – AHA 2025)
 - Influenza Vaccination after Myocardial Infarction (IAMI) trial showed that an influenza vaccine administered within 72 hours post STEMI/NSTEMI reduced all cause mortality, MI, stent thrombosis at 12 months compared to placebo
- Smoking Cessation Therapy [see Resp Lecture BONUS slides, Med Onc Slides]
- **Cardiovascular Rehab** referral (I-A – AHA 2025)
 - ↓ death, MI, hospital readmissions, and ↑ functional status and QOL
- Driving restrictions

Myocardial infarction with non-obstructive coronary artery disease (MINOCA)

Epidemiology:

- Represent 6-15% of all myocardial infarctions
- Younger, more likely female, more common in black race
- Less likely to have hyperlipidemia

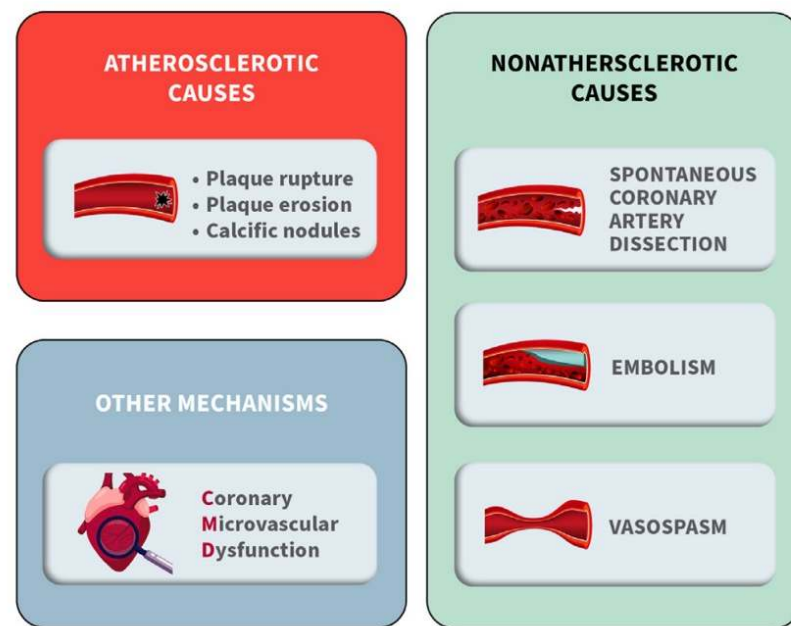
Definition: Diagnosis is made after angiography

1. Troponin elevation + one of:
 - symptoms of ischemia
 - ischemic ECG changes / new Q waves
 - new wall motion abnormality/perfusion defect
2. *No obstructive CAD* (>50% stenosis) on angiogram (also no SCAD/embolus)
3. No clinically overt specific cause for acute presentation (myocarditis, Takotsubo)
 - Also does not include type 2 MI (supply-demand mismatch e.g. sepsis, tachyarrhythmia, anemia, hypotension)

Mechanisms:

- **Plaque disruption:** rupture, erosion, calcific nodules → thrombus/embolism (may not appear on angiogram)
- **Vasospasm:** triggers include allergens (Kounis syndrome), 5-FU, cocaine, pheochromocytoma
- **Microvascular dysfunction:** vasospasm or impaired relaxation
- **Thromboembolism**
- **SCAD:** angiographic changes may be present but hard to detect

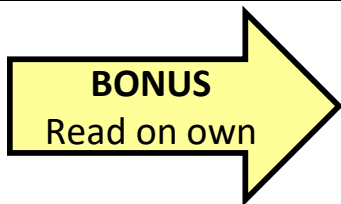
UNDERLYING CAUSES OF MINOCA



Myocardial infarction with non-obstructive coronary artery disease (MINOCA)

Prognosis: variable in studies but NOT A BENIGN ENTITY.

- Studies cited suggest up to 2X increased 5yr risk (HR) of mortality with MINOCA compared to those with obstructive CAD
- **Work-up:** algorithm provided in the Clinical Practice Update; combination of:
 - Coronary angiography with careful review of films +/- intracoronary imaging +/- provocation testing for vasospasm
 - Echocardiography: assess for Takotsubo cardiomyopathy pattern
 - Cardiac MRI: assess edema and fibrosis (LGE) pattern +/- perfusion
- **Treatment:**
 - Non-pharmacologic: cardiac rehabilitation, risk factor management, psychosocial support, avoid allergic triggers if present
 - Pharmacologic (expert opinion, this is a practice update not CPG, probably beyond the scope of a GIM exam):
 - Plaque disruption: DAPT, statins, BB, ACEi
 - Vasospasm: CCB, nitrates
 - Coronary microvascular dysfunction: BB, CCB, nitrates, ranolazine
 - SCAD: ASA + beta blockers, +/- DAPT, no routine statin unless other indication
 - Thromboembolism: anticoagulation? Closure of R>L shunt, thrombophilia consideration/hematology opinion
 - Unknown etiology (~15-20%): ACE/ARB and statins have been shown to lower MACE risk in patients with MINOCA, BB/DAPT have not been shown to reduce risk



Causes of Elevated Troponin

<i>Category</i>	<i>Specific Condition</i>	<i>Mechanism</i>	<i>Category</i>	<i>Specific Condition</i>	<i>Mechanism</i>
Ischemic (MI)	Type 1 MI	Plaque rupture (STEMI/NSTEMI)	Non-Ischemic	Myocarditis	Inflammation (viral, autoimmune, COVID)
	Type 2 MI	O ₂ supply-demand mismatch		Takotsubo	Catecholamine surge (stress)
	— Tachyarrhythmia	High HR → ↑ demand (AF, SVT)		Cardiac contusion	Blunt trauma (MVA, CPR)
	— Hypotension / Shock	↓ Perfusion pressure		Heart failure	Wall stress, chronic injury
	— Anemia	↓ O ₂ -carrying capacity		Arrhythmias	Demand-related injury
	— Hypoxia	PE, ARDS, respiratory failure		Infiltrative disease	Amyloidosis, sarcoidosis
	— Severe HTN	↑ Afterload, wall stress		Aortic stenosis	Pressure overload
	— Vasospasm	Prinzmetal angina		Aortic dissection	Involves coronary ostia
	— Coronary embolism	AF, endocarditis, prosthetic valve	Non-Cardiac	Pulmonary embolism	RV strain, hypoxia
				Pulmonary hypertension	Chronic RV pressure overload
				CKD/ESRD	↓ Troponin clearance + chronic injury
				SAH / Stroke / Seizure	Neurogenic myocardial injury



CCS 2023 Fitness to Drive Guidelines - CAD

Category		Private Car	Commercial Driver (Truck, Bus)
Coronary Artery Disease	STEMI/NSTEMI with LVEF >40%	2 weeks post d/c	1 months post d/c
	STEMI/NSTEMI with LVEF ≤ 40%	1 month post d/c	3 months post d/c
	STEMI/NSTEMI with no PCI performed	1 month post d/c	3 months post d/c
	UA (ACS without MI)	48h w/ PCI 7 d w/o PCI	7 d w/ PCI 1 month w/o PCI
	PCI in non-ACS context	48h	48h
	Asymptomatic CAD, stable angina	OK to drive	OK to drive
	CABG	1 month post d/c	3 months post d/c

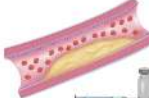
Major updates from previous guidelines:

- **LVEF is now recognized as a more important predictor** of sudden cardiac incapacitation than the type of MI
- In the previous guidelines, NSTEMI with no LV damage treated with PCI would only be restricted privately for 48 hours. This event is **now restricted minimum 2 weeks!**
- All STEMI would previously be recommended to be restricted 1 month privately. Now, a **STEMI with LVEF >40% is only restricted 2 weeks privately.**
- Elective PCI (PCI in non-ACS context) is restricted 48 hours privately (same as before) and 48 hours commercially (previously 7 days)

MCQ 1 2026

A 65F with DM2 and CCS3 angina presents with LAD and RCA disease on angiogram. Her surgeon advises that she go for CABG but she wants stents (PTCA). With regards to future risk of revascularization, What do you advise:

- a) CABG – per Freedom trial
- b) PTCA
- c) Medical treatment
- d) No difference

Multivessel Coronary Artery Disease and Diabetes n = 14,235	Percutaneous Coronary Intervention Patients after propensity score matching n = 4,301	Coronary Artery Bypass Graft Patients after propensity score matching n = 4,301
		
30-Day Death 	No Difference 2.4% vs. 2.3% p = 0.72	
All-Cause Mortality 	Favors CABG 8 Years: 27% vs. 19% Hazard Ratio: 1.39; 95% CI: 1.28-1.51	
Late Myocardial Infarction 	Favors CABG 8 Years: 16% vs. 7% Hazard Ratio: 2.32; 95% CI: 2.04-2.64	
Late Stroke 	No Difference 8 Years: 3.5% vs. 4.1% Hazard Ratio: 0.87; 95% CI: 0.71-1.08	
Late Repeat Revascularization 	Favors CABG 8 Years: 26% vs. 8% Hazard Ratio: 3.65; 95% CI: 3.24-4.34	

Cardiomyopathies & Heart Failure

Key Resources:

- 2025 CCS Guidelines – UPDATE on HF – Pharmacologic management of Heart Failure with non reduced EF (LVEF >40%) – hot off the press October 2025!
- 2024 CCS Guidelines – Hypertrophic cardiomyopathy
- 2022 CCS Guidelines – GLP1RA and SGLT2i for cardiorenal RR
- 2021 CCS Guidelines – HF update
- 2020 CCS Guidelines – Cardiac amyloidosis
- 2020 CCS Guidelines – Focused HF update (MR, SGLT2i, ARNI, amyloidosis)
- 2017 CCS Guidelines – Comprehensive HF update

MCQ 2026

A patient presents with new onset of shortness of breath. Exam reveals BP 190/100, JVP elevated, pansystolic murmur, no pleural rub. Echo shows mild MR and EF 60%. EKG suggests mild LVH. His heart failure is most likely due to?

- a) Myocarditis
- b) Mitral regurgitation
- c) Diastolic dysfunction
- d) Systolic dysfunction
- e) Pericarditis

Cardiomyopathy Guiding Principles

- 1) The **etiology** is an essential consideration → will heavily direct management and risk stratification (e.g. cardiac disease may be the first presentation for amyloid)
- 2) Most cardiomyopathy management plans include a significant component of **HF treatment**
- 3) Look for any other contributing comorbidities (OSA, smoking, DM, ETOH/drugs, iron-deficiency, arrhythmia, etc.)

HF Classification

BONUS

Read on own

Etiology Category

HFrEF (LVEF < 40%)

HFmrEF (LVEF ≥ 40%)

Divided into HFmrEF (40-50%) & HFpEF (>50%)

Ischemic Heart Disease

Myocardial infarction, ischemic cardiomyopathy

Microvascular ischemia, CAD without infarction

Hypertension

Can lead to dilated cardiomyopathy

Most common cause; LVH and diastolic dysfunction

Cardiomyopathies

Dilated (idiopathic, familial/genetic)

Restrictive (e.g., amyloidosis, sarcoidosis, endomyocardial fibrosis)

Toxins

Alcohol, anthracyclines, cocaine, methamphetamines

Less common but possible contributor

Infectious

Viral myocarditis (e.g., Coxsackie B), Chagas disease

Rare

Endocrine / Metabolic

Thyroid disorders, hemochromatosis, amyloidosis
Nutritional deficiencies: thiamine or selenium deficiency

Diabetes mellitus, obesity, metabolic syndrome

Valvular Disease

Aortic/mitral regurg or stenosis (late-stage effects)

Aortic stenosis (early), mitral stenosis

Arrhythmias

Tachycardia-induced cardiomyopathy (AF, SVT)

Atrial fibrillation → loss of atrial kick

Peripartum

Peripartum cardiomyopathy

Rare

Constrictive

Rare

Constrictive pericarditis

Aging

Less common

Very common → ↓ compliance, stiff ventricle

Genetic / Neuromuscular

Duchenne/Becker MD, lamin A/C, titin mutations causing familial dilated cardiomyopathy

Rare

Diagnostic approach

- **All patients with heart failure (regardless of LVEF):** ECG, echocardiogram, CBC, lytes, creatinine, ferritin, TSH, troponin, BNP, lipids, A1c
- **Common etiologies:** ischemic heart disease (perform ischemia testing if known CAD or risk factors), HTN/LVH, tachyarrhythmias, valvular disease
 - Non-invasive imaging to rule in/out CAD should be considered
 - Invasive (coronary angiography):
 - *Recommend* if HF **with** angina / Ischemic symptoms, likely to be good candidates for revascularization
 - Consider if: LVEF <35, at risk of CAD, **irrespective of** angina, likely good candidates for revascularization
 - Consider if: systolic HF and non-invasive coronary perfusion testing consistent with high risk
- **Workup for non ischemic cardiomyopathy**
 - Good History
 - Testing directed by history (eg) genetics if family history
 - Cardiac MRI for most with non-ischemic cardiomyopathy. (If available)
 - To help rule out infiltrative, ARVC

Acute HF Management

In general

- Establish Diagnosis ASAP
 - guidelines say <2h after contact in ED
- Start with ABCs, IV, O2, Monitor (IOM)
- Focused history: dyspnea, fatigue, edema, CP, meds
- Focused physical: congestion, valvular dz, perfusion
- CBC, lytes, urea/Cr, gluc, BNP, Troponin, **lactate**
- ECG, CXR
- Echo (w/in 48h)
- Consult CCU/Cardiology
- Consider potential etiologies or triggers for decompensation to direct care
 - Ischemia, HTN, Anemia, rapid afib, infection, med or diet non adherence, NSAIDs, PE, thyroid...
- ❖ If hypertensive or MR → short acting vasodilators (nitroglycerin drip / patches)
- ❖ Noninvasive → invasive ventilation (↓preload, afterload)
- ❖ Atrial Arrhythmia → chicken or egg?
 - ACLS if unstable
 - Achieve euvolemia
 - Beta blocker – continue if stable, do not start if in shock or +++ overloaded
 - Digoxin can help w/ rate control if soft BP
- ❖ If Cardiogenic shock → see next slide

Recognizing Cardiogenic Shock

❑ Clinical criteria:

SBP < 90 mmHg for ≥ 30 minutes

OR Requirement for vasopressors or mechanical support to maintain SBP ≥ 90

AND

Signs of end-organ hypoperfusion, such as:

- Cool extremities
- Oliguria (<30 mL/hr)
- Altered mental status
- Metabolic acidosis (↑ lactate, ↓ pH)

❑ Hemodynamics criteria:

- Cardiac index (CI) < 2.2 L/min/m²
AND
- Pulmonary capillary wedge pressure (PCWP) > 15 mmHg

Parameter	Value
CO / CI	↓ (low output)
PCWP	↑ (volume backing up)
SVR	↑ (vasoconstriction)
MAP	↓ (hypotension)
Mixed Venous Saturation (SvO₂)	↓ < 65% (reduced perfusion)
Lactate	↑ (tissue hypoxia)

Shock Grid to Guide Therapy

	Dry	Wet
Warm	(Distributive shock) Low PCWP Normal/High CI Low SVR	(Mixed shock) High PCWP Low CI Low/normal SVR
Cold	(Euvolemic cardiogenic shock) Low PCWP Low CI High SVR	(Classic cardiogenic shock) High PCWP Low CI High SVR

- *What is the pump doing?*
- *What are the filling pressures / volume status?*
- *What are the blood vessels doing?*

CI = cardiac index = CO/BSA (cardiac output/body surface area)

PCWP = pulmonary capillary wedge pressure

SVR = systemic vascular resistance

Inpatient or outpatient management?

- In your applied exam, consider safety and resource utilization when treating acute heart failure
 - For oral situations, it should be obvious if the patient needs hospitalization (i.e. they are hypoxemic or unstable)
- Risk scores exist to estimate patient mortality when presenting to the ED with HF (e.g. Emergency Heart Failure Mortality Risk Grade [EHMRG] score)
 - Freely available at MDCALC.com – gives 7day risk of mortality
 - Use of risk score and close standardized outpatient care reduces mortality (COACH-HF trial (NEJM, 2023))

Long-term Management of HFrEF

Establish etiology

Practically: Hx/PE/labs +

- Echocardiogram (r/o non-myocardial causes)
- BNP: 2017 CCS recommends BNP if diagnosis unclear *and* for prognosis
- Assess for CAD (very often coronary angiography)
- CMRI for non ischemic etiologies

Quantify degree of disability

NYHA classification (I-IV)

- Allows clinician to track progress
- Becomes important for therapy considerations

Long-term Management of HFrEF

Risk factor modification & lifestyle interventions

- Exercise (flexibility, aerobic, +/- resistance)
- Salt restriction (<2-3g/d) widely recommended (CCS, AHA, ESC)
 - NB SODIUM-HF trial 2022 showed that strict salt restriction <2g/day did not improve HF related hospital visit or CV death
- +/- fluid restriction (<2L/d)
- Smoking cessation
- EtOH avoidance
- Discuss Driving Safety (see bonus slides)

Treat HF-associated comorbidities (*see later*)

Multidisciplinary care model

Early advanced care planning discussions

Interventional therapy considerations

- ICD, CRT (*see later*)
- Surgery / percutaneous treatment of functional MR (*see later*)
- Revascularization in ischemic cardiomyopathy – controversial, see next slide

Advanced HF

- Really low EF, ++hospitalizations, NYHA IV, organ dysfunction, cardiac cachexia, high 1-year mortality, bad CPET performance...
- May need: Mechanical circulatory support (eg) VAD, Transplant workup, or palliative care

Once stabilized, reassess at least annually with clinical assessment + LVEF assessment

Revascularization and ischemic cardiomyopathy

- Controversial area where new evidence is emerging
- **STICH trial** (CABG + medical therapy vs. medical therapy) showed improved long-term mortality with revascularization with CABG compared to medical therapy alone
- Subsequent PCI trials, including the most recent **REVIVED-BCIS**, looked at patients with LVEF <35% with multivessel disease and viability on cardiac MRI, showed no improvement in outcomes/LVEF with PCI
- If choosing an option for revascularization for ischemic cardiomyopathy, CABG would likely lead to improved outcomes compared to PCI with the available evidence. However, optimizing medical therapy is paramount in ALL patients!

HFrEF Pharmacotherapy

Guideline recommended pharmacotherapy for HFrEF

“Backbone” of HFrEF therapy now **QUADUPLE Tx**

- ARNI $\leftrightarrow$ ACEI/ARB
- MRA
- SGLT2i (even if not diabetic!)
- BB [only start when euvolemic and hemodynamically stable] or

Other tips

- Goal for target doses by 3-6 months
 - No preference for uptitration in parallel vs. sequentially, no preference for order of meds started
 - “initiate standard therapies and titrate every 2-4 weeks...”
 - Avoid CCBs with LVEF < 40% (amlodipine ok for HTN)
 - Do not start BBs on NYHA IV patients
- ARNI initiation requires 36h “washout period” after ACEI use

ARNI in clinical practice:

Start ARNI: Hospitalized w new dx HFrEF

Switch to ARNI if:

- Hospitalized for HF on ACE/ARB
- Symptomatic (NYHA2+) despite max ACE/ARB

Not practical if: no drug plan
(\$7 per day vs \$0.20 per day for valsartan only)

TREAT COMORBIDITIES PER CCS HF RECOMMENDATIONS (INCL. AF, FUNCTIONAL MR, IRON DEF, CKD, DM)

DIURETICS TO RELIEVE CONGESTION (TITRATED TO MINIMUM EFFECTIVE DOSE TO MAINTAIN EUVOLEMIA)

HFrEF: LVEF $\leq$ 40% AND SYMPTOMS

Initiate Standard Therapies

ARNI or **ACEi/ARB**
then substitute **ARNI**

BETA BLOCKER

MRA

SGLT2 INHIBITOR



Assess Clinical Factors for Additional Interventions

HR >70 bpm and
sinus rhythm

- Consider ivabradine*

Recent HF hospitalization

- Consider vericiguat**

Black patients on optimal GDMT,
or patients unable to tolerate
ARNI/ACEi/ARB

- Consider combination
hydralazine-nitrates

Suboptimal rate control for
AF, or persistent symptoms
despite optimized GDMT

- Consider digoxin

Initiate standard therapies as soon as possible and titrate every 2-4 weeks to target or maximally tolerated dose over 3-6 months



Reassess LVEF, Symptoms, Clinical Risk



**NYHA III/IV, Advanced HF
or High-Risk Markers**

CONSIDER

- Referral for advanced HF
therapy (mechanical circulatory
support/transplant)
- Referral for supportive/palliative care



**LVEF $\leq$ 35% and
NYHA I-IV (ambulatory)**

Refer to CCS CRT/ICD
recommendations



**LVEF $>$ 35%,
NYHA I, and Low Risk**

Continue present management,
reassess as needed

NON-PHARMACOLOGIC THERAPIES (EDUCATION, SELF-CARE, EXERCISE)

ADVANCE CARE PLANNING AND DOCUMENTATION OF GOALS OF CARE

Long-term Management of HFrEF

See bonus
material for
more Rx info

CCS 2021 HF

All Patients						
Indication	Beta-blocker	MRA	ARNI	SGLT2 I	Vericiguat	Omecamtiv
	Clinically euvolemic Hemodynamically stable	eGFR > 30 K < 5.0 Stop K supplement as needed	eGFR > 30, K < 5.0 and SBP > 100	Clinically euvolemic No concerns of fluctuating renal function eGFR > 20	Evidence of WHF NYHA class II, III, or IV LVEF < 45 Elevated natriuretic peptide level	Currently hospitalized for HF (inpatients) or hospitalization < 1 year (outpatients) Increased NT-proBNP level On GDMT and device in accordance with guidelines
	Approaching euvoemia with good BP Rapid AF believed to be contributing to ADHF episode	History of hyperkalemia Declining renal function (overdiuresis)	SBP 95-100	Declining renal function (overdiuresis) SBP 95-100	Inpatients: SBP ≥ 100 mm Hg Off IV treatments ≥ 24 hours	Non-revascularized CAD with ischemia Omecamtiv Not available yet in Canada
	Significant volume overload Poor perfusion	AKI eGFR < 30 K > 5.0	SBP < 90 or eGFR < 30 K > 5.0	eGFR < 20 Type I diabetes History of DKA	SBP < 100 mm Hg Use of long-acting nitrates, soluble guanylate cyclase stimulators, or phosphodiesterase type 5 inhibitors Use of intravenous inotropes or implantable LVAD NT-proBNP > 8000 pg/dl	Hemodynamic or clinical instability: use of mechanical support or intravenous medication SBP < 85 eGFR < 20

Long-term Management of HFrEF

Quadruple therapy for all

And CONSIDER :

- | | |
|--|-------------------------|
| ☐ Sinus rhythm and HR >70: | Ivabradine |
| ☐ Recent HF hospitalization: | Vericiguat [^] |
| ☐ Black* pts on optimal GMT: | Hydralazine/ISDN |
| ☐ Unable to take ACE/ARB/ARNI: | Hydralazine/ISDN |
| ☐ Persistent symptoms despite Rx
above or poor rate cntrl with AFIB: | Digoxin |

Titrate drugs every 2-4 weeks over 3-6 mos then reassess:

- | | |
|--|--|
| ❖ If NYHA 1, LVEF >35, low risk: | Continue current mgmt |
| ❖ If NYHA 1-4, LVEF ≤35 ambulatory: | Check out Device therapy guidelines |
| ❖ If NYHA 3-4, high risk, advanced HF: | Advanced care plan, palliation
Refer for advanced HF therapy (LVAD etc) |

*See hidden BONUS slides
for more Rx info*

[^]Vericiguat Health Canada
approved 2023, now available

*STRONG-HF trial (Lancet 2022)
showed that a strategy of early
hospital follow-up (7-14 days)
and regular visits to initiate and
titrate HF medications was
associated with lower HF
hospitalizations and all cause
death vs. usual care*

BONUS

Read on own

SGLT2i Practical Tips

- Block SGLT2 channel (prox conv tubule) and lead to glycosuria and increase flow in DCT/CD so decreases renin secretion by JG cells (*+/- other pleomorphic effects*)
- May need to decrease diuretic dose
- Do not require dose titration
- SGLT2i are **contraindicated** for patients with true insulin deficient type 1 diabetes
- Generally, well tolerated
- Counsel and monitor patients for:
 - Genital Mycotic Infection (GMI); less commonly UTIs (bacterial and mycotic)
 - Do not start during active genital/urinary infection
 - Temporary ↓ eGFR up to 15%, resolves generally in 1-3months (in fact, *renal protective effects*)
 - **Do not start if eGFR is <20**
 - Hypoglycemia: usually not seen in the absence of concomitant insulin or secretagogues.
 - Sick Day Med List → hold SGLT2i
 - DKA 0.1% which is best detected with measurement of serum ketones (euglycemic DKA)
 - Do not start in diabetic patient with decompensated diabetes (eg polydipsia, polyuria, marked hyperglycemia – risk of DKA – they need insulin (see ENDO lecture)
 - Very rarely causes hypotension

BONUS
Read on own

Other Important HF Medication Considerations

- **Beta blockers**
 - Use bisoprolol, carvedilol or metoprolol SUCCINATE (*extended release; not available in most of Canada*)
 - NYHA IV patients should be stabilized prior to the initiation of a beta-blocker
 - Chronic beta-blocker should be continued in acute heart failure unless patient symptomatic from hypotension or bradycardia
- **Ivabradine**
 - Acts on **SINUS NODE** to reduce heart rate in patients without reducing BP or contractility (need to be in **SINUS RHYTHM**)
 - Maximize dose of beta blocker first, **use if hospitalized in last 12 months for CHF + HR > 70**
- **ACE/ARB/ARNI (Angiotensin receptor neprilysin inhibitor)**
 - ARB should be used if patient intolerant to ACEI or develops angioedema (although angioedema can still occur with ARBs – rare). **ARNI contraindicated if history of hereditary (familial) or idiopathic angioedema**
 - When switching from ACE to ARNI, **36 hour washout period** important to lower risk of angioedema. No washout required for ARB/ARNI switch
- **Vericiguat** (now approved in Canada)
 - Novel oral soluble guanylate cyclase stimulator, works by enhancing effects of NO
 - VICTORIA trial → NYHA II-IV HF (LVEF <45%) recent hospitalization or IV diuretic therapy
 - Primary outcome (CV death, HF hospitalization) reduced by 10% vs. placebo
 - Better for lower LVEFs
 - Well tolerated
 - More anemia in the vericiguat group
- **Omecamtiv mecarbil (OM)** (Mentioned in CCS 2021 guidelines, not approved by Health Canada)
 - Myosin activator studied in the GALATIC-HF trial showed a modest 2.1% reduction in composite of hospitalization for HF or CV death
 - Not approved in Canada yet and no formal recommendations

ICDs in HF

General principles:

- Primary (*low EF*) vs. secondary prevention considerations
- Primary prevention devices considered appropriate after:
 - 3m OMT
 - 3m post-revascularization
 - 40d after MI
- Patients should have expected longevity > 1 year (or considerations for VAD, transplant)
- Caution for patients with NYHA IV, not expected to improve
- **CCS 2017 HF Guidelines most prevalently used**
- Add CRT to ICD?

Indications for Primary Prevention ICD

CCS 2016 ICD Guidelines

Both ischemic (ICM) + nonischemic cardiomyopathy (NICM)

- LVEF ≤ 30 (*strong recom.*)
- LVEF 31-35% (*weak recom.*)

CCS 2017 HF Guidelines

- ICM, NYHA II-IV, LVEF $\leq 35\%$
- ICM, NYHA I, LVEF $\leq 30\%$
- NICM, NYHA II-III, LVEF $\leq 35\%$



CCS 2016 ICD, CCS 2017 HF
ICM = ischemic cardiomyopathy
NICM = nonischemic cardiomyopathy

ICDs for 2^o prevention

ICD Consideration-Secondary Prevention Ischemic and non-ischemic Population

1. Cardiac Arrest (VT-VF)
2. Sustained VT in the presence of significant structural heart disease
3. Sustained VT >48 h post MI or revascularization

Reversible Cause

YES

Treat cause

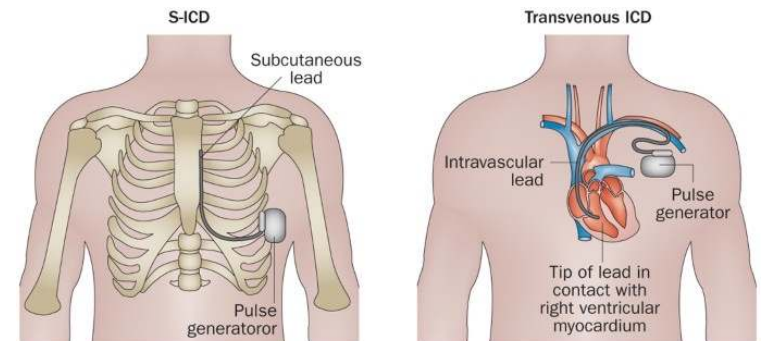
NO

ICD Indicated
Strong recommendation (1 and 2)
Weak recommendation (3)

ICD systems

Transvenous: most common, lead through subclavian vein into RV, can pace and defib.

Subcutaneous: devices sits in the axilla, lower risk of venous stenosis and endocarditis, shorter battery life, useful for younger patients. Does not pace!



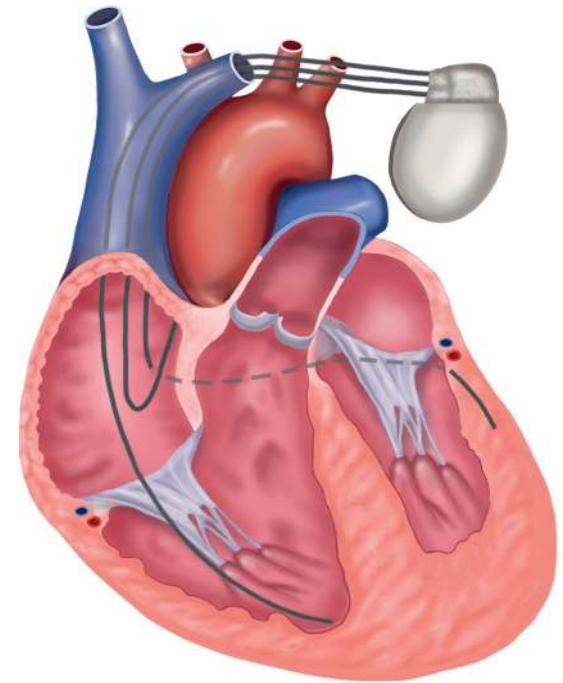
Nature Reviews | Cardiology

CCS 2016 ICD
Gold and Lewis, Nature Cardiology Reviews 2015

Cardiac Resynchronization Therapy (CRT)

General CRT Principles:

- Cardiomyopathy with LBBB = electromechanical LV/RV dyssynchrony (RV contracts before LV due to LBBB)
- Dyssynchrony = $\downarrow$ SV/CO, $\uparrow$ MR, $\uparrow$ filling pressures, $\downarrow$ functional status, $\uparrow$ hospitalizations, $\uparrow$ death
- Dyssynchrony tends to correspond to abnormal QRSd on ECG
- CRT paces RV and LV to "Resynchronize"
 - Options: CRT-D (pacing + defibrillation), CRT-P (pacing only)
- CRT has been shown to $\downarrow$ HF symptoms, hospitalizations, death
- 1/3 rule applies (*approx.*) for response rates *without* careful selection
 - 1/3 (improve) 1/3 (no major change) 1/3 (get worse)



From: Nemer DM et al. JACC: Clinical Electrophysiology. 2021 Jan 1;7(1):62-72.

CRT Indications

Slam dunk (*strong recommendation*)

- In Sinus Rhythm
- Symptoms (NYHA II-III, ambulatory IV)
- On GDMT
- LVEF $\leq 35\%$
- Typical LBBB
- QRSd $\geq 130\text{ms}$

May respond (*weak recommendation*)

- In Sinus rhythm
- Symptoms (NYHA II-III, ambulatory IV)
- On GDMT
- LVEF $\leq 35\%$
- Non-LBBB
- QRSd $\geq 150\text{ms}$

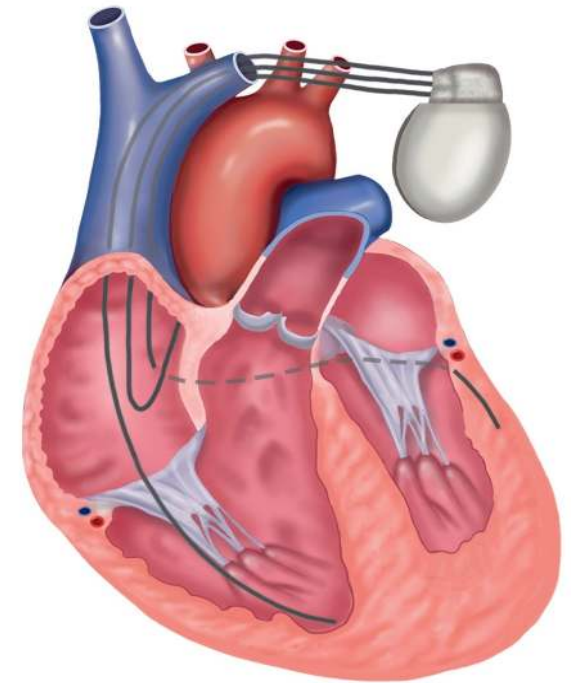
Marginal candidates (*weak recommendation*)

- Permanent AF
- Patients who require chronic RV pacing + symptomatic HFrEF

The addition of ICD therapy [CRT-D] is appropriate if ICD indications are independently met

*CRT-D = CRT-defibrillator

**CRT-P = CRT-pacemaker



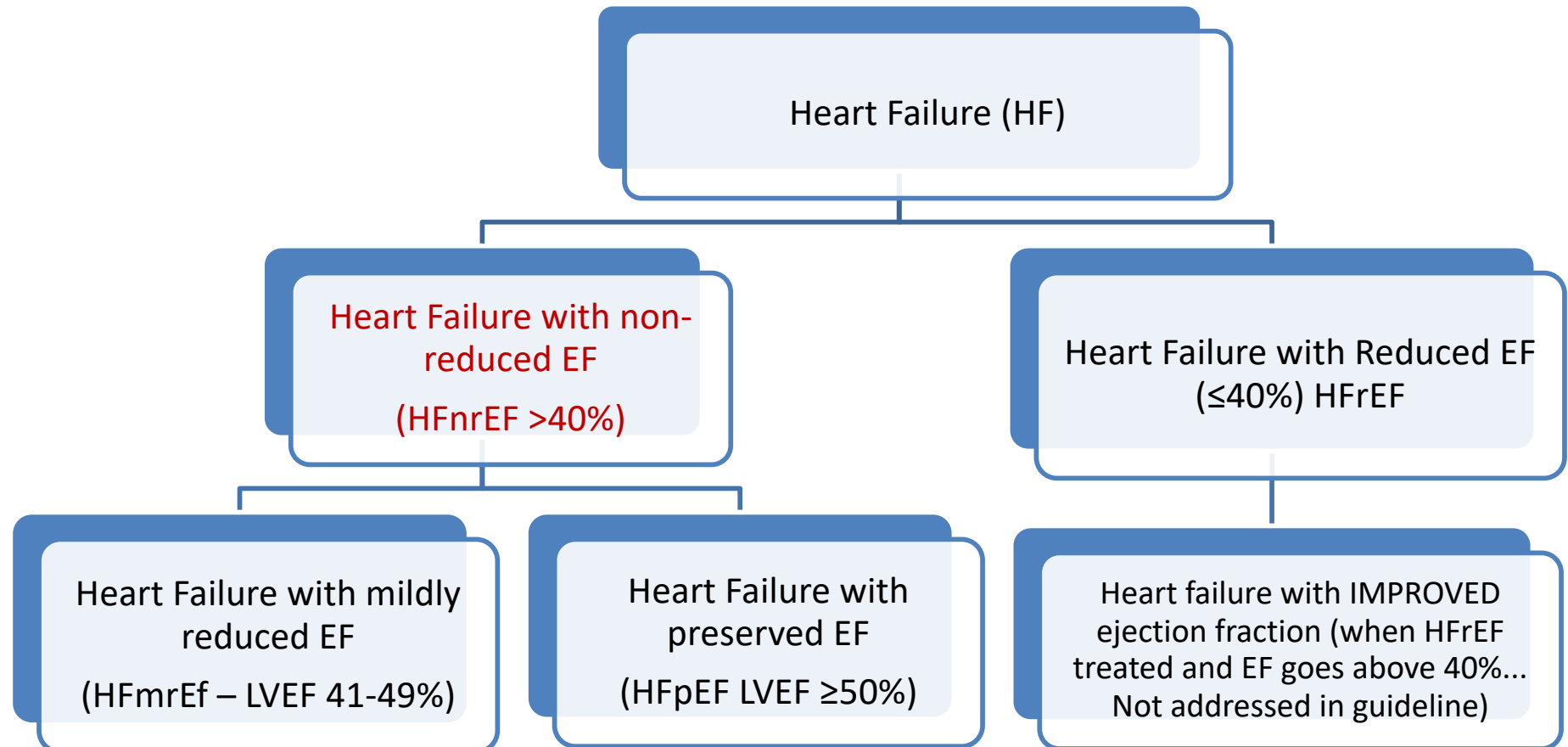
From: Nemer DM et al. JACC: Clinical Electrophysiology. 2021 Jan 1;7(1):62-72.

When Should We Worry About HFrEF Patients?

“I NEED HELP!” Consider referral to heart failure specialist if:

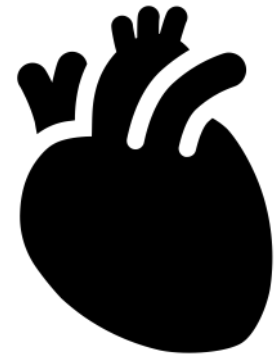
- I - IV inotropes
- N - NYHA IIIB/V or persistently elevated BNP
- E - End organ dysfunction
- E - EF $\leq$ 35%
- D - Defib shocks
- H - Hospitalizations >1
- E - Edema despite escalating diuretics
- L - Low systolic BP $\leq$ 90, tachycardia
- P - Progressive intolerance or down-titration of GDMT, Predicted mortality high risk in next 1 year

HFnrEF = New terminology Oct 2025



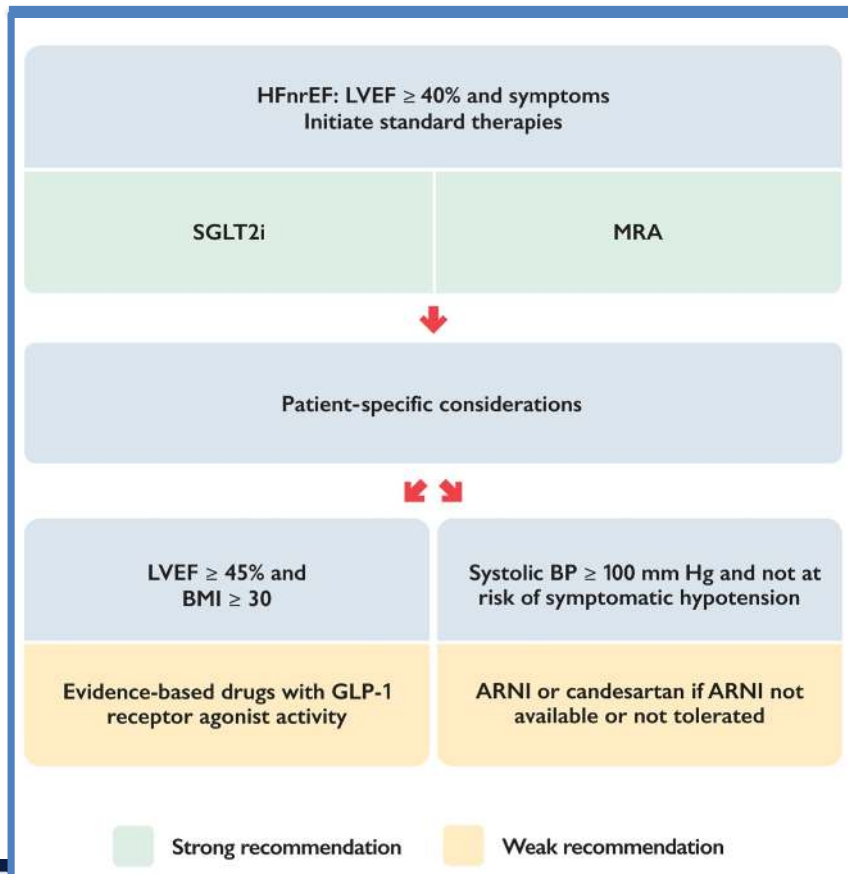
HF with non-reduced EF (HFnrEF)

- Common (now about 50% of all HF)
 - especially among older patients, females, other comorbidities (HTN, DM, OSA)
 - VERY heterogenous group vs. HFrEF
- Management principles:
 - (i) improve symptoms and (ii) treat risk factors
 - Non-pharmacologic interventions
 - Treat comorbidities – HTN, Afib, CKD, Valvular disease
 - Advanced care planning
 - 20% of HF admissions are re-admitted within 30d
 - 1 year unadjusted mortality 27% of patients admitted with HF
 - Exercise and self-care teaching
 - Pharmacologic interventions:
 - SGLT2i, MRA, ARNI, GLP1Ra (if LVEF $\geq$ 45% and BMI $\geq$ 30)
 - Loop diuretics to achieve euvolemia
 - Outcomes to reduce heart failure hospitalizations
 - to date no intervention reduces mortality



Virani S. Et al. Canadian Journal of Cardiology 41 (2025) 1857—1874

HF with non-reduced EF (HFnrEF)



Recommend SGLT2i for all (DAPA or EMPA)

- CrCl >20 , usual exclusions (eg. T1DM)

Recommend MRA for almost all (spironolactone, eplerenone, finerenone)

- CrCl >30 , K <5.4 , SBP >95 without symptoms
- Favour finerenone if diabetic kidney disease

Suggest ARNI for some

- IF low risk of hypotension

If SBP 100-120 start lowest dose 24/26mg daily

- Small reduction in Heart failure hospitalizations (HR 0.89, 95% CI 0.81 – 0.98)
- Cost considerations – unsure if cost effective at system level

Suggest GLP1RA if HF with LVEF $\geq 45\%$ and BMI ≥ 30
to $\downarrow$ HF hospitalizations and improve QOL (STEP-HFpEF trial)



BONUS

Read on own

Trial Summary: HF with non-reduced EF (HFnrEF)

- Guideline-based recommendations:
 - BP control, based on HTN Canada guidelines
 - Consider candesartan **CHARM-Preserved trial – ↓ HF hospitalizations but otherwise negative trial for MACE*
 - SGLT2i for all (even if not diabetic) to ↓ HF hospitalizations
 - Consider MRA (specifically **non steroidal MRA!**)
 - TOPCAT trial spironolactone – overall negative trial for MACE; ↓ HF hospitalizations but overall 'all cause' hospitalizations the same!
 - ***NEW* FINEARTS-HF trial (NEJM 2024):** finerenone, a novel non-steroidal MRA, ↓ the composite of worsening HF events and CV death compared to placebo, driven by reducing HF events, with increased risk of hyperkalemia

BONUS
Read on own

Trial summary: GDMT in HFnrEF

- **CCS 2022 strongly recommends SGLT2i in this population to reduce HF hospitalizations**
 - Based on consistent data from EMPEROR-Preserved and DELIVER trials (RCTs studying empa 10mg daily, dapa 10mg daily vs. placebo in HFpEF with LVEF>40%, NYHA II-IV) showing reduction in HHF
 - No reduction in mortality or composite kidney outcome however
 - Limitations: not for T1DM, dialysis or severe reduced eGFR <20-25 ml/min/1.73m²
- **MRA, ARB and ARNI to be used to reduce HF hospitalization**
 - MRA/ARB recommendations are consistent with CCS 2017 HF guidelines, based on TOPCAT/CHARM-Preserved trials
 - PARAGON-HF (ARNI vs. valsartan in HFpEF with LVEF>45%) did not show any benefit to ARNI overall, however subgroup with LVEF 45-49% may derive benefit so ARNI could be considered for patients with HFnrEF and LVEF <50%
- ***NEW*** Role for GLP-1 agonists – **officially in guidelines with CCS HFnrEF 2025, based upon evidence below**
 - **STEP-HFpEF** trial (NEJM, 2023) showed semaglutide improved *symptoms* and weight loss in patients with HFpEF and obesity
 - **The pooled analysis of four GLP-1 agonist trials in HFpEF (Lancet 2024)** showed reduction in the primary outcome of CV death and hospitalization for HF, driven by hospitalization for HF
 - **SUMMIT** trial (NEJM 2024) showed treatment with **tirzepatide led to a lower MACE** or worsening heart failure and improved health status in patients with heart failure with preserved ejection fraction and obesity.

(see Endo slides, they are recommended if Atherosclerotic CVD in T2DM to reduce MACE)

CCS 2017 HF
ACC/AHA 2022 HF

Special HF Populations and conditions

- **Diabetes:**

- Metformin (if eGFR ≥ 15 **dose adjust see Endo Lecture**) may be considered a first-line agent for T2DM in HF, don't forget **SGLT2i if eGFR ≥ 20 to reduce HHF**
- GLP1RA reduce all cause CV mortality/MACE in T2DM w Atherosclerotic CVD or multiple RFs but not specifically indicated for Heart Failure mgmt., use as indicated by CCS/CDA guidelines elsewhere
- AVOID: saxagliptin (*but other DPP-4 inhibitors ok*), thiazolidinediones



- **Exercise, Smoking/ETOH, and Education**

- Regular exercise, self-teaching around comorbidities (fluid restriction, daily weights), and smoking/ETOH cessation all directly recommended in HF guidelines



- **Iron Deficiency:**

- Ferritin < 100 or ferritin $100 - 299$ + Tsat $< 20\%$
- Consider therapy with IV iron

- **Sleep Apnea**

- CPAP for patients with OSA, as per other guidelines
- NO recommendation for ventilation in the setting of Central SA

- **Chronic Kidney Disease**

- Continue GDMT for patients on chronic dialysis
- Best data for ACEi/ARBs, BBs
- MRAs safe
- **SGLT2i recommended if GFR ≥ 20 AND CHF** (CCS, KDIGO) even if no diabetes to Prevent **composite** of decline in eGFR, progression to ESRD, kidney death, all cause mortality, nonfatal MI, hosp for HF – KDIGO 2024
(see next slide for drugs and dosing and eGFR cutoffs)

- **Management of CHF in Black Patients**

Frequently asked @ IMR in past : do you give ACE/ARB/ARNI to Black Patients? Or start with Hydralazine/ISDN?

- ANSWER: In presence of CAD and LV dysfunction (or LV dysfunction from non-ischemic etiologies) still treat with ACEi/ARB/ARNI. Add hydralazine/nitrate combination to patients who have ongoing symptoms despite ACEi/ARB/ARNI use.
- A-Heft trial showed ISDN-Hydralazine helpful in this population but NOTE patients were already on Ace or ARB and Beta Blocker at trial enrollment.

Prescribing notes:

SGLT2i Doses and eGFR Cutoffs in HF and CKD

SGLT2i	HF dosing	eGFR cutoff	Notes
canagliflozin	100 mg/d	>30	No trial of canagliflozin in people without diabetes =NOT recommended/mentioned in CCS 2025 HFmrEF guideline
dapagliflozin	10 mg/d	≥25	Studied in patients without diabetes.
empagliflozin	10 mg/d	≥20	

Text of CCS 2022 Cardiorenal guideline caution re: individual SGLT2i eGFR cutoffs and initial dosing

Diuretics in Heart Failure

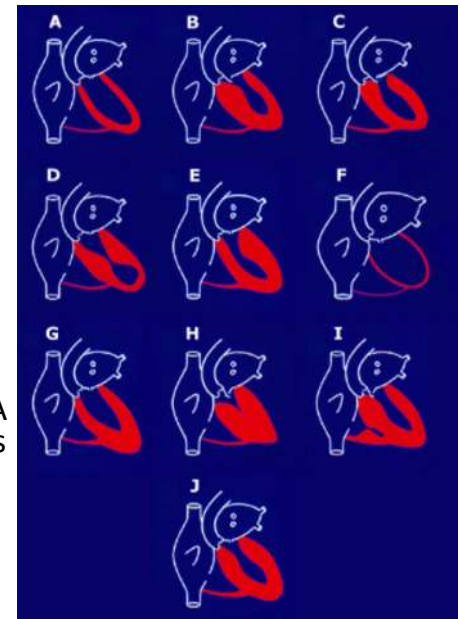
- Ubiquitously used, less rigorously studied



- ARNI and SGLT2i may reduce diuretic requirements when introduced – monitor carefully
- CHF Exacerbation: Continuous infusion vs bolus furosemide?
 - Favour continuous infusion to more quickly achieve diuresis, but more studies needed given small and heterogenous studies (Cochrane review 2020)
- Metolazone (TZD diuretic) is generally the second diuretic to be added to augment diuresis
 - Monitor for hypokalemia, hyponatremia, contraction alkalosis, renal function due to brisk diuresis associated with these agents used with loop diuretics
- Contraction metabolic alkalosis (low urine chloride, high urine sodium) can occur on high dose loop / TZD diuretics
 - = Consider adding ACETAZOLAMIDE – helps with alkalosis + RCT of IV acetazolamide 500 daily in addition to standard therapies for CHF exacerbation showed pts more likely to be ‘decongested’ in 72h (ADVOR trial, Mullens et al, NEJM 2022)

Hypertrophic Cardiomyopathy

- Mostly, genetic variants that code for proteins of the sarcomere
- Most common phenotype is **asymmetric septal hypertrophy**
- Often associated with:
 - Dynamic LV outflow tract obstruction (LVOTO)
 - Systolic anterior motion (SAM) MV → eccentric MR
 - Papillary muscle abnormality
- Presentations: Chest pain, dyspnea, syncope, arrhythmia, stroke, HF, SCD
- Broad management principles:
 - Family screening/genetics for **1st degree relatives**
 - Sports participation: “shared decision making” in guidelines. Recently published LIVE-HCM (JAMA 2023) showed that vigorous/competitive exercise in young (mean age 39) patients with HCM was not associated with increased mortality/syncope/ICD shocks compared to nonvigorous exercise.
 - **OAC for ANYONE with AF and HCM (CHADS65 does not apply, nor do CV guidelines)**
 - ***NEW* CCS 2024 HCM Practice Update:** Consider longer duration Holter monitoring (e.g. 7-14 days) for patients with HCM at high risk of AF (severe LA dilation, high burden atrial ectopy, unexplained palpitations, embolic events)
 - Consideration for ICD if:
 - Sustained VA or prior cardiac arrest (Class I)
 - FMHx of SCD, LV wall thickness >30 mm, unexplained syncope, apical aneurysm, LVEF <50% (Class IIa)
 - Extensive LGE on MRI or NSVT on Holter monitoring (Class IIb)



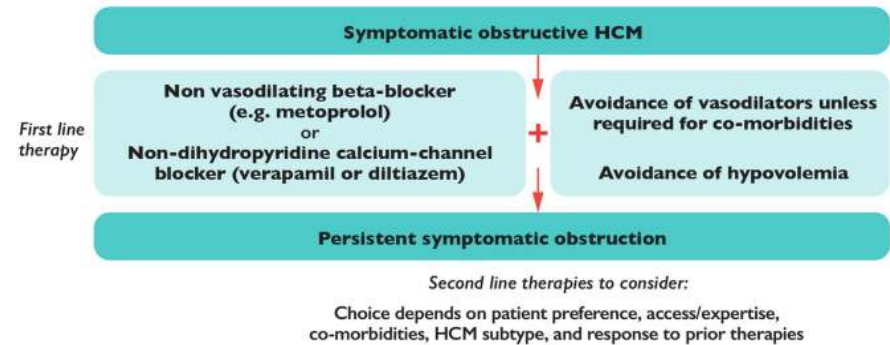
From: UpToDate®

AHA HCM 2020
CCS HCM 2024
AHA HCM 2024

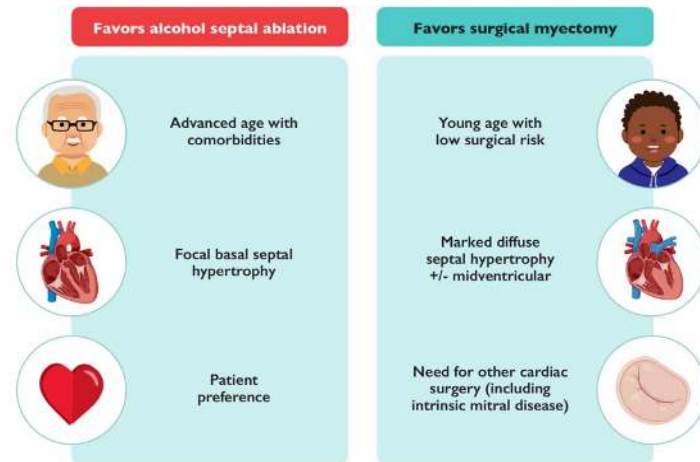
Hypertrophic Cardiomyopathy

For LVOT obstruction management:

- Avoid hypovolemia/stop diuretics/stop vasodilators
- **First line: B-blockers/non-dihydropyridine-CCBs**
- **Second line:**
 - Disopyramide/mavacamten (cardiac myosin inhibitors) if still symptomatic despite BB and/or CCB
- **For refractory symptoms of LVOTO:**
 - septal myomectomy/ETOH ablation



Persistent symptomatic obstruction considered for invasive therapy



PROS



CONS



Drug therapy (generally attempted prior to invasive therapy)

Disopyramide	Cardiac myosin inhibitors (Mavacamten)
• Long experience • Antiarrhythmic effect can be useful for atrial fibrillation • Non-invasive	• RCT showing significant improvement of symptoms, exercise capacity, QoL, imaging markers, and biomarkers • Once daily use • Non-invasive
• Recurrent shortage in recent years • Anticholinergic side effects • Only short-acting available in Canada: taken 3 times a day • QT prolongation • Efficacy based on observational data	• Shorter experience • Risk of systolic dysfunction requiring need for echocardiography every 1-3 months • Cost • Pediatric trials pending

Invasive septal reduction therapy (generally reserved when medical therapy fails)

Alcohol septal ablation	Surgical myectomy (+/- mitral intervention)
• Less invasive than surgery, preferred in older patients with co-morbidities • Sustained efficacy for LVOT obstruction	• Likely most effective and durable therapy for obstructive HCM • Useful for diverse septal morphologies • Concomitant correction of mitral valve anomalies contributing to obstruction/symptoms
• Access to local high volume centre • Risk of AV block requiring pacemaker implantation ($\leq 10\%$) • Feasibility depends on septal morphology and coronary anatomy • Not used in pediatric population	• Access to local high volume centre • Invasive therapy • Risk of AV block requiring pacemaker implantation ($\leq 5\%$)

BONUS

Read on own

HCM

(check out physical exam videos at internalmedicinereview.ca)

Principles for physical exam

- An underfilled LV (decreased preload) will cause the mitral valve to be close to the LVOT, causing worse obstruction
- Increased afterload "opens" the LVOT, increases ventricular volume, decreasing gradient/improves obstruction = reduced murmur
- REDUCED murmur
 - Bradycardia (eg beta blockade) gives more diastolic filling time and increases ventricular volume, which improves obstruction = reduced murmur
 - Passive leg raise (↑ venous return) – reduced murmur
 - Handgrip (↑ afterload)- reduced murmur
- INCREASED MURMUR
 - Valsalva or standing up - ↓ venous return - increased murmur
 - Reduce afterload (ACEi) – increased murmur
- Treatment involves adequate volume status, beta-blockers +/- disopyramide, **avoiding** afterload reducing agents (ACEi) and preload reducing agents (nitrate, diuretics)

Cardiac Amyloidosis

- **CA** is usually a restrictive cardiomyopathy
- Can be inherited or acquired
- Most commonly – **AL** (*cancer-associated*) vs. **ATTR** (*wild type vs. hereditary = slowly progressive more common to clinically present in older men*)
- Presents with HF (usually HFpEF), presyncope/syncope, atrial arrhythmia (Afib, sometimes Ventricular arrhythmias), bradyarrhythmia, higher rates of AS as well
- Patients frequently also have extracardiac manifestations:
 - Autonomic dysfunction
 - Orthostatic hypotension
 - Gastroparesis
 - Sexual dysfunction
 - Sweating abnormalities
 - Neuropathy
 - Carpel tunnel syndrome
 - Renal insufficiency/Nephrotic syndrome

Suspect if pts with HF and any one of:
"normotensive" LVH
"low flow, low gradient AS" w EF >40
Unexplained sensorimotor neuropathy
/ dysautonomia
Bilateral Carpal Tunnel history

Cardiac Amyloidosis

Evaluation involves

- ECG – low voltage, pseudoinfarction pattern
- Echo – LVH, diastolic dysfunction
- +/- CMRI
- BW – S/U PEP, serum free light chains → AL
- Tc-99m-PYP scan → ATTR (both wild type + hereditary)
- Genetic testing → hereditary ATTR
- +/- biopsy

Therapy (nuanced from standard HF therapy)

- Diuretics (high doses, often multiple)
- Cautious use / avoidance of:
 - BB/CCB reduce cardiac output, particularly if restrictive physiology)
 - ACEI/ARB (exacerbate dysautonomia/orthostasis),
 - digoxin (reduce inotropy)
- OAC for AF (*again, regardless of CHADS65*)
- ATTR → tafamidis or inotersen or patisiran +/- liver transplant
- AL → chemotherapy +/- autologous stem cell transplant
- ICD decisions difficult → EP problem



CCS 2023 Fitness to Drive Guidelines - HF

Category		Private Car	Commercial Driver (Truck, Bus)
Heart Failure	NYHA I	OK to drive	LVEF $\geq$ 30% OK
	NYHA II	OK to drive	LVEF $\geq$ 30% OK
	NYHA III	OK to drive	NO DRIVING
	NYHA IV	NO DRIVING	NO DRIVING
	Home inotropes	NO DRIVING	NO DRIVING
	LVAD	2 months after implant if NYHA 1-2	NO DRIVING
	Heart transplant	6 weeks after discharge + NYHA 1-2 + stable immunosuppression + followed annually	6 months post discharge + NYHA 1 + LVEF $\geq$ 50% + followed annually

Major updates from previous guidelines:

- **Functional status (NYHA class) is weighted higher than LVEF**
- **For private driving**, only significant change is that patients with **LVAD can drive** 2 months after implant if NYHA 1–2 (previously disqualified)
- **For commercial driving**, **LVEF threshold lowered from $\geq$ 35% to $\geq$ 30%** for NYHA I-II
- **Recommendations for post heart transplant commercial now requires LVEF $\geq$ 50%** (previously 35%)

MCQ 2 2026

A patient presents with new onset of shortness of breath. Exam reveals BP 190/100, JVP elevated, pansystolic murmur, no pleural rub. Echo shows mild MR and EF 60%. EKG suggests LVH. His heart failure is most likely due to?

- a) Myocarditis
- b) Mitral regurgitation
- c) **Diastolic dysfunction**
- d) Systolic dysfunction
- e) Pericarditis

Answer: C. New onset shortness of breath and elevated JVP suggest volume overload and congestion. LVH on ECG is suggestive of longstanding and chronic hypertension, a risk factor for HFpEF and a contributing factor to diastolic dysfunction. LVEF of 60% is indicative of preserve EF and lack of systolic dysfunction. Lack of pleural rub is reassuring that this is likely not an inflammatory process such as myocarditis or pericarditis.

Valvular Heart Disease

Key Resources:

- ACC/AHA 2020 Valve Guidelines
- CCS 2020 HF Update

PHYSICAL EXAM – High Yield for Applied Scenarios – see BONUS SLIDES for info on distinguishing etiology of murmurs clinically.

MCQ 3 2026

33 y/o M who presents with a murmur. Brisk carotid upstroke. Systolic murmur heard best at the LLSB, that increases with valsalva, quieter with passive leg raise. What is the patient not at risk for in the next 5 years?

- a) Sudden cardiac death
- b) Stroke
- c) Aortic dissection
- d) CHF

Valve Disease - Guiding Principles

Valves disease is a complicated topic

- Know the **physical exam** findings (JAMA RCE/physical exam) – *more resources at the end of this section*
- Know WHEN TO REFER for CONSIDERATION of intervention (Class I guidelines). Class II recommendations have been added for completeness, knowing the principles around these recommendations is helpful but it is unlikely you will need to know the specific numbers for Class II recommendations.
- **Canadian guidelines are old (2004); **ACC/AHA 2020 Valvular Heart Disease Guidelines** more up to date and testable*
- Don't need to know PV disease or much about TV intervention

Valves management involves:

- | | |
|------------------------------|--|
| Surveillance | → duration changes based on risk and severity of pathology |
| Pharmacotherapy | → temporizes a mechanical problem, treats symptoms
- NB. indications for endocarditis prophylaxis |
| Surgery | → effective, durable, generally more carries risk |
| Catheter-based interventions | → less durable, less data, usually less risk |

Valve Disease - Guiding Principles

Transthoracic echocardiography is the initial diagnostic test for all valvular disease

General AHA staging system similar for all valve disease

- Stage A → “at risk”
- Stage B → “progressive”
- Stage C → “severe disease, but asymptomatic”
- Stage D → “severe disease, with symptoms”

Severe disease matters most (*and usually guides intervention*)

- Symptoms: angina, dyspnea, heart failure, syncope → guides treatment
- Surrogate markers of impending badness (e.g. LV dilatation, pulmonary HTN, new Afib)
- Repeat assessment (e.g. echo) q6-12m for **severe asymptomatic** valvular disease

Valve type (bioprosthetic vs. mechanical)

- For any patient whom VKA is contraindicated → bioprosthetic
- Younger patients (<50y) → mechanical
- Older patients (>65y) → bioprosthetic
- Patients 50-65y → individualized decision making

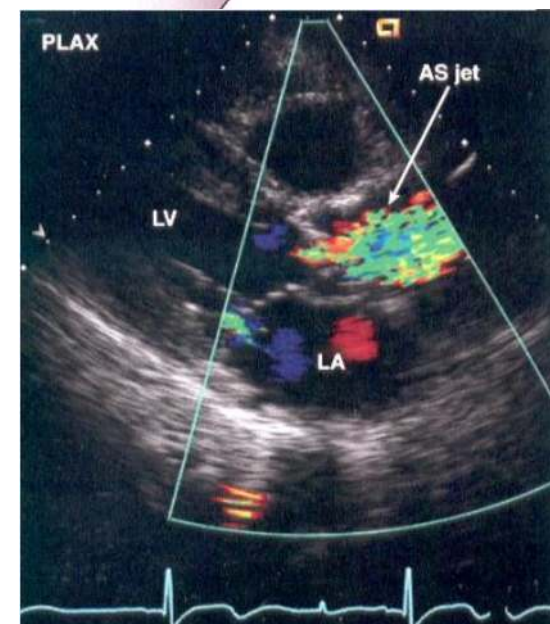
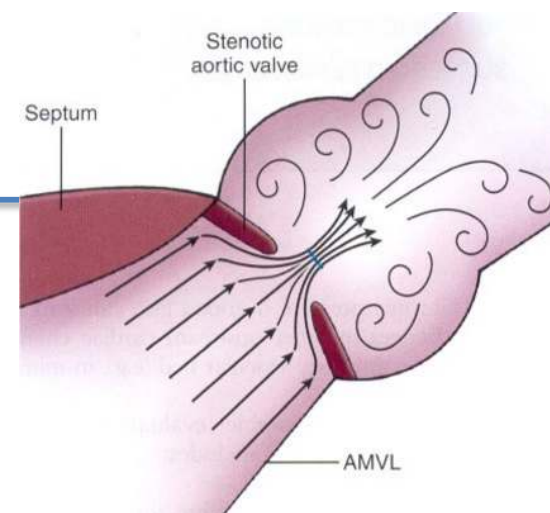
Pharmacotherapy for VHD

Very few recommendations for medical therapy in valve disease (primarily an interventional disease)

- **Aortic stenosis** → Treat HTN as per standard guidelines; treat lipids per guidelines for prevention of atherosclerosis (*not to prevent hemodynamic progression of AS – no evidence for this*); use ACEI/ARBs post-TAVI
- **Aortic regurgitation** → Treat HTN (preferably with ACE/ARB); symptomatic AR and/or LV systolic dysfunction + prohibitive surgical risk → GDMT with ACE/ARB or ARNI
- **Mitral stenosis** → Anticoagulation (VKA) indicated if → prior embolic event *OR* LA thrombus *OR* AF. Control tachycardia with betablocker/CCB can help symptoms (AF or sinus tachycardia) as this reduces the gradient through the valve
- **MR, TR** → Treat HF as per standard guidelines; vasodilator therapy is NOT recommended for asymptomatic primary MR and normal LV function

Aortic Stenosis

- **Etiology:** bicuspid (young), rheumatic (developing countries), calcific (old)
 - Bicuspid AV associated with aortopathy ***, heritable, screen 1st degree family members
- Prolonged asymptomatic period, then rapid deterioration with onset of symptoms (**angina, syncope, HF**)
- Patients with severe AS are *often* **afterload dependent** (caution with vasodilators/afterload reducers, e.g. ACEi)
- **Severe AS Criteria** (*mostly from echo now*):
 - Mean Gradient ≥ 40 mmHg
 - Max jet velocity ≥ 4 m/s
 - AVA < 1.0 cm²



Low-flow, low-gradient AS with low LVEF (<50%)

D2	Symptomatic severe low-flow, low-gradient AS with reduced LVEF	Severe leaflet calcification/fibrosis with severely reduced leaflet motion	AVA $\leq 1.0 \text{ cm}^2$ with resting aortic $V_{\max} < 4 \text{ m/s}$ or mean $\Delta P < 40 \text{ mm Hg}$ Dobutamine stress echocardiography shows AVA $< 1.0 \text{ cm}^2$ with $V_{\max} \geq 4 \text{ m/s}$ at any flow rate	LV diastolic dysfunction LV hypertrophy LVEF $< 50\%$	HF Angina Syncope or presyncope
D3	Symptomatic severe low-gradient AS with normal LVEF or paradoxical low-flow severe AS	Severe leaflet calcification/fibrosis with severely reduced leaflet motion	AVA $\leq 1.0 \text{ cm}^2$ (indexed AVA $\leq 0.6 \text{ cm}^2/\text{m}^2$) with an aortic $V_{\max} < 4 \text{ m/s}$ or mean $\Delta P < 40 \text{ mm Hg}$ AND Stroke volume index $< 35 \text{ mL/m}^2$ Measured when patient is normotensive (systolic blood pressure $< 140 \text{ mm Hg}$)	Increased LV relative wall thickness Small LV chamber with low stroke volume Restrictive diastolic filling LVEF $\geq 50\%$	HF Angina Syncope or presyncope

DON'T PANIC!!

Bottom Line: if the ventricle is weak (*LV systolic dysfunction*) → will not be able to mount gradients / velocities in the severe range, even if valve is anatomically severe

- Be suspicious when → AVA < 1.0 (looks severe) with non-severe gradients/ $V_{\max}$
- Check the LVEF!
- Correct answer is usually → do a dobutamine stress echo (allows the LV to squeeze better to see if the gradients increase) or calcium score of the valve.

“Paradoxical” low-flow, low-gradient AS

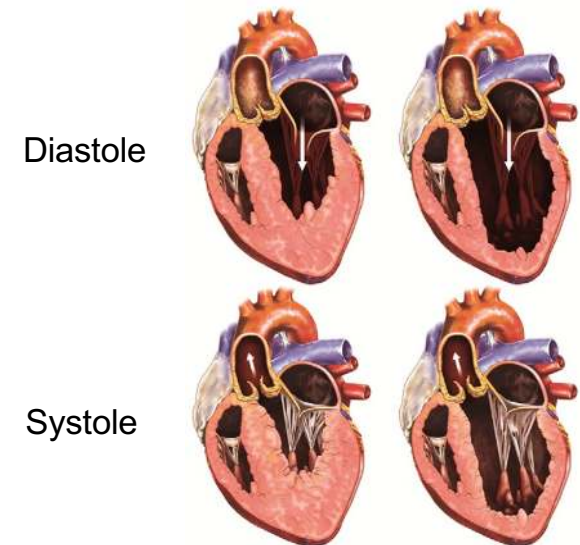
D3	Symptomatic severe low-gradient AS with normal LVEF or paradoxical low-flow severe AS	Severe leaflet calcification/fibrosis with severely reduced leaflet motion	AVA $\leq 1.0 \text{ cm}^2$ (indexed AVA $\leq 0.6 \text{ cm}^2/\text{m}^2$) with an aortic $V_{\text{max}} < 4 \text{ m/s}$ or mean $\Delta P < 40 \text{ mm Hg}$ AND Stroke volume index $< 35 \text{ mL/m}^2$ Measured when patient is normotensive (systolic blood pressure $< 140 \text{ mm Hg}$)	Increased LV relative wall thickness Small LV chamber with low stroke volume Restrictive diastolic filling LVEF $\geq 50\%$	HF Angina Syncope or presyncope
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**The paradox is that the LVEF is normal, but stroke volume is low.
How can this happen?**

Typically this occurs in a patient who has a **small LV cavity** with concentric LV hypertrophy. Therefore, the **stroke volume is low because the LV cavity is small**, and the ejection fraction is normal ($>55\%$) and often hyperdynamic ($>65\%$) to maintain cardiac output. These patients will have a LOW stroke (SV index $< 35 \text{ mL/m}^2$) despite a normal LVEF.

To further evaluate these patients if you are unsure if the valve is severe, obtain a **cardiac CT for calcium scoring of the valve** (>2000 in men, >1300 in women indicates severe). **Dobutamine stress echo will NOT be helpful here** as the ventricle is already squeezing well (LVEF is normal).

Paradoxical LFLG AS Classic LFLG AS



Aortic Stenosis - Intervention

Class I indications for replacement*:

- Severe, symptomatic AS
- Severe, asymptomatic AS with LV dysfunction (LVEF <50%)
- Severe, asymptomatic AS undergoing other CV surgery
- Symptomatic low-flow, low gradient AS with LV dysfunction (LVEF<50%)
- Symptomatic low-flow, low gradient AS with LVEF >50% (“paradoxical” low-flow, low-gradient aortic stenosis)
 - *if* AS most likely cause of symptoms

AVATAR trial (Circulation 2021) showed ↓ all cause death/MI/stroke/HHF with an early surgery strategy in pts w severe ASYMPTOMATIC AS vs conservative strategy.
Small trial published AFTER latest guideline iteration: take home for GIM = earlier referral if progressive asymptomatic AS

Aortic Stenosis - Intervention

- **Surgical AVR:**

- Mechanical valve if <50 y and no VKA contraindication.
- Bioprosthetic if >65 y or VKA contraindication.
- Age 50–65: shared decision-making.

- **Transcatheter AVI (TAVR):**

- Indicated for intermediate/high/prohibitive surgical risk.
- Preferred if >80 y or life expectancy <10 y.
- Consider age 65–80 vs. SAVR
- Contraindicated if comorbidities preclude benefit; palliative care if life expectancy <1 y.
- All TAVI valves are bioprosthetic → endocarditis prophylaxis required.

NEW – RHEIA Trial (ESC 2024): In women with severe symptomatic AS, TAVI vs. SAVR:

- ↓ re-hospitalization (valve/procedure complications, worsening HF)
- ↓ atrial fibrillation at 1 y (3.3% vs. 28.8%)
- Shorter LOS (4 vs. 9 d)
- ↑ pacemaker need (8.8% vs. 2.9%)

SAVR vs TAVR?
= Team-based decision,
factoring values, anatomy,
valve durability, longevity

BONUS

Read on own

Aortic Stenosis - Intervention

Class IIa (reasonable):

- Asymptomatic severe AS, low surgical risk + ↓ exercise tolerance or SBP rise <10 mmHg on stress test
- Asymptomatic *very severe* AS ($V_{max} > 5$ m/s), low surgical risk
- Asymptomatic severe AS, low surgical risk + BNP >3× ULN
- Asymptomatic severe AS, low surgical risk + $V_{max} \uparrow \geq 0.3$ m/s/yr

Class IIb (may be considered):

- Asymptomatic severe AS + progressive LVEF decline <60% on ≥3 serial studies
- Moderate AS undergoing other cardiac surgery

“Critical” aortic stenosis with an aortic velocity >5m/s is helpful to know to intervene even if the patient is asymptomatic (recall severe AS is >4m/s).

To know someone with severe AS is truly asymptomatic can be a challenge. Stress testing or BNP can be helpful.

Progressive increases in aortic velocity, or decreases in LVEF while still in the normal range, signal a higher risk patient, hence these class II recommendations.

Aortic Regurgitation

- **Acute AR**
 - Dissection
 - Endocarditis
 - Trauma
 - Prosthetic valve dysfunction
- **Chronic AR (*many, but remember these*)**
 - Primary aorta problems: dilatation (associated with autoimmune conditions, syphilis, Marfan, bicuspid, HTN, others ...), dissection, trauma
 - Primary valve problems: degenerative (calcific), bicuspid, rheumatic, endocarditis, VSD
- Severe AR is defined using specific echocardiographic parameters that **you should not need to know**

Aortic Regurgitation - Intervention

Class I indications for surgery*:

- Severe, symptomatic AR
- Severe, asymptomatic AR with $LVEF \leq 55\%$, *if no other cause for LV dysfunction identified*
- Severe, asymptomatic AR undergoing other CVSx

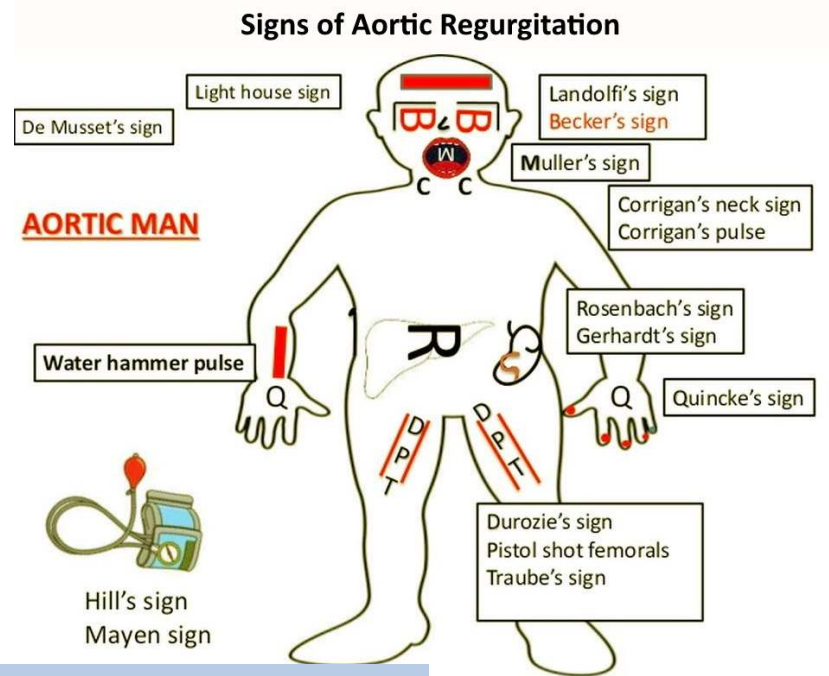
*May require AVR + ascending aortic replacement if associated aortopathy (*more info to come*)

BONUS
Read on own

Aortic Regurgitation – Physical Exam

Murmur: Early diastolic high-pitched, blowing decrescendo murmur
best heard at **left 3rd ICS** at end-expiration

It can lead to the **Austin Flint murmur** which is a low-pitched **mid-to-late diastolic rumble** at apex. This is from regurgitant jet hitting anterior mitral leaflet.



See online physical exam videos @
www.internalmedicineireview.ca for more on the exam for aortic regurgitation and peripheral signs of the same.

BONUS

Read on own

Aortic Regurgitation - Intervention

Class IIa indications for surgery (reasonable):

- Asymptomatic, severe AR with LVEF >55%, if LV is severely enlarged (cut off LV end *systolic* dimension >50mm or indexed to body surface area >25mm/m²)
- Moderate AR undergoing other cardiac surgery

Class IIb indications for surgery (can be considered):

- Asymptomatic, severe AR with LVEF >55% but progressive decline in LVEF on three serial studies to 55-60% or progressive increase in LV dilation (LV end *diastolic* dimension >65mm)

Aortic regurgitation causes a volume load on the left ventricle.

Systolic dysfunction is a devastating consequence of volume loading (which is why it is a class I recommendation to intervene).

Volume loading also causes LV dilation through remodeling, hence why LV enlargement is included in class IIa and IIb indications for intervention.

Mitral Stenosis

- Etiology is almost all rheumatic (other – MAC, radiation, autoimmune ...)
 - Often associated with AF
 - *OAC with VKA* if i) rheumatic MS and AF; ii) rheumatic MS and prior embolic event; iii) rheumatic MS and LA thrombus (**N.B. ii and iii do not require AFIB**)
 - INVICTUS trial: **VKA** vs. rivaroxaban in pts with rheumatic heart disease and AF
 - VKA had ↓ stroke/systemic embolism/MI/death (i.e. DOAC harmful in this setting!) NEJM 2022
 - Management considerations for AF and heart rate
 - MS does not like high HRs → loss of diastolic filling time
 - MS does not like AF → loss of atrial kick
- **Severe MS:**
 - **MV area $\leq 1.5 \text{ cm}^2$** (very severe = $\leq 1 \text{ cm}^2$)
 - Pulmonary artery systolic pressure $> 50 \text{ mmHg}$
 - Diastolic pressure half time (PHT) $\geq 150 \text{ ms}$

Echo criteria for severe MS in guidelines... but we don't think you need to memorize!

Mitral Stenosis - Intervention

2 types of interventions – percutaneous vs. surgical

Percutaneous mitral balloon commissurotomy (PMBC) indicated if (*Class I*):

- Severe, symptomatic MS + favourable valve anatomy + can be performed at a “Comprehensive Valve Centre”
 - **CONTRAINDICATED** if: i) LA thrombus (need preop TEE) ii) $\geq$ moderate MR

MV surgery (commissurotomy +/- repair *OR* replacement) indicated if (*Class I*):

- Severe, symptomatic MS + acceptable surgical risk + **contraindicated/failed PMBC**
- Severe MS and other cardiac surgery planned

BONUS

Read on own

Mitral Stenosis - Intervention

- **Percutaneous mitral balloon commissurotomy (PMBC)** is reasonable if (*Class IIa*)*:
 - Asymptomatic, severe MS with pulmonary hypertension (PA systolic pressure > 50mmHg by echocardiography or right heart catheterization)
- **Percutaneous mitral balloon commissurotomy (PMBC)** can be considered if (*Class IIb*)*:
 - Asymptomatic, severe MS with new atrial fibrillation
 - Asymptomatic, severe MS with mean gradient >15 mmHg or wedge pressure >25 mmHg with exercise
 - Highly symptomatic patients (NYHA III-IV) with severe MS with unfavourable anatomy or high risk for surgery (as a palliative attempt to try to relieve symptoms)

An obstructed mitral valve leads to LA dilation, atrial arrhythmias and pulmonary hypertension.

Therefore, think of pulmonary hypertension and atrial fibrillation as the “end organ” effects of mitral stenosis!

*All recommendations require the PMBC be performed at a “Comprehensive Valve Centre”.
PMBC remains contraindicated if LAA thrombus or moderate or more MR

Mitral Regurgitation

Acute MR

- **Presentation:** shock, pulmonary edema, soft murmur (rapid pressure equalization)
- **Causes:**
 - Ischemia → papillary muscle dysfunction
 - Chordal rupture (e.g., MVP → flail leaflet)
 - Endocarditis
 - Trauma

Chronic MR

- **Primary (degenerative = leaflet problem):**
 - Leaflet: MVP (myxomatous, fibroelastic), rheumatic, endocarditis
 - Annulus: calcification
 - Chordae: trauma, infection, idiopathic
 - Papillary muscle: trauma
- **Secondary (functional = “around” leaflet problem):**
 - **Ventricular functional:** eg dilated cardiomyopathy → leaflet tethering + annular dilation → malcoaptation
 - **Atrial functional:** AF → atrial enlargement → annular dilation → malcoaptation
- **Severe MR:** defined by echo (details not required).

Severe MR is defined using specific echocardiographic parameters that **you should not need to know**



Mitral Regurgitation - Intervention

PRIMARY MR – *“the goal of therapy is to correct MR before onset of LV systolic dysfunction”*

Class I indications for surgery (repair when possible vs. replacement) for **PRIMARY MR**:

- Severe, symptomatic 1° MR irrespective of LVEF
- Severe, asymptomatic 1° MR + LV systolic dysfunction (**LVEF $\leq$ 60%,** LVESD $\geq$ 40mm)

BONUS

Read on own

Mitral Regurgitation - Intervention

Class IIa indications for **PRIMARY MR (reasonable)**:

- Asymptomatic patients with severe MR with preserved systolic function (LVEF $\geq 60\%$, LVESD $< 40\text{mm}$) can undergo mitral valve repair with $> 95\%$ success and $< 1\%$ mortality at a Comprehensive Valve Centre
- Severely symptomatic (NYHA III-IV) patients with primary MR with high or prohibitive surgical risk can undergo transcatheter edge to edge repair if favorable anatomy and life expectancy greater than 1 year.

Class IIb indications for **PRIMARY MR (can be considered)**:

- Asymptomatic, severe MR with preserved systolic function with progressive increase in LV size or decrease in LVEF on 3 or more serial echocardiograms

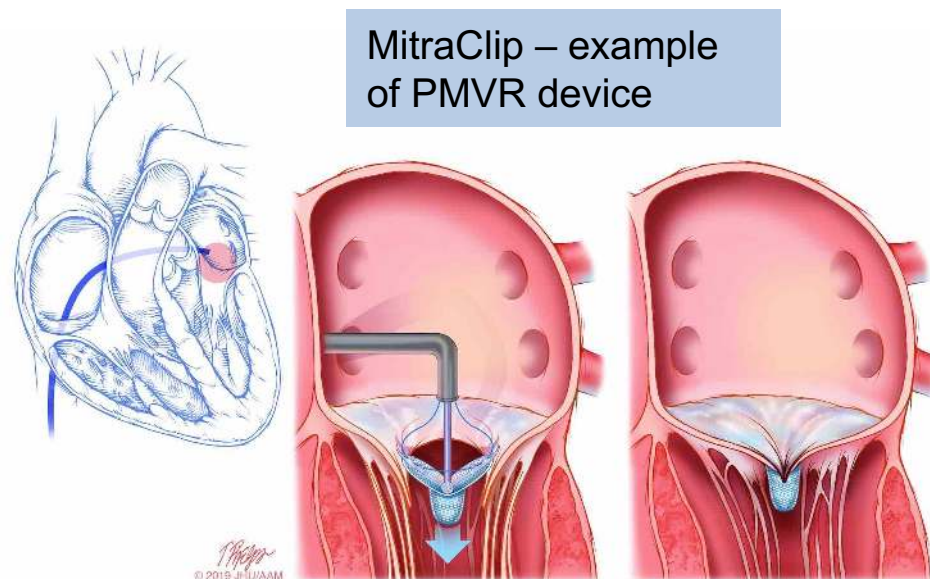
Mitral Regurgitation - Intervention

Treatment of Secondary MR

- “We recommend that maximally tolerated GDMT, including cardiac resynchronization therapy (CRT) and revascularization where appropriate, be implemented **before** consideration of percutaneous mitral valve repair (PMVR) for patients with HFrEF and severe FMR (Strong Recommendation, High-Quality Evidence).
(CCS 2020 HF Update)
- We suggest that patients with symptomatic HF (HFrEF) despite maximal GDMT and severe mitral regurgitation be evaluated for PMVR (Weak Recommendation, Moderate-Quality Evidence).

AHA 2020 Guidelines → congruent with CCS

- focus on GDMT
- **no class I** indication for surgery/intervention for 2° MR



From: Johns Hopkins Clinical Connection - Johns Hopkins
Medicine

BONUS

Read on own

Mitral Regurgitation - Intervention

- **Class IIa** indications for **SECONDARY MR** (reasonable):
 - **Transcatheter edge-to-edge repair:** Severe secondary MR LVEF <50% with NYHA II-IV symptoms on maximum GDMT with LVEF 20-50%, LVESD $\leq$ 70 mm and PA systolic pressure $\leq$ 70 mmHg
 - **Surgical repair:** Severe secondary MR at the time of CABG
- **Class IIb** indications for **SECONDARY MR** (can be considered):
 - **Surgery:**
 - Severe secondary MR with LVEF $\geq$ 50% and NYHA III-IV symptoms despite GDMT
 - Severe secondary MR with LVEF <50% with NYHA III-IV symptoms despite GDMT
 - May consider surgical replacement over repair (annulus dilation from LV remodeling/dilation may make the repair very challenging)

Tricuspid Regurgitation

TR can be primary or secondary as well

- Primary: lead-related, trauma, IE, rheumatic, carcinoid, CTDz, ...
- Secondary: Pulm HTN-related, RV-related, RA-related

Severe TR is defined using specific echocardiographic parameters that **you should not need to know**

Class I indications for surgery:

- Severe TR in patients undergoing left sided valve surgery

Class IIa indications for surgery:

- Addition of an annuloplasty ring for moderate or more TR at the time of left sided valve surgery
- Severe primary or secondary TR if right sided heart failure and no pulmonary hypertension to reduce symptoms and risk of heart failure hospitalization

Take Away Message for TR Interventions:

Multiple trials (TRILUMINATE trial, NEJM 2023; TRI-FR trial, ESC 2024; TRISCEND II trial, NEJM 2024) demonstrate that use of novel devices such as TriClip and procedures such as transcatheter tricuspid-valve replacement (TTVR) may be superior to medical therapy alone and help improve symptoms and HF hospitalization.

Antithrombotic Therapy After Valve Replacement

Mechanical Valves *** NO DOACs ***

- Lifelong therapy with warfarin; add ASA if other antiplatelet indication
- Goal INR varies with: type of valve, valve position, patient risk factors
 - ***NEW*** INR 1.5-2.0 ON-X aortic valve (useful if INR monitoring/compliance issues, needs low dose ASA)
 - INR 2.5 (2-3) for current generation AVR and no other risk factors
 - INR 3.0 (2.5-3.5) for any MVR *or* old AVR (ball-in-cage) *or* AVR with risk factors (RF = AF, prior clot, LV dysfxn, hypercoagulable state)
 - Bridging for invasive procedures is reasonable when the INR is subtherapeutic based on an individualized assessment of thrombosis and bleeding (see perioperative medicine lecture)

Bioprosthetic Valves

- Lifelong therapy with ASA 75-100mg daily
- Initial 3-6 months post-implantation
 - Surgical bioprosthetic valve replacement → consider VKA (INR 2.5) in addition to ASA
 - TAVI → may consider DAPT (clopidogrel) or VKA (INR 2.5) if low risk bleed (ASA monotherapy likely safer per guideline text)

Bioprosthetic valve (within 3 months) + new onset AF → VKA

Infective Endocarditis

- Will be covered in another section (*Infectious Disease*)
- **Know:**
 - Class I Indications for surgery
 - Indications for prophylaxis against infective endocarditis



CCS 2023 Fitness to Drive Guidelines - Valves

Category		Private Car	Commercial Driver (Truck, Bus)
Valvular Heart Disease	Medically treated AS, AI, MS, MR, TR	AS: OK if NYHA 1-2 MS, MR, AI, TR: OK if NYHA 1-3	Only OK if in the lowest risk group: NYHA 1, no syncope, LVEF $\geq$ 50% AS, AI: as above MR: also no pHTN or systemic embolism (?error) MS: no LVEF requirement TR: no RV dysfunction, symptomatic RV failure, RV arrhythmias
	Post TAVI or SAVR for AS, AR	1 month if stable QRS duration, no high grade AV block (2 nd type II or 3 rd) if no PPM and NYHA I-III AR requires as above + LVEF $\geq$ 50%	3 month if stable QRS duration, no high grade AV block (2 nd type II or 3 rd) if no PPM and NYHA I AR requires as above + LVEF $\geq$ 50%
	Post TEER for MR or TR	48h post d/c if NYHA I-III	1 month post d/c if NYHA I and LVEF $\geq$ 50%
	Post balloon valvuloplasty for MS	48h post d/c if NYHA I-III	1 month post d/c if NYHA I
	Post TMVR, TTVR or surgical valve replacement	1 month post d/c if NYHA I-III	3 months post d/c if NYHA I and LVEF $\geq$ 50%

MCQ 3 2026

33 y/o M who presents with a murmur. Brisk carotid upstroke. Systolic murmur heard best at the LLSB, that increases with valsalva, quieter with passive leg raise. What is the patient not at risk for in the next 5 years?

- a) Sudden cardiac death
- b) Stroke
- c) Aortic dissection**
- d) CHF

Answer: C. Presence of a systolic murmur at the apex which is louder with less preload and softer with increased preload are suggestive of left ventricular outflow tract obstruction which can be seen in HOCM. Patients with HCM are at higher risk of sudden cardiac death from ventricular arrhythmias, higher risk of stroke and CHF. The murmur is not related to the aortic valve.

Aortopathy

Key Resources:

- ACC/AHA 2022 Aortic Disease Guidelines
- CCS 2014 TAD Guidelines
- CCS 2023 The Aortic Team Model and Collaborative Decision Pathways for the Management of Complex Aortic Disease

MCQ 4 2026

You see a 55 year old man in ER with acute onset of chest pain. Which of the following will most likely rule in acute aortic dissection?

- a) Sudden onset chest pain
- b) Severe hypertension
- c) Aortic insufficiency
- d) Side to side pulse deficit
- e) Focal neurological deficit

Thoracic Aortic Dissection

- Often presents as a “tearing” or “ripping” pain radiating to the back.
- Can present as several “mimics”:
 - ACS (e.g. STEMI) – usually inferior STEMI
 - Valvular heart disease (aortic regurgitation leading to heart failure)
 - Pericardial effusion, cardiac tamponade
 - Syncope
 - Stroke, focal neuro symptoms
 - Renal failure, limb ischemia, spinal cord injury
- Risk factors include **hypertension**, vasculitis, valvular heart disease (esp. Bicuspid AV), drugs (cocaine), collagen disorders* (Marfan, Ehlers-Danlos)
- Look for
 - focal neuro deficit (LR+ 6.6-33)
 - Pulse deficit/differential BP (LR+ 5.7)
!!! CHECK BILATERAL BPs !!!
 - Enlarged aorta or mediastinum on CXR (LR+ 2.0)
- **CT scan (angio / aorta protocol)** is best first-line imaging modality
 - Others → MR, TEE

*See online “hypermobility” physical exam video at
www.internalmedicinereview.ca

CCS 2014 Thoracic Aortic Disease; JAMA RCE

Thoracic Aortic Dissection

- **Management**

- Preferentially use IV meds (easy on / easy off), in ICU setting with arterial line monitoring
- BBs first line (or CCBs if BB contraindication/intolerance) → vasodilators (nitroprusside, ACEI)
- Control HR
 - Target HR < 60-80
 - *IV labetalol good 1st line option (both HR and BP)*
- Control BP
 - DO NOT control BP before HR (hypotension → compensatory incr. HR → more shear stress)
 - Target BP <120 systolic (or to lowest BP that maintains adequate perfusion)
- Control pain with as needed analgesics (will help with hemodynamics)
- Further management depends on type of dissection (Stanford Classification – Type A or B)
 - **Type A** → dissection involves ascending aorta → refer for **urgent surgical intervention**
 - **Type B** → dissection does not involve ascending aorta → medical management → HR and BP control
 - **Exception:** if the type B dissection causes malperfusion (e.g. gut ischemia, leg ischemia), endovascular management will be necessary. Rupture is another rare indication for surgery

Thoracic Aortopathy/Aortic Aneurysm

Imaging in Aortic Dilatation/Aneurysm

- **At diagnosis:**
 - **TTE** → assess aortic valve anatomy & function.
 - **CTA/MRA** → image entire aorta (MRI preferred if <50 to avoid radiation).
- **Follow-up:** use the *same modality* each time. Prefer MRA due to frequency of screening to avoid radiation.
 - **Loeys-Dietz:** annual MRA (or CTA).
 - **Marfan / other heritable disease:** MRA (or CTA) every 3 yrs.
 - **Bicuspid / degenerative aortopathy:** CTA/MRA q. 3–5 yrs.
 - **TTE:** yearly in Loeys-Dietz & Marfan; every 1–3 yrs in bicuspid/degenerative disease.
- **Increase frequency** if aorta is within 5 mm of surgical threshold or grows >2.5–3 mm/yr.
- If clinical features of connective tissue disease (Marfan, Loeys-Dietz, Ehlers-Danlos), family history of thoracic/peripheral/intracranial aneurysms, or onset age less than 60, refer for **genetic testing**
 - If gene positive, family members who are also positive should be screened with an echocardiogram
 - If no culprit gene identified, first degree relatives should be screened with an echocardiogram
- Treat **hypertension**
 - Target: Lower may be better (AHA 2022 = 130/80 and CCS 2023 mention this but do not provide specifics, **Hypertension Canada 2025 now target SBP<130** in people at high CV risk.
 - Antihypertensive choice:
 - ACC/AHA and CCS recommend beta blocker followed by ARB for BP control (IIa recommendation)
 - **Marfan:** use a beta blocker or losartan (class I recommendation) regardless if the patient has HTN
- Other recommendations:
 - **Smoking cessation** (class I)
 - **Statins** and/or antiplatelets if evidence of atherosclerosis (class IIa), statins may be considered for primary prevention if no atherosclerosis (class IIb)

FDA 2018: avoid fluoroquinolones if aortopathy, may ↑ risk of aneurysm rupture

ACC/AHA 2022 Aortic Disease
CCS 2014 Thoracic Aortic Disease
CCS 2023 Complex Aortic Disease

Thoracic Aortopathy/Aortic Aneurysm

Size criteria for operative management of asymptomatic aneurysm

	ACC/AHA 2022 - Ascending Aorta (cm)	CCS 2014
Degenerative	5.5 (5 if Expert Centre, IIa)	5.5
Bicuspid aortic valve (BAV)	5.5*	5.5
Marfan syndrome	5** (root, not ascending)	5
Familial aortopathy	5***	4.5
Loeys-Dietz	Complicated. Generally > 4.5 but will vary depending on the genetic variant.	4.2
Vascular Ehlers-Danlos	Complicated. Requires expert surgeon due to fragile aorta. No threshold given in ACC/AHA guidelines.	4.2
Turner syndrome	Uses aortic size index (ASI, or aortic diameter/BSA) as these patients have short stature. Threshold 2.5 cm/m ² .	4.2
Undergoing cardiac surgery	5 (4.5 if Expert Centre, IIa)	4.5

Nuances from ACC/AHA 2022 Aorta Guidelines (note all require surgery by experienced surgeons at a Multidisciplinary Aortic Centre i.e. Expert Centre):

***BAV:** aneurysm/root threshold of **5 cm** is reasonable (class IIa) if other risk factors for dissection present such as rapid progression >0.3cm/yr, family history of dissection, coarctation, "root phenotype" aortopathy or can be considered (class IIB) if none are present

****Marfan:** root threshold of **4.5 cm** is reasonable (class IIa) if high risk features (rapid progression >0.3cm/yr, family history of dissection, diffuse aortic root and ascending aorta, marked vertebral artery tortuosity)

*****Familial:** ascending aorta threshold of **4.5 cm** is reasonable (class IIa) if family history of aortic dissection/sudden death <50 age, rapid growth (0.5cm in 1 year, 0.3cm/yr in 2 years)

BONUS
Read on
own

Thoracic Aortopathy/Aortic Aneurysm

- **The Aortic Team** is a concept described in the CCS 2023 Aortic Update, referring to a specialized team to manage patients with aortopathy, involving cardiology, cardiac surgery, vascular surgery and interventional radiology. Variations of an Aortic Team will exist at different institutions.
- Where to refer?
 - Generally planned, not complex, ascending aorta procedures can be done at a local cardiac centre
 - **More complex patients should be managed with an Aortic Team** (tertiary or quaternary centre) e.g. outpatient planned but complex like an arch replacement, or complex acute presentations like a Type B dissection with malperfusion
- All patients with identified aortic disease, gene positive aortopathy or “at risk” for aortic disease are recommended to be referred to an Aortic Team
 - Rationale: the Aortic Team will help to optimize and monitor patients prior to any intervention, as well as monitor for complications post intervention
 - The guidelines do not make distinction on type of aortic disease, i.e. if an elderly patient with a mildly dilated ascending aorta of 4.2cm, should this patient be followed by an Aortic Team? Not clear. They may still glean benefit from risk factor management and surveillance but this could also be done locally.

Abdominal Aortic Aneurysm Pearls

- **Screening:** one-time U/S for men 65–80 (CTFPHC 2017)
In practice: also screen those with strong family history of aortopathy – per US/European guidelines
- **Medical therapy:** only smoking cessation reduces rupture risk. Manage BP (<130/80 per AHA, <130 Hypertension Canada), use statin/ASA if atherosclerotic disease. Avoid fluoroquinolones (FDA 2018).
- **Surveillance (U/S first-line; CT if inadequately visualized):**
 - 3.0–3.9 cm → q3 yrs
 - 4.0–4.9 cm men / 4.0–4.4 cm women → annually
 - ≥5.0 cm men / ≥4.5 cm women → q6 mo
- **Repair thresholds:**
 - Men ≥5.5 cm, women ≥5.0 cm, or if symptomatic.
 - Reasonable if growth >0.5 cm in 6 mo.
- **Other factors:** family history, connective tissue disorder → refer early to vascular surgery.

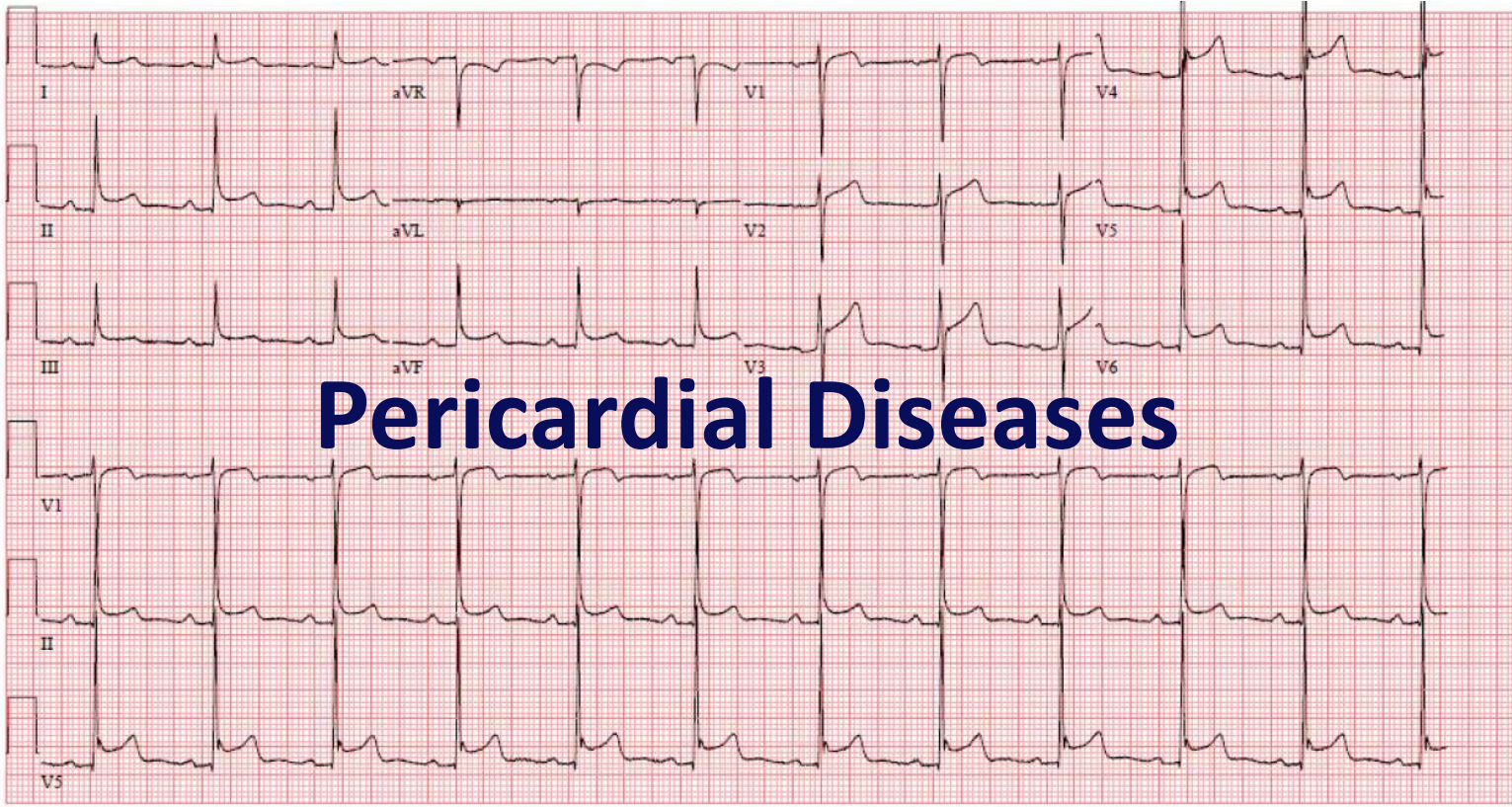
MCQ 4 2026

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- a) Sudden onset chest pain
- b) Severe hypertension
- c) Aortic insufficiency
- d) Side to side pulse deficit
- e) **Focal neurological deficit**

Answer: E. Pulse deficit: LR 5.7, focal neurological deficit LR 6.6

Clinical Finding	Positive LR
Sudden onset of pain	1.6
Severe pain	2.0
Tearing/ripping quality of pain	1.2
Migrating pain	1.4
Pulse deficit or BP differential	5.7
Focal neurologic deficit (e.g. stroke symptoms)	6.6 to 33.0



Pericardial Diseases

Key Resources:

- ESC 2015 Guidelines
Pericardial Disease

Acute Pericarditis

- Inflammation of the pericardium
- Characterized by **at least 2 of**:
 - Pleuritic chest pain
 - Friction rub
 - Diffuse ST-segment elevation +/- PR depression, without reciprocal ST-segment depression
 - Presence of new/worse pericardial effusion
- Additional supporting evidence
 - Inflammatory markers (CRP)
 - Inflammation on CT, MR
- Troponin can be (+): “*Myo-pericarditis*”
 - Myocardial involvement
 - Higher risk
- **Etiologies**: most cases idiopathic, consider TB if risk factors
 - **Vascular** – post MI, Dressler’s
 - **Infectious** - Coxsackie, echovirus, adenovirus, flu, parvo, TB, fungal ...
 - **Toxin (drug)**- procainamide, hydralazine, INH, minoxidil, dilantin
 - **Autoimmune** – RA, SLE ...
 - **Metabolic** - uremia, dialysis, hypothyroidism
 - **Iatrogenic** – Radiation, post CV Surg
 - **Neoplastic** - mesothelioma, breast/lung/melanoma mets, leukemia, lymphoma

When to Admit Pericarditis

Risk factors in pericarditis:

- Immunocompromised
- Trauma
- Oral anticoagulation therapy
- Myopericarditis (Troponin elevation)
- Fever $T > 38^{\circ}\text{C}$
- Subacute onset
- Significant effusion ($>20\text{mm}$) or cardiac tamponade
- Hemodynamic instability
- Lack of improvement after 7d appropriate therapy

Treatment of Acute Pericarditis

- **First episode:**

High dose NSAID 1-2 weeks (as needed until pain/CRP resolves)
+ Colchicine x3 months

- **Recurrence:**

High dose NSAID x2 weeks (as needed until pain/CRP resolves)
+ Colchicine x6 months

- **Post-MI**

Use ASA instead of NSAIDs (High dose = ASA 650 po QID)

- **Pregnancy**

- **Discuss with OB / Cardiology**
- **Ibuprofen preferred Tri 1,2 – NSAIDs** relatively contraindicated in trimester 3 / after 32 weeks (risk of PDA in infant).
Low dose ASA can be used in tri 3.
- Colchicine appears safe for this indication in pregnancy and breastfeeding (ESC, ACC consensus)
- Prednisone does not cross placenta but due to risks below generally avoid if possible

- **Steroids settle inflammation acutely but increase recurrence risk**

- **avoid** unless immune-mediated etiology or clearly non-responsive/contraindications to ASA/NSAIDs.
- Should still give colchicine in addition
- Use low doses (0.25-0.5mg/kg/d)
- Strongly consider taper

Other considerations:

- Reassess frequently
- GI protection
- Exercise restriction
- For advanced cases – IVIG vs. anakinra vs. azathioprine
- Last line of defense - pericardiectomy

If NSAID Intolerant (eg) allergy – pick colchicine over prednisone in MCQ!

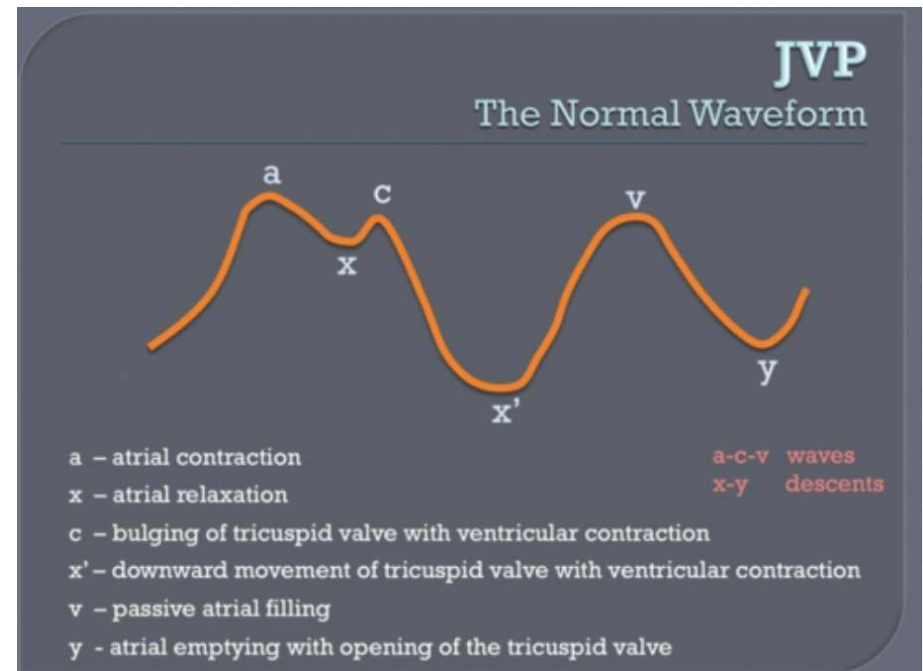
Pericardial Constriction and Tamponade

- **Constriction** = fibrous, non-compliant pericardium limits expansion of cardiac chambers
 - Rapid early filling of ventricles then abrupt cessation as they hit the “stretch limit” of pericardium → **rapid Y descent**, square root sign on cardiac cath
 - **+ve Kussmaul’s sign** and often no pulsus paradoxus
- **Tamponade** = accumulation of fluid in pericardial space increases the intrapericardial pressure and overcomes intracardiac pressure leading to impaired filling
 - **Pulsus paradoxus, blunted Y descent** on JVP, Beck’s Triad (hypotension, distended JVP, muffled heart sounds)

Constriction vs. Tamponade vs. Restriction

Discussed (*tested*) together because of some overlapping clinical findings

- Constriction and tamponade are pericardial problems characterized by enhanced **ventricular interdependence**
 - **Increased R heart filling = Reduced L heart filling**
- Patients with tamponade are *often* unstable
- Patients with constriction normally present with heart failure/congestion
- Patients with restriction have a myocardial problem that can resemble constriction but **no ventricular interdependence**
Use history, clinical exam, and other clues to help differentiate



	Cardiac Tamponade	Constrictive pericarditis	Restrictive cardiomyopathy
Pathophysiology	Accumulation of pericardial fluid continuously compresses all chambers throughout cardiac cycle, impairing filling	Fibrous pericardium limits expansion of cardiac chambers, and shields them from intrathoracic pressure (ITP) changes	Myocardial stiffness limits expansion of ventricular chambers; normal pericardium that transmits ITP changes
Ventricular Interdependence	Present (discordant RV/LV pressure changes with inspiration)	Present (discordant RV/LV pressure changes with inspiration)	Absent (normal, concordant RV/LV pressure changes due to ITP transmission)
Kussmaul's sign (inspiratory ↑ JVP [or does not drop])	NO (ITP drop transmits to RV via pericardium, so RV can fill)	YES (RV cannot expand anywhere further beyond early diastole)	YES (stiff RV cannot expand beyond early diastole)
Elevated JVP (what is the RA seeing?)	Absent/blunted Y descent (RA can't empty into RV, even in early diastole)	Prominent Y descent (RA rapidly empties into underfilled RV)	Prominent Y descent (RA rapidly empties into underfilled RV)
Pulsus paradoxus (inspiratory ↓ BP >12mmHg)	YES (ITP change exaggerates ventricular interdependence)	USUALLY NO (mechanism debated; ?related to lack of ITP transmission)	NO (no ventricular interdependence)
Other features	Beck's triad (low BP, high JVP, muffled heart sounds)	Pericardial knock (rapid early diastolic filling) +/- friction rub	Manifestations of systemic disease (amyloid, sarcoid)

Arrhythmias

Key Resources:

- +++ CCS AF Guidelines [*NB. Most recent = 2020 update*]
- ACC/AHA 2023 AF Update
- CCS 2020 Syncope Update
- ACC/AHA 2018 Pacing Guidelines
- ESC 2021 Pacing Guidelines
- CCS VT Guidelines
- ACLS Guidelines

MCQ 5 2026

You are called to see a patient in the ER who had presented with an acute myocardial infarction. They have now developed rapid heart rate at 150 bpm. ECG is shown. BP 70/40. Next step in management?

- a) Verapamil IV
- b) Metoprolol IV
- c) Electrical cardioversion
- d) Lidocaine
- e) Procainamide



Screening for Atrial Fibrillation (AF)

- General Population
 - pulse-based screening or rhythm-based screening at all routine health assessments in people >65yrs
 - follow-up with ECG assessment if “irregularly irregular”
- Cardiac Implantable Electrical Device (PPM, ICD)
 - interrogate all high atrial rate episodes for possible AF
- Non-lacunar Embolic Stroke of Unknown Source
 - “at least 24h of ambulatory ECG monitoring”
- Longer monitoring if AF is still suspected but not proven

AF Etiology & Initial Investigations

Major Guideline: CCS 2020

Major Considerations:

- Always look for an **etiology/precipitant** (EtOH, drugs, withdrawal, ischemia, PE, valves, thyroid disease, OSA, infection, sleep deprivation, acute pulmonary disease ...)
- Basic Workup for new AF:
 - Document rhythm
 - ECHO – assess LV size and function, LA size, valve ..
 - CBC, lytes (Ca, Mg), Cr, Coags, TSH for all
 - LFT before amiodarone prescription
 - A1C, FBG, Fasting lipid profile as part of comprehensive cardiac risk assessment
- Always assess patient AF-related symptoms and quality of life (CCS-SAF), assess patients with AF for frailty, cog impairment, dementia, depression

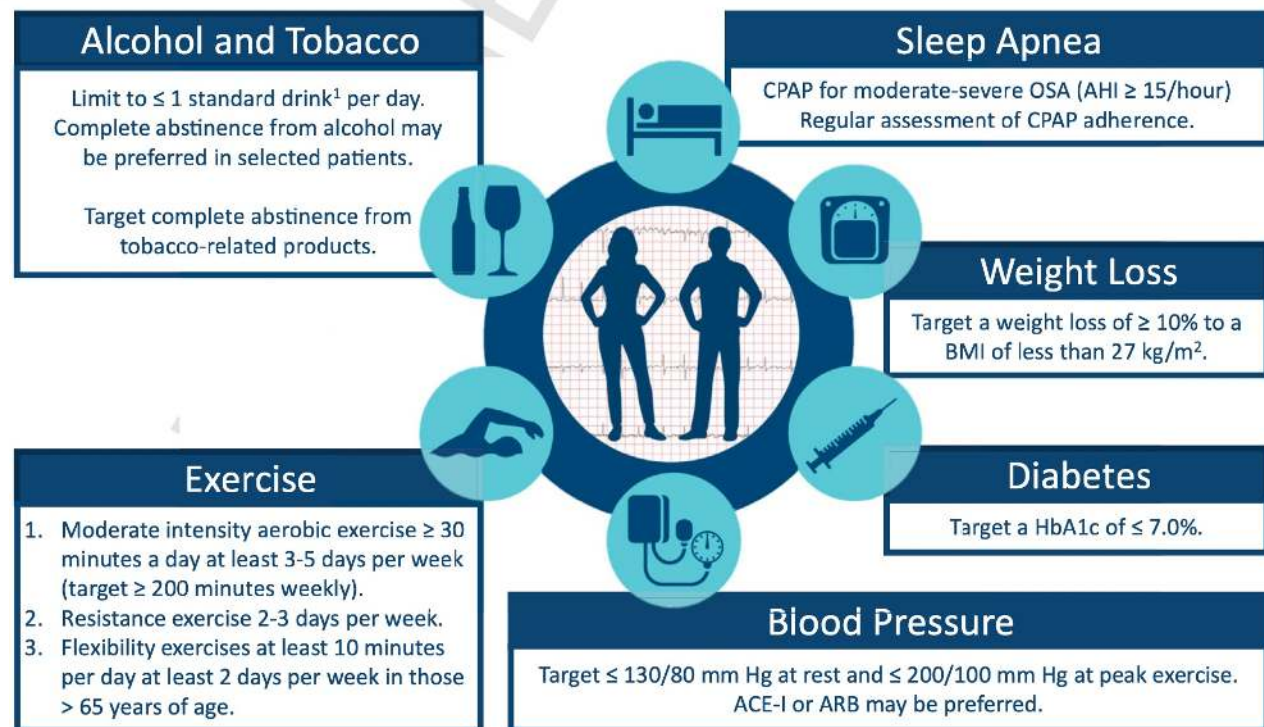
CCS – SAF Scale – Symptoms of AF

Class 0	Asymptomatic
Class I	Minimal symptoms or 1 episode <i>without syncope or CHF</i>
Class II	Symptoms have minimal effect: - Mild awareness if in persistent/permanent AF - Rare episodes (“less than a few per year”) in those w/ paroxysmal AF.
Class III	Symptoms have moderate effect on QOL: - Moderate awareness most days with persistent/perm AF - More common episodes (“more than every few months”) or more severe symptoms in pts with paroxysmal AF
Class IV	Symptoms have severe effect on QOL: - Syncope due to AF and / or - CHF due to AF and / or - Unpleasant symptoms in pts with persistent/perm AF and / or - Frequent and highly symptomatic episodes in pts w paroxysmal AF

AF: Prevention and Treatment

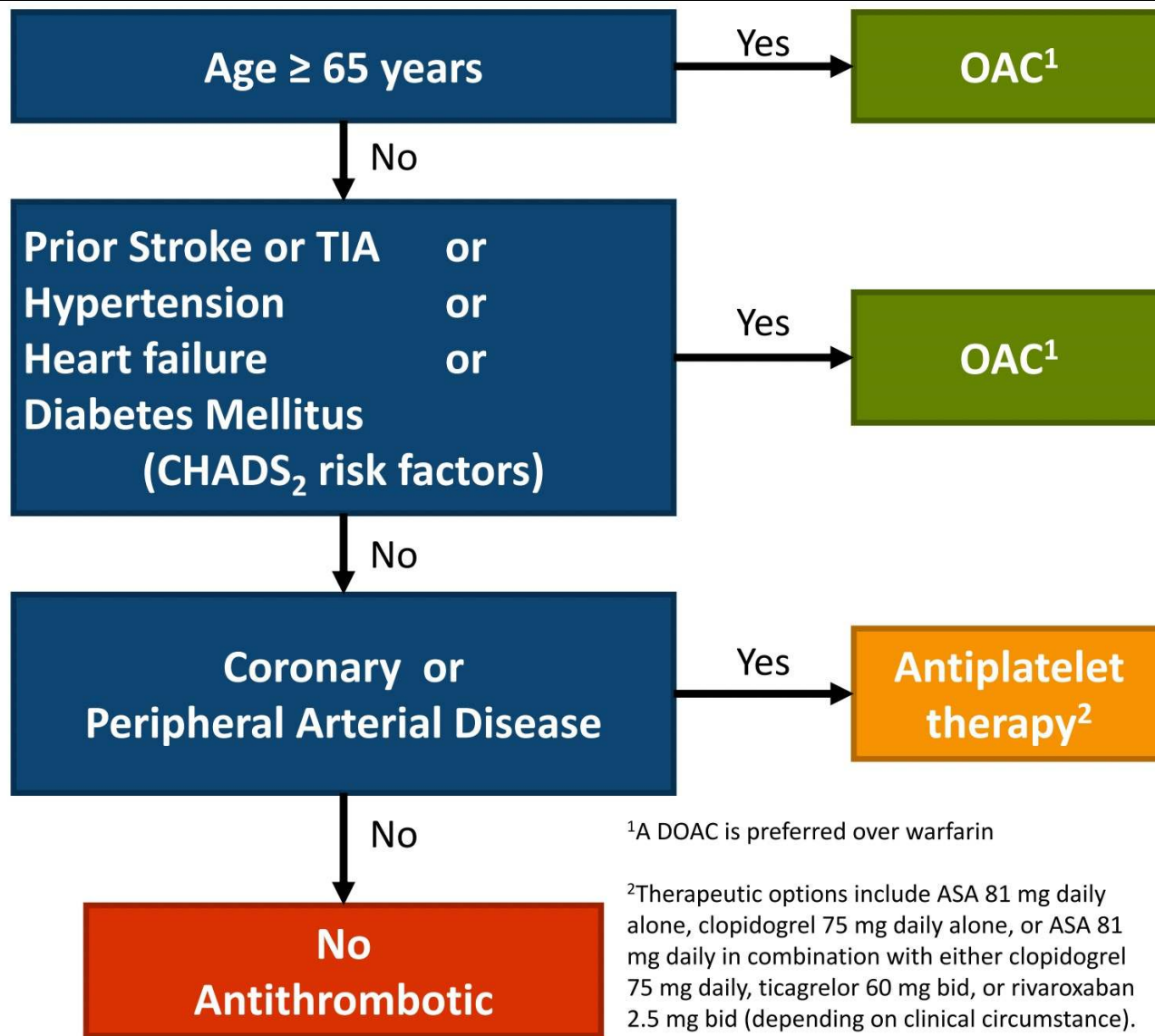
Major Considerations cont'd:

- Prevention = modifiable risk factor management
- Achieve rate control with b-blocker, CCB, or digoxin
- AF with pre-excitation (WPW):
→ DC cardioversion (or procainamide)
- Rhythm control preferred if **QoL impaired** (symptomatic despite rate control), **concurrent HF**, or **hemodynamically unstable** (DC cardioversion)

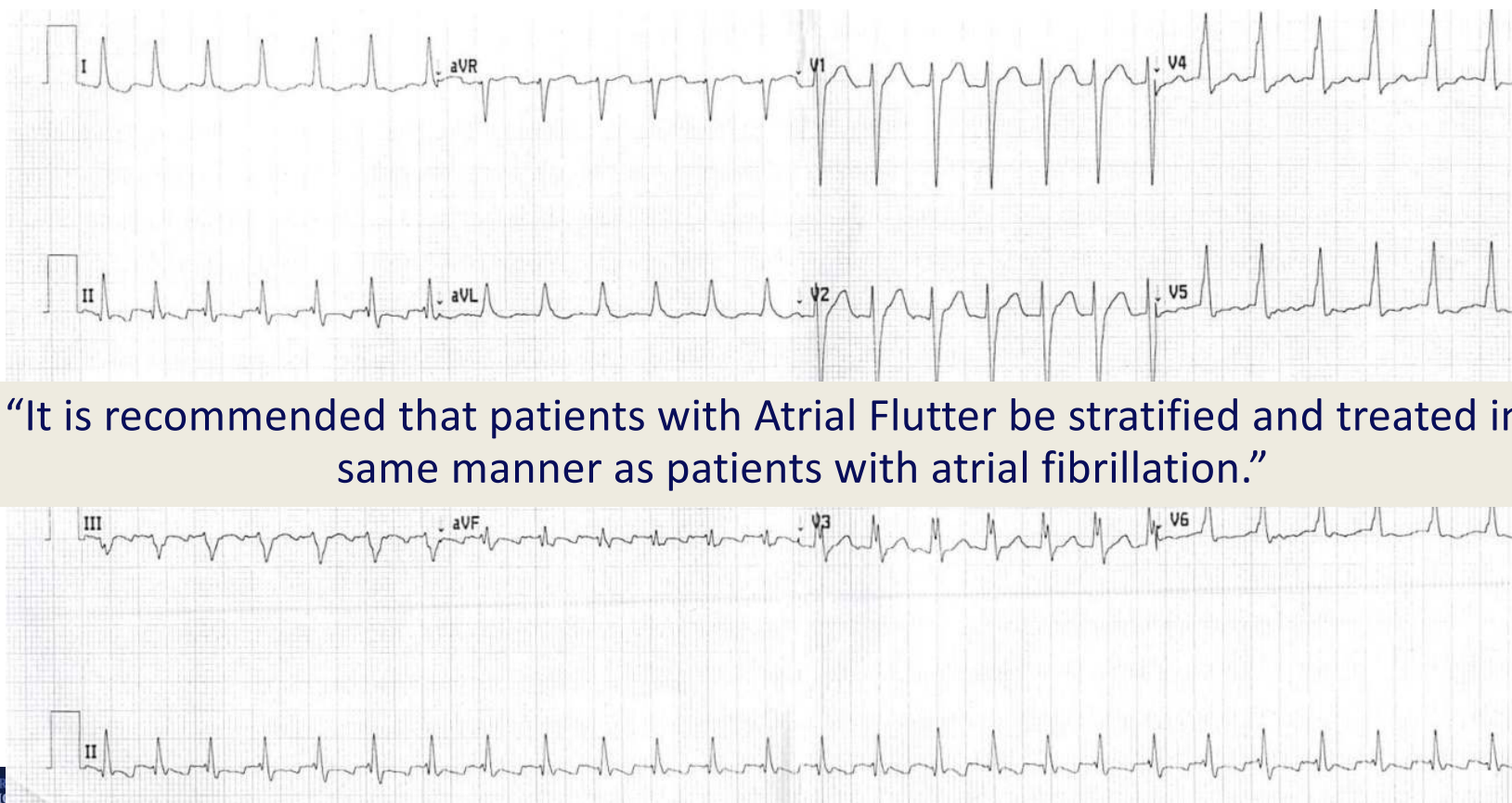


¹defined as containing 14 g of alcohol; 44 mL (1.5 fluid oz.) of 80-proof liquor, 148 mL (5 fluid oz.) of wine or 355 mL (12 fluid oz.) of beer

This cutoff of 130/80 is not mentioned in the CCS text anywhere, only in this figure! We suggest just go with HTN Canada guidelines for resting BP target.



What about Atrial Flutter (AFL)?



“It is recommended that patients with Atrial Flutter be stratified and treated in the same manner as patients with atrial fibrillation.”

Anticoagulation in AF/AFL

- DOACs are 1st line for almost everyone
 - VKA (i.e. warfarin) should be used instead of DOAC for **valvular** AF (CCS 2016, 2018 and 2020 definition):
 - Mechanical heart valves
 - Rheumatic mitral stenosis
 - Moderate-severe non-rheumatic mitral stenosis
 - Warfarin should also be considered (class IIa) in patients with new onset AF $\leq$ 3 months post-valve replacement (surgical or percutaneous)

FRAIL-AF (Circulation, 2023) looked at elderly (>75) frail patients with AF who were already on VKA and randomized to change to DOAC vs stay on VKA.

- *Surprisingly, more bleeding was observed in those who were switched to DOAC (15.2%) compared to those who continued VKA (9.4%), trial stopped early.*
 - Criticisms – small trial (n=1330), bleeding events were self reported and the primary outcome was driven by minor bleeding, not major bleeding. No formal guideline recommendations have incorporated this trial.
 - No difference in ischemic/embolic events
 - Take Away #1 – if elderly frail patient stable on VKA can continue
 - Take Away #2 – caution in extrapolating results of other RCTs to populations excluded from enrolment (frail elderly)

Anticoagulation in CKD/ESRD

Calculate CrCl using Cockcroft – Gault at baseline, and at least annually:

- Stage 3 CKD (eGFR >30) – A/C as usual
- Stage 4 CKD (eGFR >15 <30) – A/C as usual (CCS)
 - “Non-randomized data supports use of OAC for AF”
 - “a DOAC is preferred over VKA”
 - “Apixaban and rivaroxaban are approved for use with stage 4 CKD”
- Stage 5 CKD (eGFR <15) : “ we suggest that such patients not routinely receive anticoagulation therapy or antiplatelet therapy for AF”
 - ***NEW* ACC/AHA 2023 Afib Update:** in stage 5 CKD it “might be reasonable” to prescribe for patients with “elevated risk of stroke” warfarin (INR 2-3) or DOAC (apixaban 2.5-5mg BID or rivaroxaban 15mg daily) → go with CCS for now

REFERENCE: Dosing per manufacturer

- Apixaban
 - 5mg PO BID
 - 2.5 mg PO BID if 2/3: ≥80 years, ≤ 60kg, creatinine ≥133umol/L
- Dabigatran
 - 150 mg PO BID
 - 110 mg PO BID if age > 75 years or eCrCl 30-49 mL/min
- Edoxaban
 - 60 mg PO daily
 - 30 mg PO daily if CrCL 30-50 mL/min, ≤ 60kg, or concomitant use of potent P-glycoprotein inhibitors
- Rivaroxaban
 - 20 mg PO daily
 - 15 mg PO daily if CrCl 30-49mL/min

RENAL-AF RCT (Circulation 2022): AF w CHADS₂≥2 on dialysis randomized to apixaban vs VKA (n=154)
Major (+ clinically relevant) bleeding rate high : 32% Apixa, 26% VKA at 1 yr
Embolic/stroke rate low: 3% both groups Hence anticoagulating this group controversial!

Stroke reduction in AFIB with CAD/PAD (OAC and Antiplatelets)

SAPT =	Single Anti Platelet Therapy
DAPT =	Dual Anti Platelet Therapy (ASA + P2Y inhibitor)
OAC =	Oral anticoagulant (DOAC preferred)
Dual Pathway Therapy =	SAPT* + OAC** * clopidogrel is preferred P2Yi **OAC regimen = normal dose (adjust for CrCl) edoxaban, apixaban, dabigatran BUT rivaroxaban only 15 mg PO daily (10 mg in patients with CrCl 30-50 mL/min) or VKA INR 2-2.5.
Triple Therapy =	DAPT^ + OAC~ ^ASA + Clopidogrel preferred to ↓ bleed ~OAC regimen = rivaroxaban 2.5 mg PO BID , or apixaban 5 mg BID (adjust for age 80, Wt <60, Cr>133) or VKA INR 2-2.5.

Stroke reduction in AFIB with CAD/PAD (OAC and Antiplatelets) CCS 2020 summary slide

Scenario	CHADS	Initial Treatment	Durations	Notes
Stable CAD/PAD (no recent ACS or PCI in past 12 months)	0	SAPT (*consider ASA + Rivaroxaban 2.5mg bid)	∞	*COMPASS trial, "consider" per CCS
	≥1	OAC	∞	Prefer DOAC
PCI, elective / low risk thrombotic events. (no ACS)	0	DAPT	See CAD slides for durations	
	≥1	Dual Pathway	1-12 months post PCI, then OAC ∞	All pts should receive ASA 160 load at time of PCI if ASA naive
PCI for ACS / high risk thrombotic events	0	DAPT	See CAD slides for durations	
	≥1	Triple therapy 1-30 days → Dual pathway	1-12 months post PCI, then OAC ∞	
PCI, no ACS (elective PCI)	0	DAPT	See CAD slides for durations	
	≥1	Dual pathway (OAC + clopidogrel)	1-12 months post ACS then OAC ∞	OAC = Apixaban 5mg bid per guidelines

Who is HIGH RISK Post-PCI?



Patient Factors

- Age (> 65 years)
- Low body weight (< 60 kg)
- Hypertension
- History of bleeding (esp. within 1y)
- Prior Stroke or intracranial bleed
- Excess alcohol consumption
- Labile INR (TTR <60%)



Patient Factors

- Diabetes mellitus
- Current smoker
- CKD (eGFR < 60 mL/min)
- Prior acute coronary syndrome
- Prior stent thrombosis

Clinical Presentation

- Acute coronary syndrome

Factors that Increase Risk of Bleeding



Concomitant use of:

- antiplatelet use
- NSAIDs
- prednisone



Laboratory

- Anemia (hemoglobin <110 g/L)
- Abnormal liver function
- CKD (eGFR < 60 mL/min)

Factors that Increase Risk of Ischemic Coronary Events



Angiographic factors

- Multi-vessel disease
- Multiple (≥ 3) stents implanted
- Stenting of a bifurcation lesion
- Total stent length > 60 mm
- Left main or proximal LAD stenting
- Chronic occlusion intervention
- Bioabsorbable vascular scaffold

Mitigate bleeding risk:

Avoid prasugrel/ticagrelor

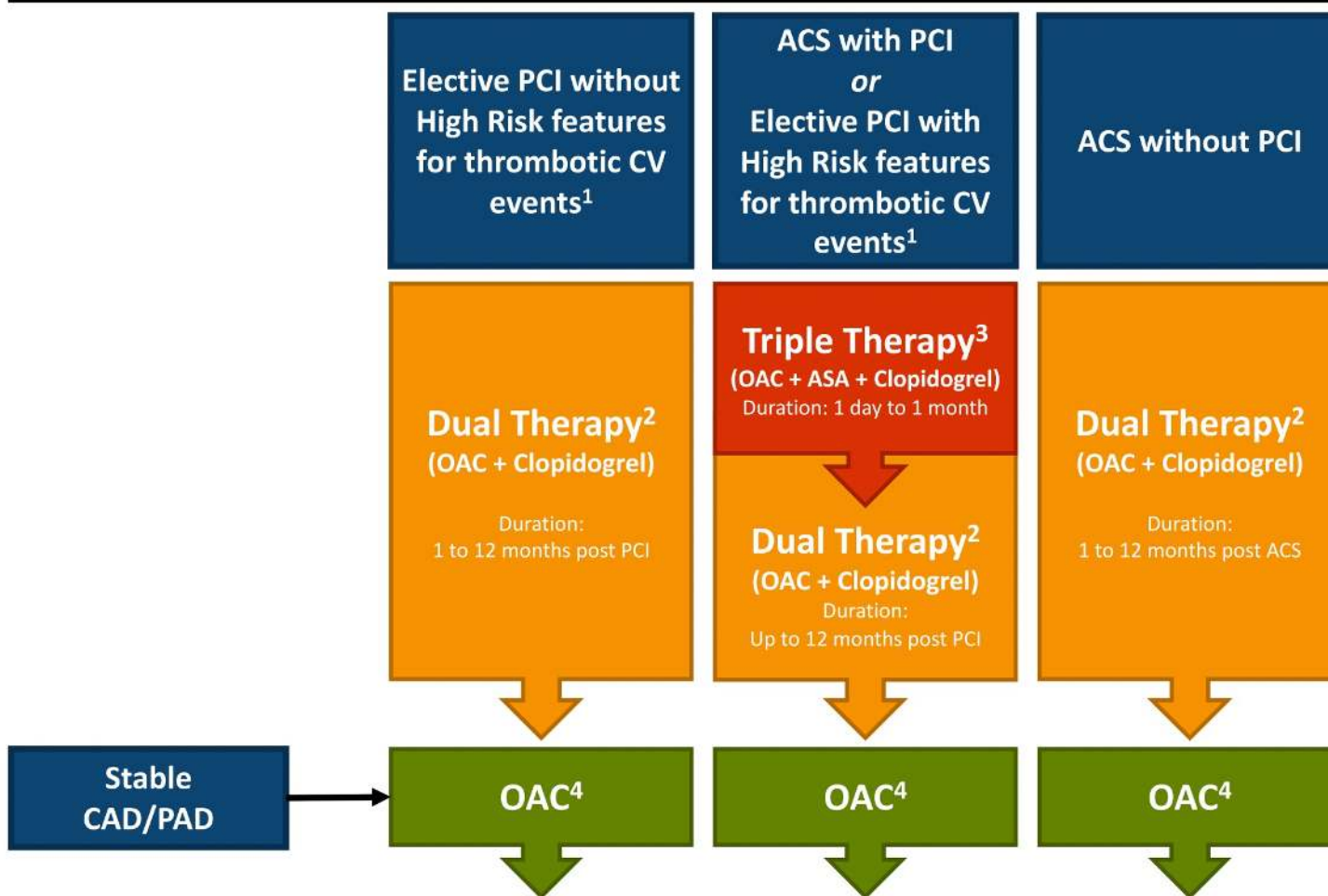
– that's why they choose clopidogrel in guideline table

Consider PPI

Avoid other NSAIDs

if VKA target INR 2-2.5

AF Patients with Coronary or Vascular Disease and an Indication for OAC (Age $\geq$ 65 years or CHADS₂ $\geq$ 1)



AF with Vascular Disease

(don't worry, we break it down in next 4 slides)

AF – OAC in Special Populations

Liver Disease	We recommend that OAC not be routinely prescribed in Child-Pugh class C or liver disease associated with significant coagulopathy
Cancer	Individualize OAC treatment in pts with active cancer. Consider DOAC> VKA.
Congenital Heart Disease	OAC for most patients with AF and HCM even if CHADS = 0 OAC if CHADS65 risk factors
Frail Elderly	OAC for most frail elderly – individualizing recommendations if high risk bleed
Thyrotoxicosis	OAC for “most patients during acute thyrotoxicosis until euthyroid state is restored”
Secondary Atrial Fibrillation =clearly provoked by transient/ reversible risk factor such as sepsis, post non-cardiac surgery	No OAC (however, if suspected high recurrence risk consider OAC) *NEW* ACC/AHA 2023 AF Update: counsel patients about recurrence risk, arrange follow-up including thromboembolic risk stratification and surveillance of recurrence. The benefit of anticoagulation is uncertain
Pregnancy (2021 CCS Pregnancy Heart Disease Update)	No DOACs (cross placenta). Consider LMWH, Warfarin (Tri2-3) in unique circumstances (see OB Med lecture for more details). Anticoag should be considered for pregnant women with AF and (1) structural heart disease or (2) no structural heart disease but CHADS ≥1
Atrial high rate episodes (AHRE) (often represent subclinical atrial fibrillation detected by a pacemaker; could also be atrial tachycardia)	*NEW* ACC/AHA 2023 AF Update: >24h AHRE with CHA ₂ DS ₂ -VASc of 2+ (correlates to risk of CHADS-65 of 1+) → OAC with shared decision making 5 min-24 hours of AHRE with CHA ₂ DS ₂ -VASc of 3+ (correlates to risk of CHADS-65 of 2+) → OAC with shared decision making (based on novel data from ARTESIA trial (NEJM 2024)

1. Valvular AF (any duration), or
2. NVAf Duration < 12 hours and recent stroke/TIA, or
3. NVAf Duration 12-48 hours and CHADS₂ ≥ 2, or
4. NVAf Duration > 48 hours

**Therapeutic OAC for ≥3 weeks
before cardioversion**

Alternate: TEE to
exclude LA thrombus

1. Hemodynamically unstable acute AF¹, or
2. NVAf Duration < 12 hours and no recent stroke/TIA, or
3. NVAf Duration 12-48 hours and CHADS₂ 0-1

**Initiate OAC as soon as possible
(preferably prior to cardioversion)**

Cardio- version of AF

DOAC preferred

CARDIOVERSION²

ANTICOAGULATION FOR 4 WEEKS POST CARDIOVERSION

**LONG-TERM ANTICOAGULATION BASED
ON THE “CCS ALGORITHM” (“CHADS-65”)**

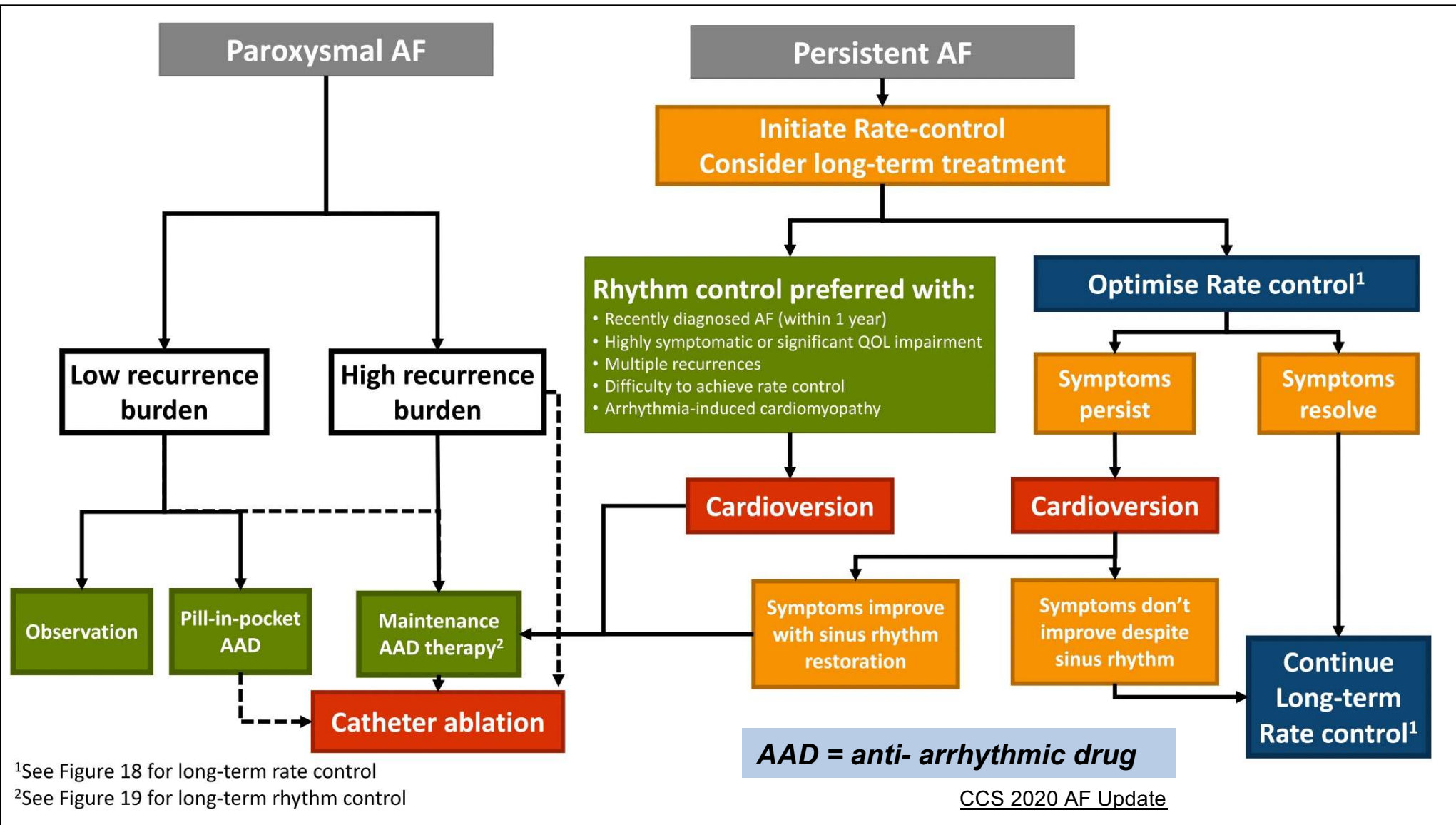
“Recent stroke” is defined
as stroke within the last 6
months

¹Hemodynamically unstable acute AF is defined as AF causing hypotension, cardiac ischemia, or pulmonary edema

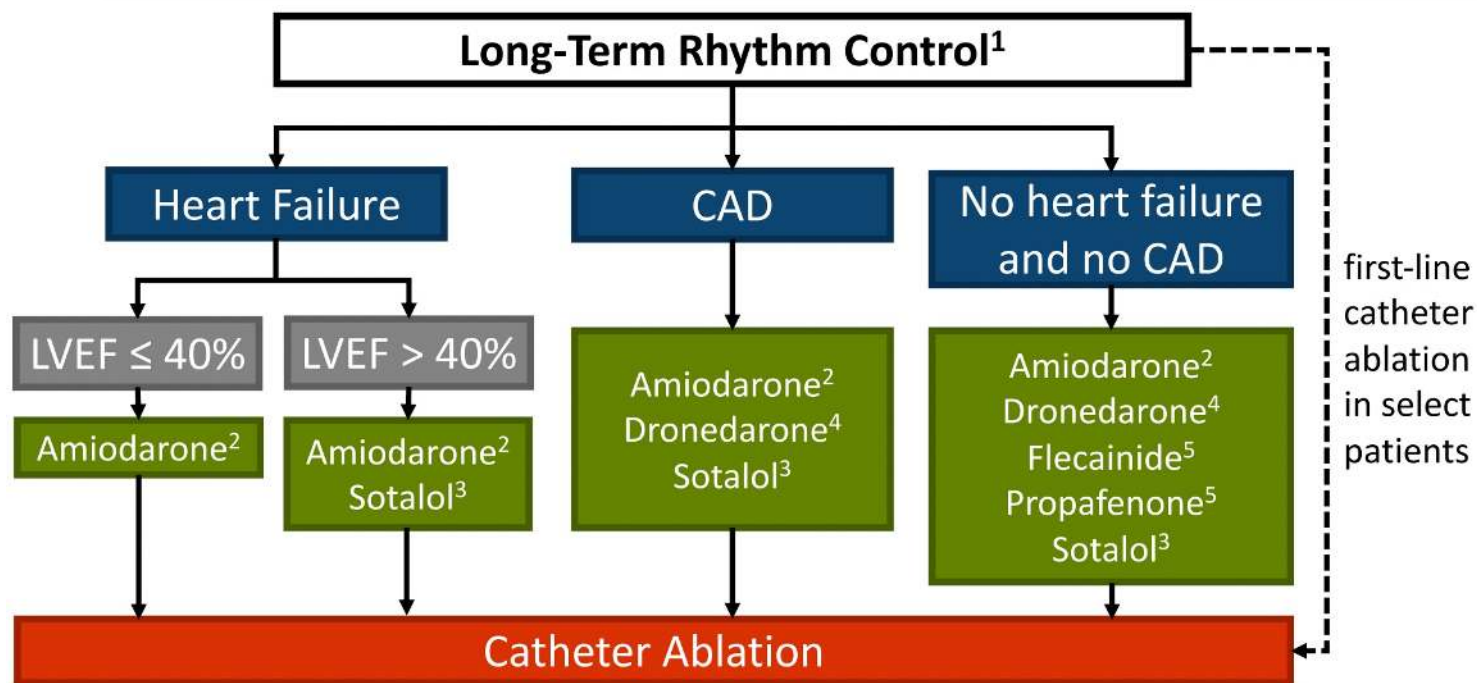
²Pharmacological or electrical cardioversion

Rate vs. Rhythm Control

- Until recently, we taught that for most patients, rate = rhythm control (AFFIRM Trial)
 - Target resting HR <100
 - **CCS 2020:** “we suggest a rhythm control study be considered for most stable patients with **recent onset AF**” [recent: <1yr – reduced CV death and stroke, mod-qual evidence]
 - Relevant Trials → **EAST-AFNET Trial (2020), EARLY-AF (2021)**
 - “we recommend that synchronized cardioversion may be used for sinus rhythm restoration in hemodynamically stable patients with recent onset Atrial Fib” → then pick AAD based upon patient characteristics
- NB. In the ED / Acute and unstable setting – still go with ACLS!
- ***EAST-AFNET Trial (2020) also supports rhythm control in CVD***
 - ***NEW* ACC/AHA 2023 AF Update** early rhythm control recommended:
 - Recommended for reduced LVEF
 - Reasonable for symptomatic AF to improve symptoms, recent diagnosis (<1 yr) to reduce progression, in combination of atrial fibrillation and HF



CCS 2020: Long Term Rhythm Control Choices



NEW ACC/AHA 2023 AF

Update: catheter ablation is recommended for patients for continued rhythm control if antiarrhythmic drug is not tolerated, ineffective or not desired.

Also first line for patients with few comorbidities to reduce progression from paroxysmal to persistent AF

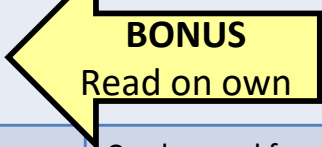
¹Consider AF symptom burden, possibility of adverse drug reactions and patient preference

²Consider alternative AADs or ablation rather than long-term amiodarone (significant risk of extra-cardiac side-effects)

³Sotalol should be used with caution in patients with high-risk features for torsade de pointes (≥ 65 years, women, reduced renal function, concomitant potassium-wasting diuretics). Sotalol is not recommended for patients with left ventricular hypertrophy.

⁴Dronedarone should be used with caution in combination with digoxin

⁵Class IC agent should be combined with AV-nodal blocking agent. Use caution for patients with left ventricular hypertrophy.

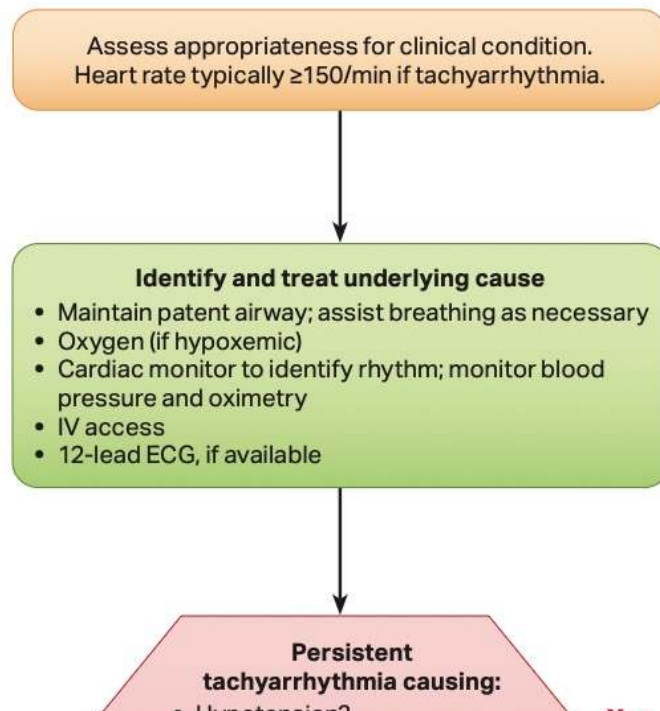
Anti Arrhythmic Drug	Use Acute or long term?	Prescribing Notes	Efficacy	Watch out for...	 BONUS Read on own
Amiodarone (Class III)	Acute and long term	IV or oral. Avoid if long QT. Not recommend for acute cardioversion – onset >8h. 10-12g load then ≤ 200mg daily maintenance	Variable, ~50%. Not first line due to toxicities.	IV: risks – ↑QT, ↓BP, bradycardia, phlebitis Long term Toxicity: transaminitis, pneumonitis, thyroid, derm	Can be used for rate control in ICU setting if c/I to 1st line rate control. ✓ LFT, Thyroid q6 mos ✓ CXR annually
Dronedarone (Class III)	Long term	Paroxysmal or Persistent AF only, not PERMANENT AF. Don't give w/ Digoxin or if Long QT.	<50%, but can ↓hospitalization	↑QT, Hepatotox. LFTs q 3 months x 1 yr then q 6mos	Not for CHF, LVEF<40% (↑mortality), AV conduction delay, ↑QTc
Flecainide or propafenone (class Ic)	Acute and long term	IV not available in Canada Oral 'pill in the pocket' – need to give BB or ND CCB 30 min prior to prevent 1:1 conduction.	~60% convert within 3 hrs of oral dose	1:1 av conduction – first dose given in monitored setting.	Not appropriate if structural heart disease or ISCHEMIC CAD, EF <40, liver failure, CrCL<35.
Ibutilide (Class III)	Acute	Pre load MgSO4 1-4g; monitor for 4 h post infusion.	AF – 30-50% A Flutter – 50-75%	Torsades de Pointes (TdP), non sust VT	Avoid in ACS, reduced EF, long QT!
Procainamide (Class Ia)	Acute	1g load over 60min	AF – 50-60% A flutter – 30%	Hypotension, PVCs, VT during infusion	Not for BRUGADA
Sotalol (BB at low dose, Class III at high dose)	Long Term	Baseline QT <440	~40%-50%	QT prolongation, TdP – repeat ECG 48-72h on therapy for QT	Do not use if QT long, CrCl<40, EF <40%, other RF for TdP
Vernakalant – atrial selective AAD.	Acute	Recent onset <48h Afib**	AF – 52-69% A flutter – doesn't work	Hypotension, brady after cardioversion. Abn taste, paresthesia, nausea common.	Avoid if hypotensive, in CHF, bradycardic, recent ACS or severe aortic stenosis

BONUS
Read on own

Other Highlights of AF Guidelines

- **Left Atrial Appendage Occlusion (Percutaneous > Surgical) for stroke prevention**
 - Only for patients who absolutely cannot tolerate OAC (e.g. – cerebral amyloid angiopathy and high CHADS AF)
 - Update on the PRAGUE-17 trial four years post LAA closure device vs. DOAC showed that LAA closure was non-inferior to DOAC with respect to embolic and bleeding risk, had significantly less non-procedural bleeding
 - ***NEW*** Above recommended is echoed in the ACC/AHA 2023 AF Guideline Update: LAAO reasonable for patients with AF and moderate to high risk of stroke ($\text{CHA}_2\text{DS}_2\text{-VASc}$ 2+ = CHADS-65 of 1+) and irreversible contraindication to anticoagulation
 - ***NEW*** Patients with high risk of bleeding (without absolute contraindication) can be considered for LAAO
- **AF with WPW:** Electrical cardioversion if unstable, IV procainamide or ibutilide or electrical cardioversion if stable (restore sinus rhythm preferred > rate control, avoid AV nodal blocking agents in patients with evidence of ventricular pre-excitation)
- Periprocedural Anticoagulation for AF → Periop Medicine Lecture
- Timing of OAC initiation after stroke → Neurology Lecture
- Management of Bleeding on OAC → Heme Lecture

Tachyarrhythmia - ACLS



Doses/Details

Synchronized cardioversion:

Refer to your specific device's recommended energy level to maximize first shock success.

Adenosine IV dose:

First dose: 6 mg rapid IV push; follow with NS flush.
Second dose: 12 mg if required.

Antiarrhythmic Infusions for Stable Wide-QRS Tachycardia

Procainamide IV dose:

20-50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases $>50\%$, or maximum dose 17 mg/kg given. Maintenance infusion: 1-4 mg/min. Avoid if prolonged QT or CHF.

Amiodarone IV dose:

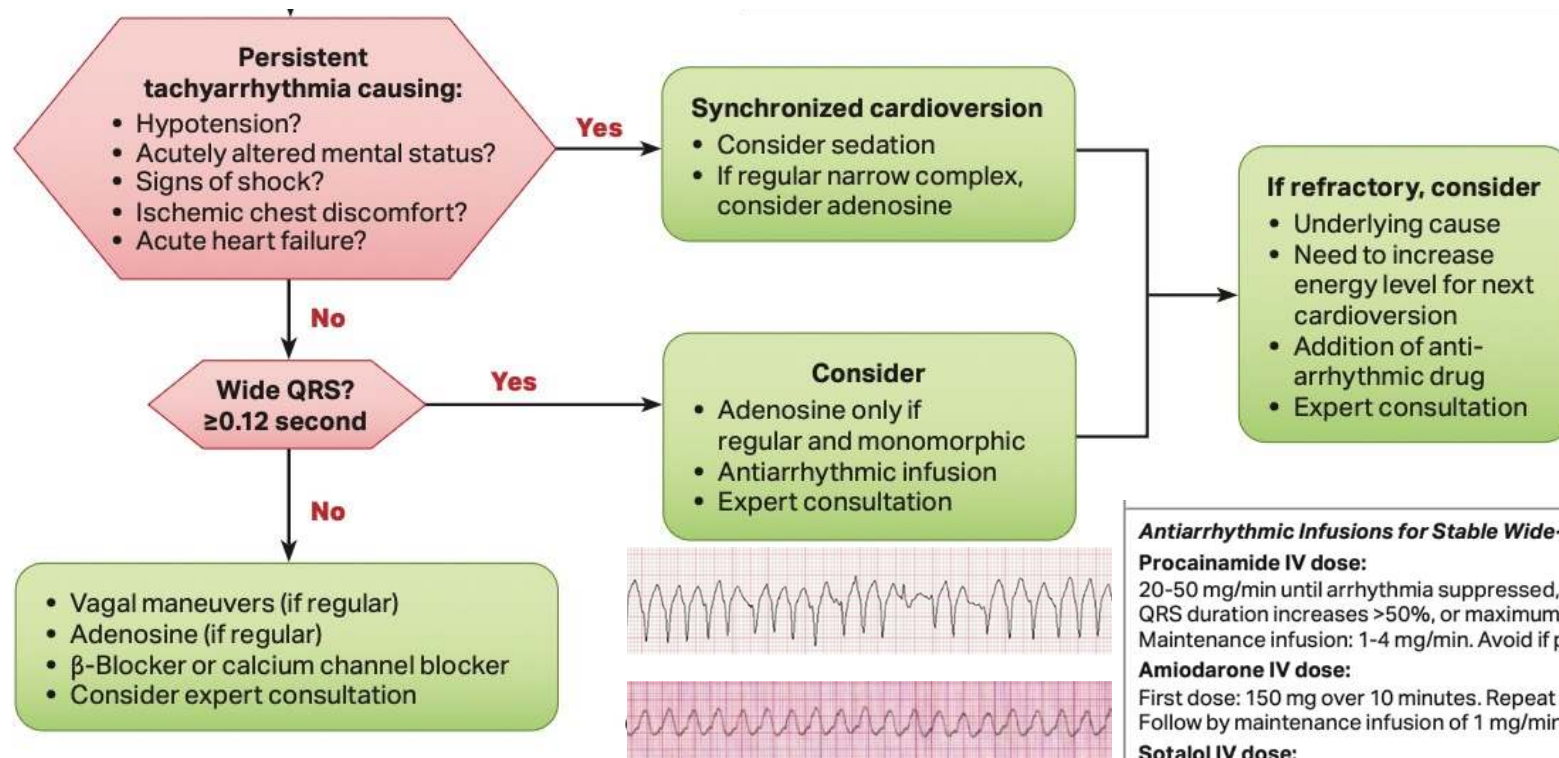
First dose: 150 mg over 10 minutes. Repeat as needed if VT recurs. Follow by maintenance infusion of 1 mg/min for first 6 hours.

Sotalol IV dose:

100 mg (1.5 mg/kg) over 5 minutes. Avoid if prolonged QT.

Synchronized cardioversion

Tachyarrhythmia - ACLS



Antiarrhythmic Infusions for Stable Wide-QRS Tachycardia

Procainamide IV dose:

20-50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases >50%, or maximum dose 17 mg/kg given. Maintenance infusion: 1-4 mg/min. Avoid if prolonged QT or CHF.

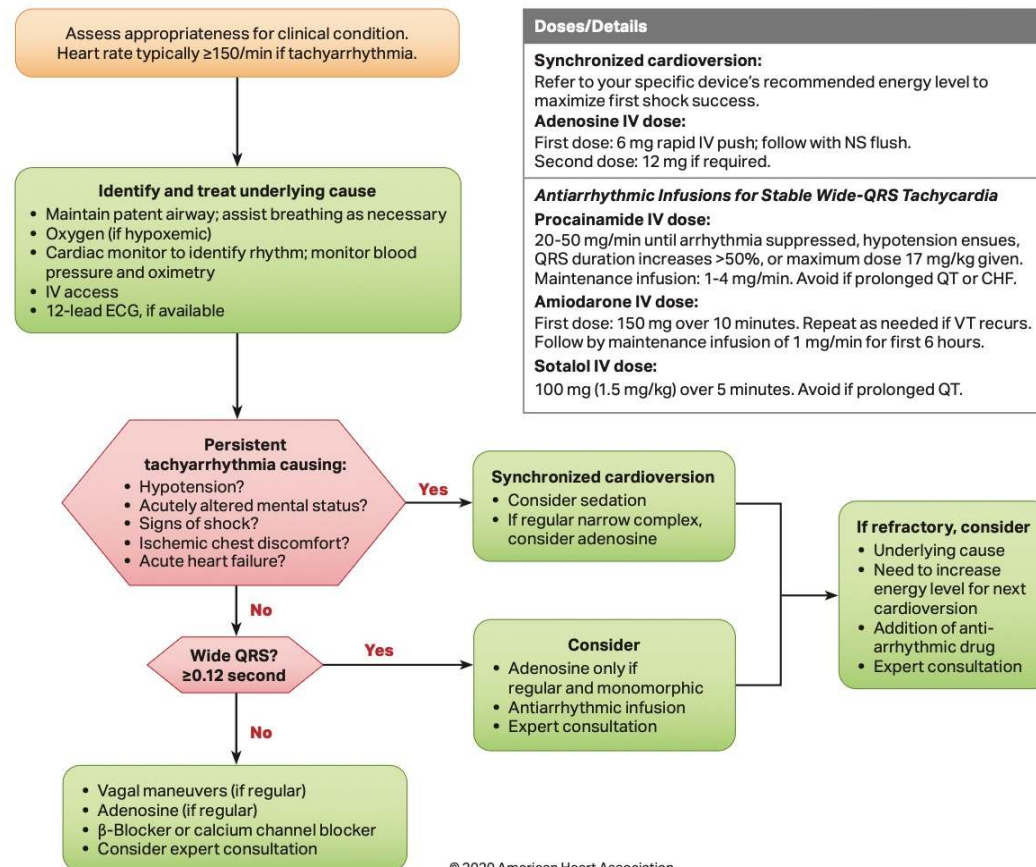
Amiodarone IV dose:

First dose: 150 mg over 10 minutes. Repeat as needed if VT recurs. Follow by maintenance infusion of 1 mg/min for first 6 hours.

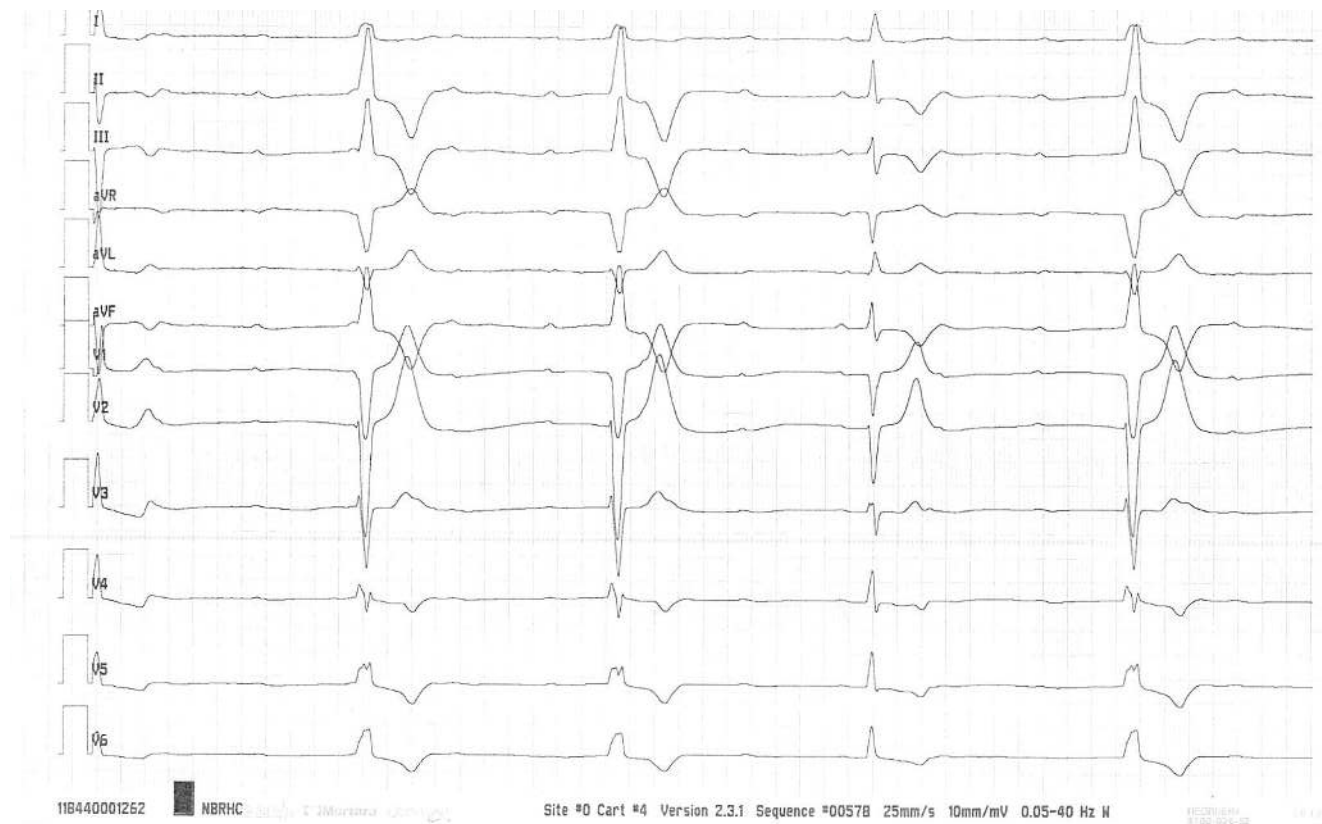
Sotalol IV dose:

100 mg (1.5 mg/kg) over 5 minutes. Avoid if prolonged QT.

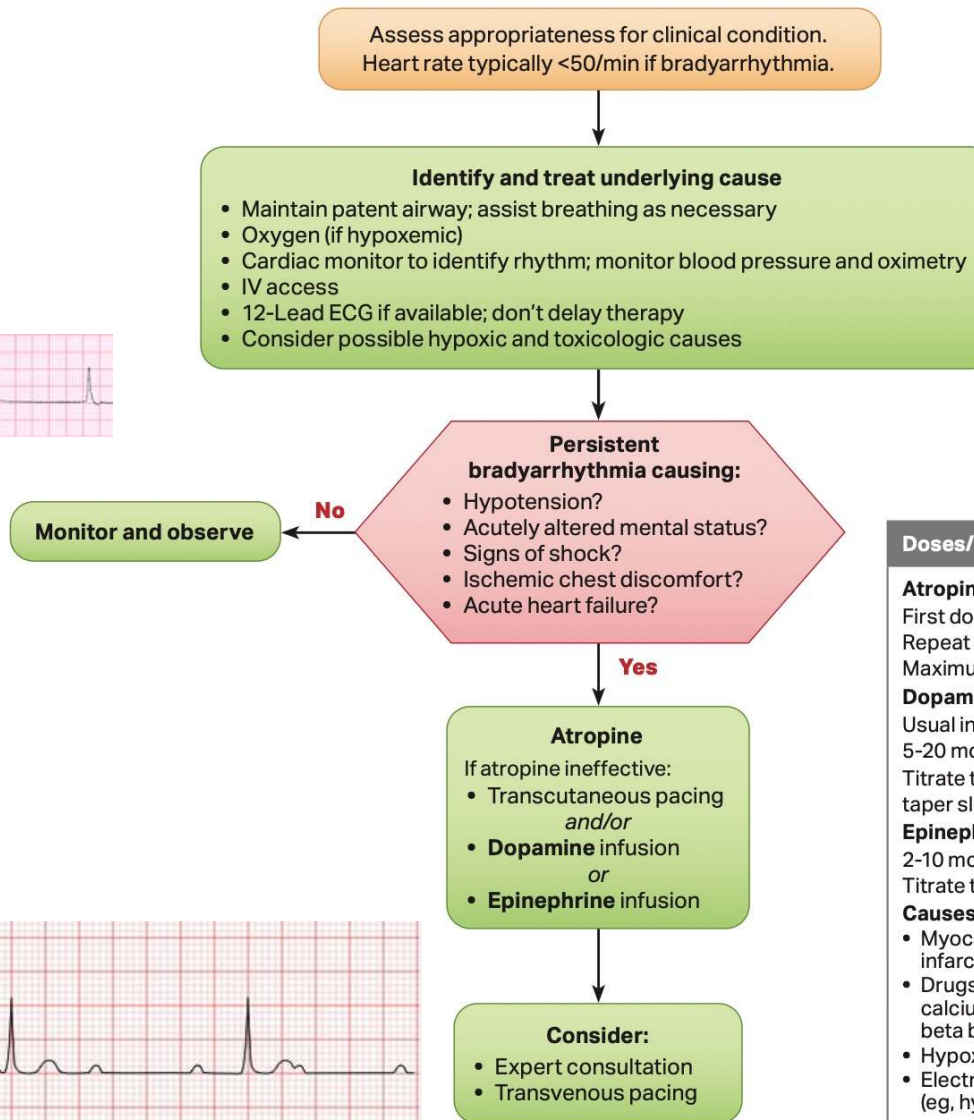
Tachyarrhythmia - ACLS



Bradyarrhythmias



Bradycardia ACLS



Doses/Details

Atropine IV dose:

First dose: 1 mg bolus.
Repeat every 3-5 minutes.
Maximum: 3 mg.

Dopamine IV infusion:

Usual infusion rate is
5-20 mcg/kg per minute.
Titrate to patient response;
taper slowly.

Epinephrine IV infusion:

2-10 mcg per minute infusion.
Titrate to patient response.

Causes:

- Myocardial ischemia/infarction
- Drugs/toxicologic (eg, calcium-channel blockers, beta blockers, digoxin)
- Hypoxia
- Electrolyte abnormality (eg, hyperkalemia)

Bradyarrhythmia & Pacing

Core principles when considering PPM therapy

- **Pacemakers are class I therapy for:**
 - Symptoms
 - Prevention of sudden death in asymptomatic patients
- **Other considerations**
 - Allow for uptitration of GDMT with significant (usually mortality) benefit
 - Syncope (narrow indications)
- **In general, PPMs are indicated for have symptoms related to bradyarrhythmia *OR* high-grade conduction system disease with risk of death (latter which does not require symptoms)**

Tips for brady w/u:

- In younger patients, think about OSA, Lyme, Sarcoid, electrolytes, thyroid, genetic/inherited
- Detection strategies should be dictated by symptom frequency
- Cardiac imaging in almost all patients
- If exercise-induced symptoms → EST
- if "reflex" symptoms → Tilt, CSM
- Not great indications for EP study in most brady

Pacemakers (*Summary of Indications*)

Sinus Node Dysfunction → *need symptoms*

- Sinus bradycardia, sinus pauses (no specific cutoff)
- Requirements for GDMT (i.e. needs BB for LV dysf)
- Tachy-brady syndrome (e.g. pAF with RVR with ++brady in sinus)
- Chronotropic incompetence (HR does not increase with exertion)

Acquired AV Block → *do not need symptoms*

- 2nd degree AVB Mobitz type 2
- 3rd degree AVB
- High grade AVB
- Permanent AF and symptomatic bradycardia
- Symptomatic AVB (spontaneous or from required drug therapy)

***If in doubt, a symptomatic patient with a bradycardic rhythm should prompt a consult from Cardiology/Electrophysiology**

NO PPM FOR:

- Asymptomatic 1st degree AVB
- Asymptomatic Mobitz I
- Asymptomatic 2:1 AVB at level of AVN
- Reversible, non-recurrent causes
- Vagally-mediated bradycardia

Special circumstances:

- For sarcoid, neuromuscular diseases, if PPM is needed can also consider ICD due to risk of VT/sudden death

Pacemakers after Myocardial Infarction – *R.C. Classic*

- **NB.** *Should undergo a waiting period in the MI setting before PPM*
- **NB.** *ESC makes it easier – “implantation of permanent pacemaker is indicated with the same recommendations as the general population when AVB does not resolve within a waiting period of at least 5d post-MI”*

ACC/AHA explicit recommendations

- **Temporizing measures**
 - Atropine administration reasonable
 - Temporary pacing for medically refractory symptoms or hemodynamic significance related to SND or AVB
- **Permanent pacemaker**
 - Mobitz II AVB
 - High grade AVB
 - Alternating BBB
 - 3rd degree AVB
- **Class III recommendations**
 - Transient AVB that resolves
 - New BBB or isolated fascicular block

Other special circumstances for pacing:

- Cardioinhibitory carotid sinus syndrome
- Asystolic syncope during tilt testing
- Post-cardiac surgery
- Post-TAVI (high degree AV or 3o heart block a common complication post procedure)

Pacemaker function

Pacemakers can be:

- **Single-chamber** (usually ventricular; atrial-only rare)
- **Dual-chamber** (atrial + ventricular)
- **Resynchronization devices** (atrial, RV, plus LV via coronary sinus)
- **Device choice** depends on the patient. Transvenous systems are most common, but leadless devices exist.

Pacemaker settings:

1. **Chamber paced:** A = atria, V = ventricle, D = both
2. **Chamber sensed:** A, V, D both, O = none sensed
3. **Response:** I = inhibited, T = triggered, D = both, O = none
4. **Optional “R”** = rate response (rate increases with activity; useful for non-sedentary patients with single-lead ventricular pacemakers)

V

V

I

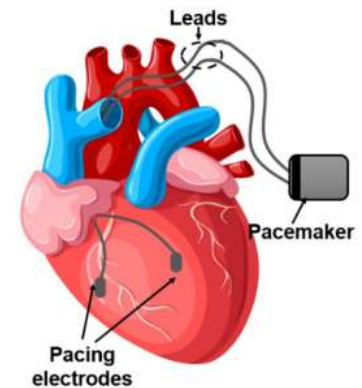
R

Chamber paced

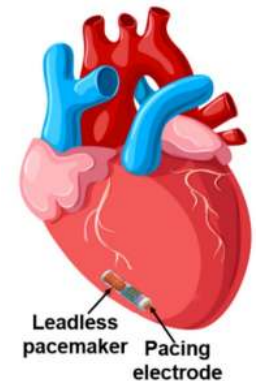
Chamber sensed

Chamber response

Rate response



(a)



(b)

Pacemaker function

- **Common pacemaker modes:**

- **DDD** – both chambers sensed and paced. If a native P wave occurs, the device will be “inhibited” and not pace the atria. If that P wave is not conducted down the AV node, the absence of a QRS after a native P wave will “trigger” the pacemaker to pace the ventricle. This is the “D” or “dual” function of the third letter. This setting is useful for someone with complete heart block and intact sinus node function.
- **VVI** – only the ventricle is sensed and paced. If a native QRS comes through, the device will be “inhibited,” hence the “I.”
- **AAI** – only the atria is sensed and paced. This is useful for someone with sick sinus syndrome who has intact AV node conduction. Pacing the atria avoids rare deleterious effects of RV pacing (rarely can produce LV dysfunction). AAI is often paired with DDD (AAI-DDD) and something called “mode switching,” which will switch the pacemaker to DDD in which if the AV node starts to fail (i.e. the PR interval gets too long before a QRS comes through).



DDD



VVI



AAI

BONUS

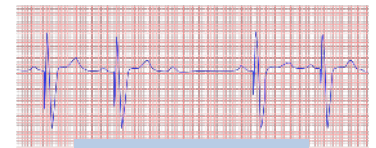
Read on own

Pacemaker malfunction – beyond GIM exam, but info requested by your colleagues

- **Failure to capture:** there is a pacemaker spike but **no P-wave or QRS**. This may occur because the lead has failed or moved. Lead revision is often necessary.
- **Oversensing:** the pacemaker senses **something else** as cardiac activity (e.g. cautery) and will be **inappropriately inhibited**, therefore **not pacing the patient**.
- **Undersensing:** the pacemaker does not sense cardiac activity and **will not be appropriately inhibited**, leading to **inappropriate pacing**.
- Fixes for these issues involve changing in programming and/or lead revision.
- **Magnets:** a magnet will not “disable” a pacemaker. It will make it go into an “**asynchronous**” mode, most often **VOO**. This will be at the pacemakers “magnet rate,” most often 100 bpm. This is useful if a patient is undergoing surgery and cautery near the chest may be detected by the device as cardiac activity. The patient will “oversense” this and not pace, which would be bad for someone who is dependent! Turning the device temporarily to VOO avoids this problem as the device will not “see” or “sense” the cautery, and won’t become inhibited!
 - N.B. for an ICD, a magnet will disable the “tachytherapies” or shocks, which is useful if a patient is being shocked inappropriately until the device can be reprogrammed.



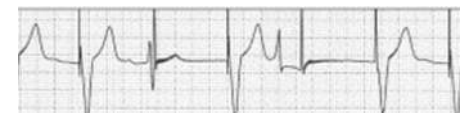
Failure to capture



Oversensing



Undersensing



VOO

Ventricular Arrhythmias

- Know your ACLS Algorithm
- Beta-blockers and amiodarone are the go to drugs, but remember amiodarone **prolongs QT**
- Device therapy is important secondary prevention consideration (*see Heart Failure/ICD section*)
- Never forget **DRIVING RESTRICTIONS** (*for both arrhythmias + devices*)



Ventricular Arrhythmias

COMMON OSCE SCENARIO!

Don't forget to look for etiology after patient cardioverted! Great table in CCS guidelines

- If unstable -> ACLS!! Defibrillation, ABCs, look for etiology (ischemia, long QT, etc.)
- Electrical storm (≥ 3 episodes in 24h)
 - Beta Blocker, preferably non-selective
 - IV amiodarone
 - Consider sedation / anesthesia
 - Consult heart rhythm expert early
 - Treat acute HF, if present
- If stable, sustained (>30 sec) VT
 - DC cardioversion vs. procainamide
 - Amiodarone IV 150mg then 900mg/24h infusion
 - Lidocaine
- Polymorphic VT/VF
 - Normal QT $\rightarrow$ acute ischemia (ACS Tx and amio or lido) vs. no ischemia (amio)
 - Prolonged QT $\rightarrow$ IV Mg, overdrive pacing, non-selective BB, lidocaine if refractory

Prescribing tips / Back Pocket Doses:

AMIO: 300mg IV initial push if unstable

Lidocaine 100mg IV push If unstable

Procainamide 10mg/kg IV over 20 min for stable monomorphic VT

Beta Blocker at maximal doses for everyone post sustained VT/VF.

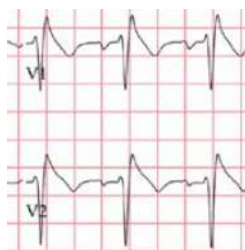
- ***NEW* CCS 2024 Post Cardiac Arrest Clinical Practice Update:**
 - In patients with Out of Hospital Cardiac Arrest (OHCA) who remain in polymorphic VT/VF after 1 defibrillation attempt, either amiodarone or lidocaine can be administered
 - In patients with OHCA with a refractory shockable rhythm (no termination after 3 shocks), double sequential or vector change defibrillation can be considered (DOSE VF trial, NEJM 2022)

BONUS

Read on own

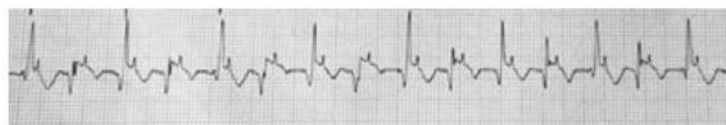
Channelopathies and ventricular arrhythmias

- Brugada syndrome, long QT syndrome and catecholaminergic polymorphic VT (CPVT) are three channelopathies which carry risk of sudden cardiac death. Detailed knowledge of these conditions is beyond the scope of an IM exam.
- ICDs are indicated in the setting of sudden cardiac arrest and expected meaningful survival greater than 1 year, or recurrent ventricular arrhythmias not controlled with medical therapy.
- **Channelopathy pearls:**
 - **Brugada syndrome** is an autosomal dominant mutation, most commonly in sodium channels (SCN5A in 20-30% of patients, but many more mutations exist). Type 1 pattern is the only clinically relevant pattern. Types 2 and 3 only matter if they turn into Type 1 with fever or procainamide challenge. In an asymptomatic patient with only an inducible Type 1 pattern (i.e. no Brugada without provoking agent), no ICD is indicated (unless there is a strong FHx of SCD, shared decision making). In an asymptomatic patient with a spontaneous Type 1 pattern (i.e. persistent, not induced), EP study can be considered for risk stratification. Aside from cardiac arrest, ICD is indicated for syncope presumed to be from a ventricular arrhythmia with Type 1 ECG. Quinidine is useful to reduce recurrent ICD shocks.
 - **CPVT** involves a mutation in the ryanodine receptor. *Bidirectional VT* can be provoked by exercise, diagnosed by stress testing. Treatment is nadolol. ICD only indicated if sudden cardiac death.



Brugada Type 1
Pattern (Dome in
V1/V2)

Bidirectional VT



BONUS
Read on own

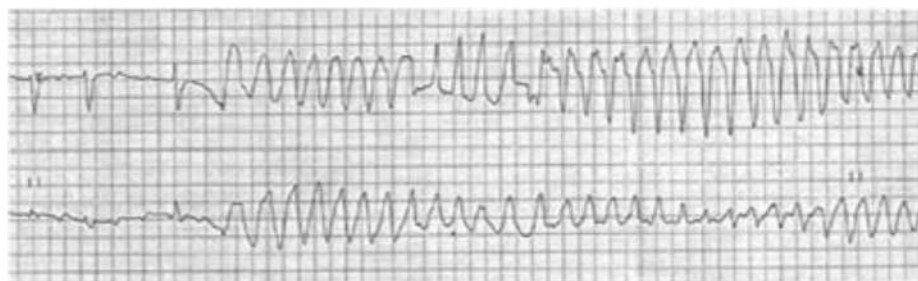
Channelopathies and ventricular arrhythmias

- **Channelopathy pearls:**

- **Long QT syndrome** involves mutations in potassium channels (LQTS1/2) and sodium channels (LQTS3). Various triggers (exercise=1, loud noises=2, sleep=3) and patterns. A QTc over 500 ms is long and needs an explanation! Differential includes a bundle branch block (widens the QRS so would extend the QTc), drugs, electrolytes, medications, congenital. **Beta blockade is recommended if QTc is 470 ms or more to reduce risk of ventricular arrhythmias.**
 - Torsades occurs in the setting of QT prolongation is triggered by an “R-on-T” event, in which a premature ventricular contraction hits the T-wave, producing polymorphic VT.
 - Acute management involves stopping offending medications and correction of electrolytes (Target Mg = 1.0, K >4). Increasing the patient’s heart rate (shortens the QT interval and makes it less likely for a PVC to hit the QT) may be necessary acutely, which can be achieved with temporary pacing (overdrive pacing) or **isoproterenol (beta agonist)**. Lidocaine can be helpful as it does not lengthen, and actually may shorten, the QTc.
 - If the QT interval cannot be corrected, device therapy is often necessary (dual chamber ICD which can be used to pace the patient atrially at 80 bpm to prevent bradycardia/PVCs, and also act to treat polymorphic VT.
 - Why beta blockers if we are preventing bradycardia with overdrive pacing? They likely suppress ventricular ectopy, reducing the “R-on-T” likelihood and have been shown in studies to reduce the risk of cardiac arrest



Long QT 1 Pattern



Torsades de pointes initiation (R-on-T)

BONUS

Read on own

Ventricular Arrhythmias- CCS 2020 update

Position Statement on the Mgmt of VT/VF in Presence of SHD

In All Patients

*Structural Heart Disease (SHD) = abnormal myocardium and scar

Initial Evaluation of Sustained VT/VF
(once you've resuscitated them!)

***NEW* CCS 2024 Post Cardiac Arrest Update:** Primary PCI (i.e. "activate cath lab") or thrombolysis is indicated on presentation for comatose patients with cardiac arrest and STEMI.

If ST elevation is not present, coronary angiography can be delayed unless there are high risk features (hemodynamic instability, high risk ECG changes, recurrent ventricular arrhythmias suspected related to ischemia, per COACT NEJM 2019, TOMAHAWK NEJM 2021)

History	Clinical Evaluation	ECG
- Precipitating factors - Symptoms during and preceding VT/VF - Symptoms of heart failure - Nature of structural heart disease (if known) - Medications (prescribed and other) - Drugs, alcohol, stimulants - Family history - Other pertinent medical history	- Hemodynamic assessment - Heart failure assessment	During VT (if stable) - After restoration of underlying rhythm - Ischemia/infarction changes - Conduction disease - QT interval
	Laboratory Investigations	
	- CBC - Electrolytes, renal function, liver function - TSH - Troponin - BNP/NTproBNP - Toxicology, drug levels (if suspected)	Echocardiography
		Transthoracic*
		ICD Interrogation
		If present

In Selected Patients

Cardiac Magnetic Resonance Imaging (CMR)	Coronary Evaluation	Positron Emission Tomography (PET)
- In patients with suboptimal images or equivocal findings on echocardiography or other imaging modalities - For further evaluation of inflammation or scar/fibrosis	- Coronary angiography in patients with VT/VF and acute STEMI (emergent) - In patients with polymorphic VT/VF and suspicion of CAD - In selected patients with monomorphic VT in whom CAD is suspected	In patients with suspected inflammatory processes (particularly sarcoidosis), to guide decisions regarding anti-inflammatory/immunosuppression

Approach to WCT: Tips to Help Differentiate VT from SVT w/ Aberrancy



1. Look for a “smoking gun”
 - AV dissociation → P waves marching, capture beats, fusion beats
2. Look at the axis, QRS polarity in aVR
 - NW axis, upright in aVR point towards VT
3. Does the QRS look like a typical bundle branch block?
 - Is there concordance in the precordial leads?
 - Is there RSr' (atypical “bunny ears”) in V1 if RBBB-ish
 - Is there a Q wave in V6 if LBBB-ish
4. +/- Consider the width of the QRS

Does rate help?

→not really

Does regularity help?

→maybe, but not reliable

Does QRSd help?

→ probably

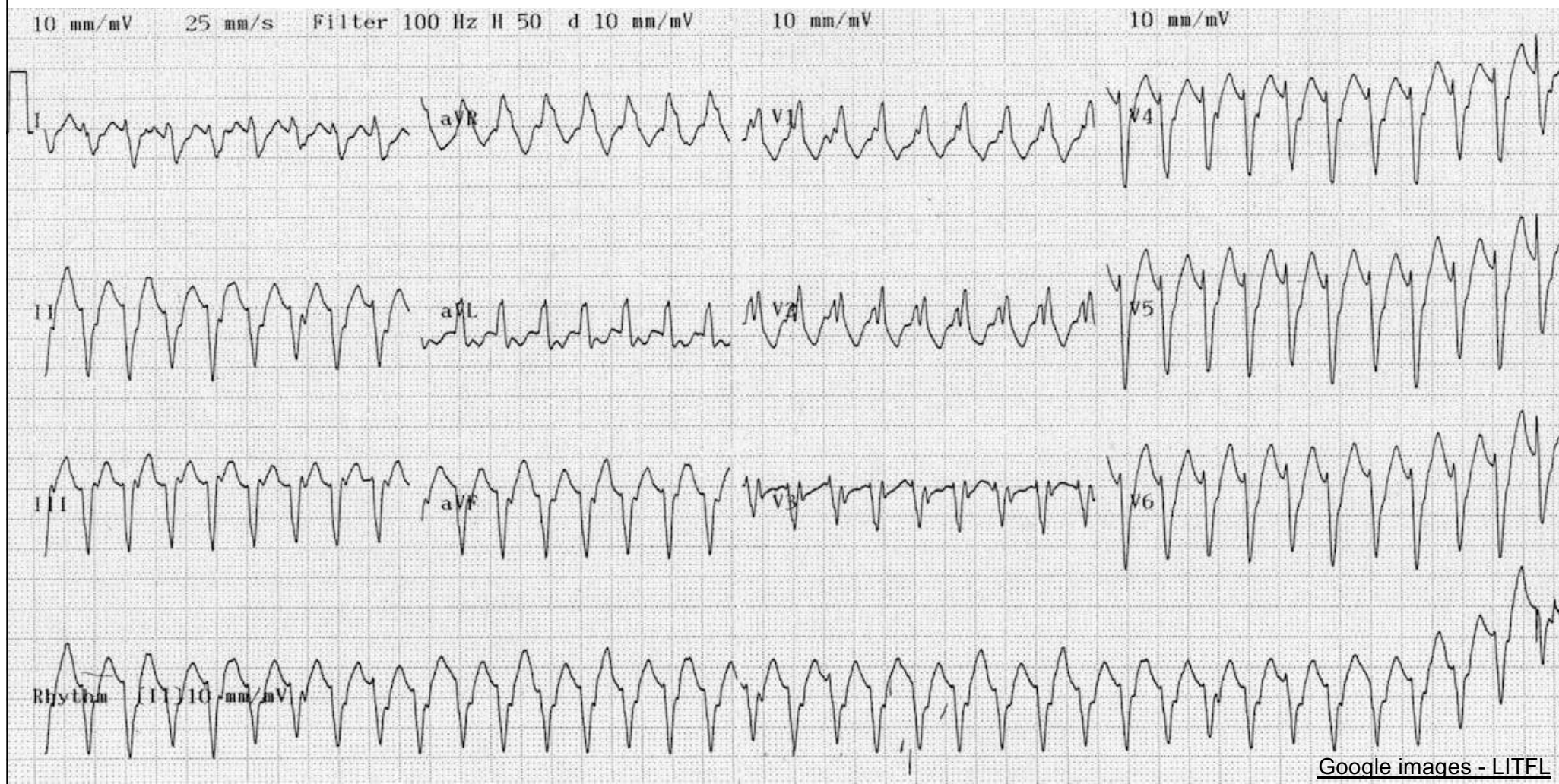
Does previous ECG help?

→ oh yea

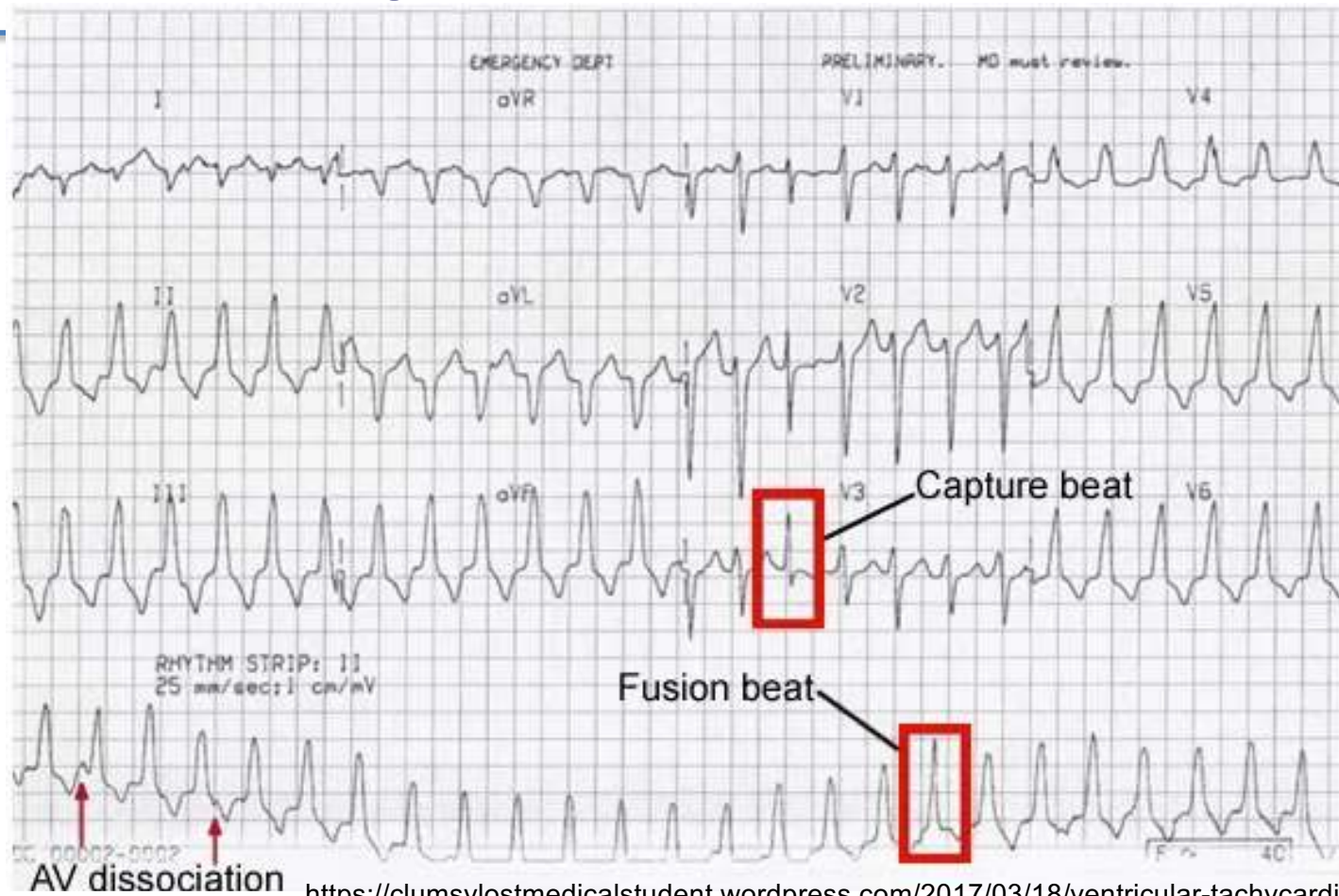
Does age / med Hx help?

→ DEFINITELY...but that's cheating

Positive QRS in aVR



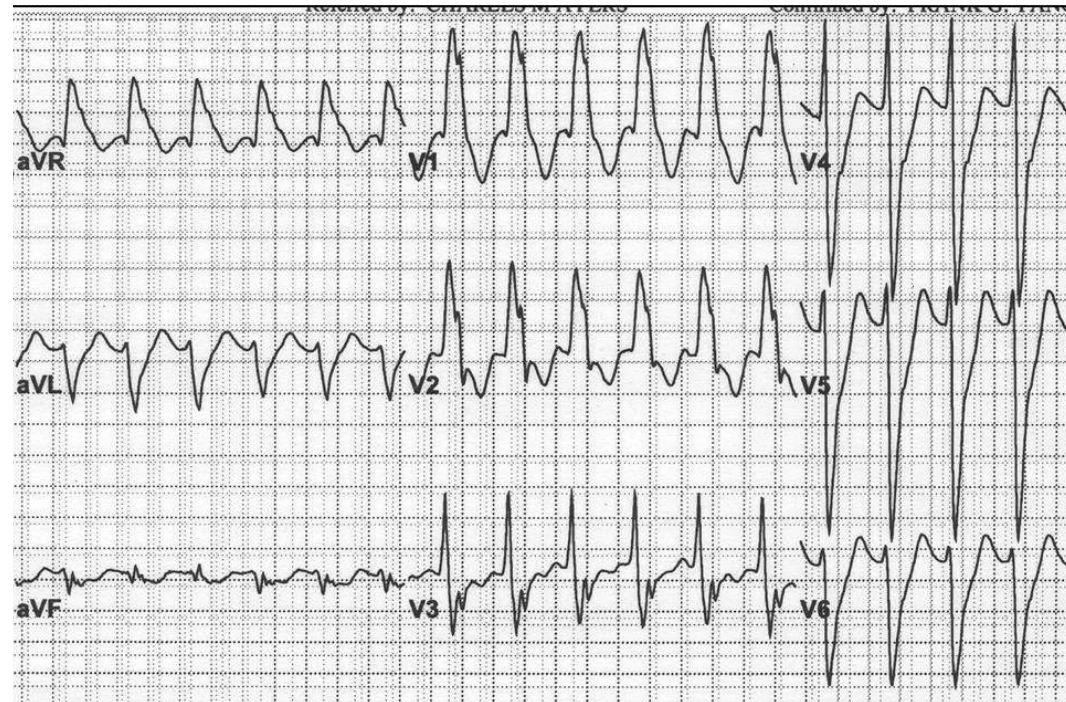
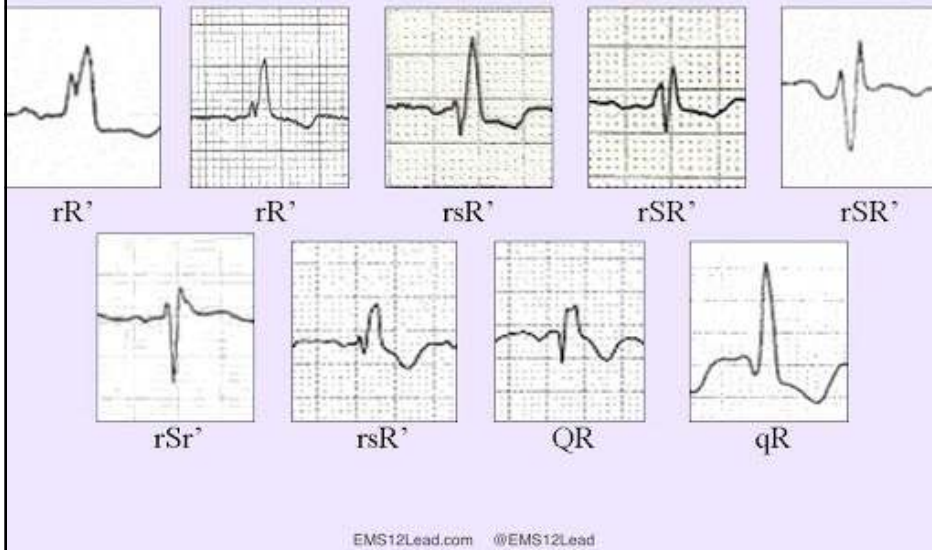
Capture and Fusion



<https://clumsylostmedicalstudent.wordpress.com/2017/03/18/ventricular-tachycardia-ventricular-fibrillation/>

“Rabbit Ears”

Various RBBB Morphologies Lead V1



Cardiac Arrest - ACLS

• Drug therapy:

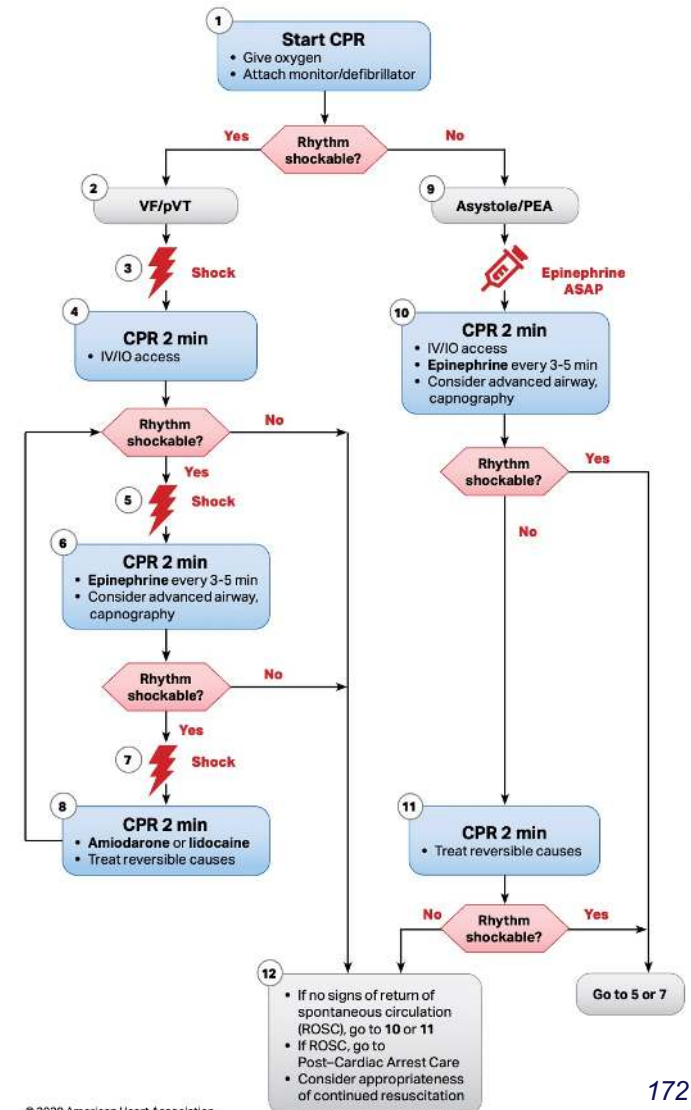
Epinephrine IV/IO dose: 1 mg every 3-5 minutes

Amiodarone IV/IO dose: First dose: 300 mg bolus. Second dose: 150 mg.

Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

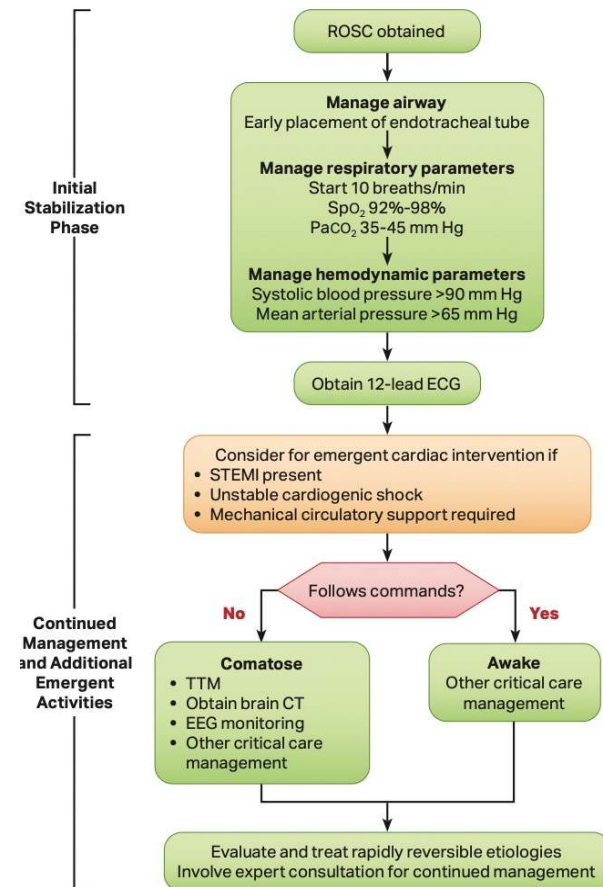
• Reversible causes:

- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

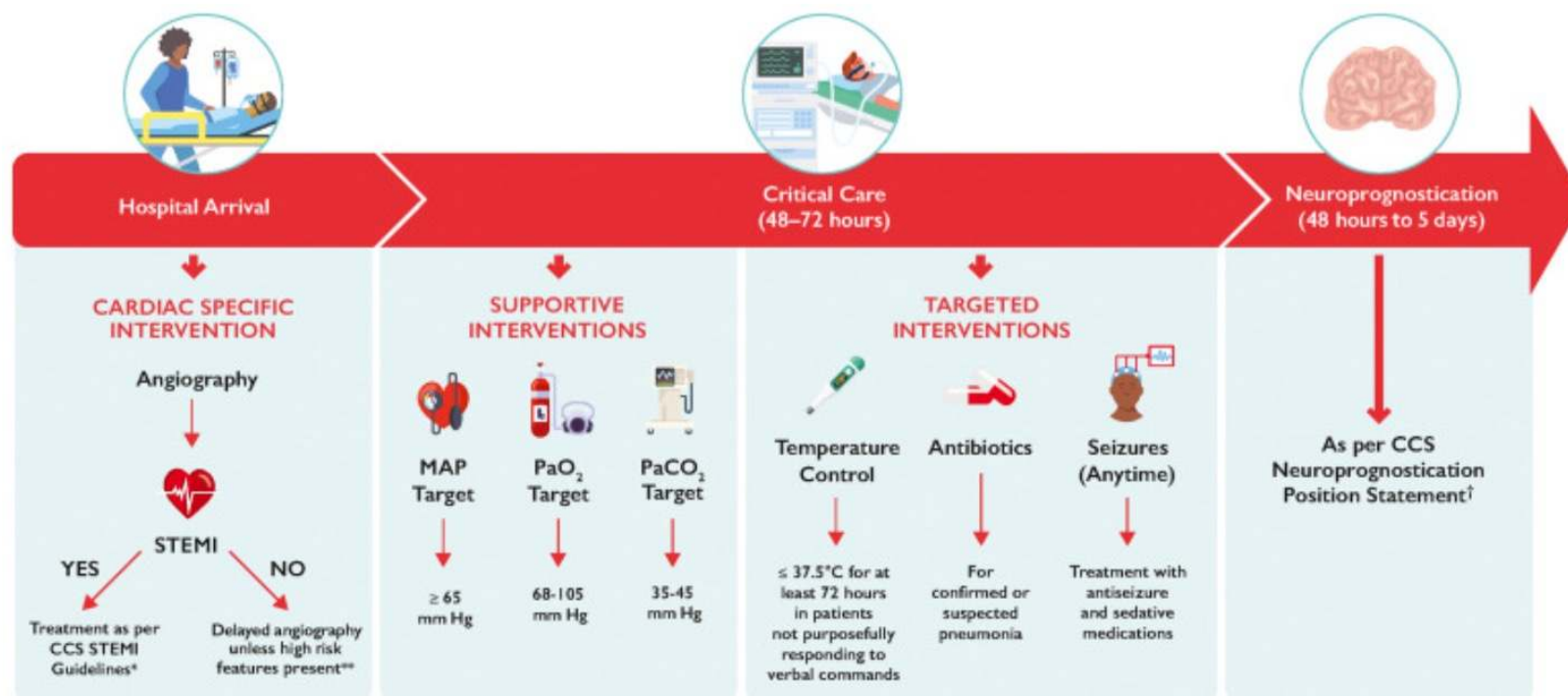


Post Cardiac Arrest Care - ACLS

- **Initial stabilization phase:**
 - Airway management: Waveform capnography or capnometry to confirm and monitor endotracheal tube placement
 - Respiratory parameters: Titrate Spo2 92%-98%; start at 10 breaths/min; titrate to Paco2 of 35-45 mm Hg
 - Hemodynamic parameters: Titrate fluid/vasopressors/inotropes to target MAP > 65 mmHg
- **Once in the critical care unit:**
 - TTM: If patient is not following commands, start TTM as soon as possible; begin at 32-36°C for 24 Hours
 - Maintain normoxia, normocapnia, euglycemia
 - Provide continuous or intermittent electroencephalogram (EEG) monitoring
 - Provide lung-protective ventilation



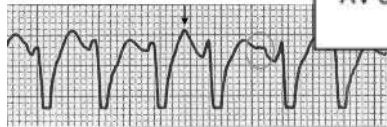
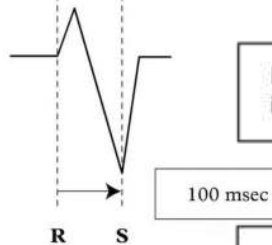
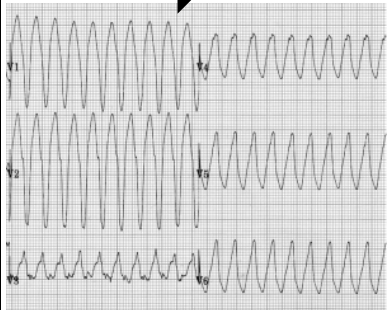
Post Cardiac Arrest Care - ACLS



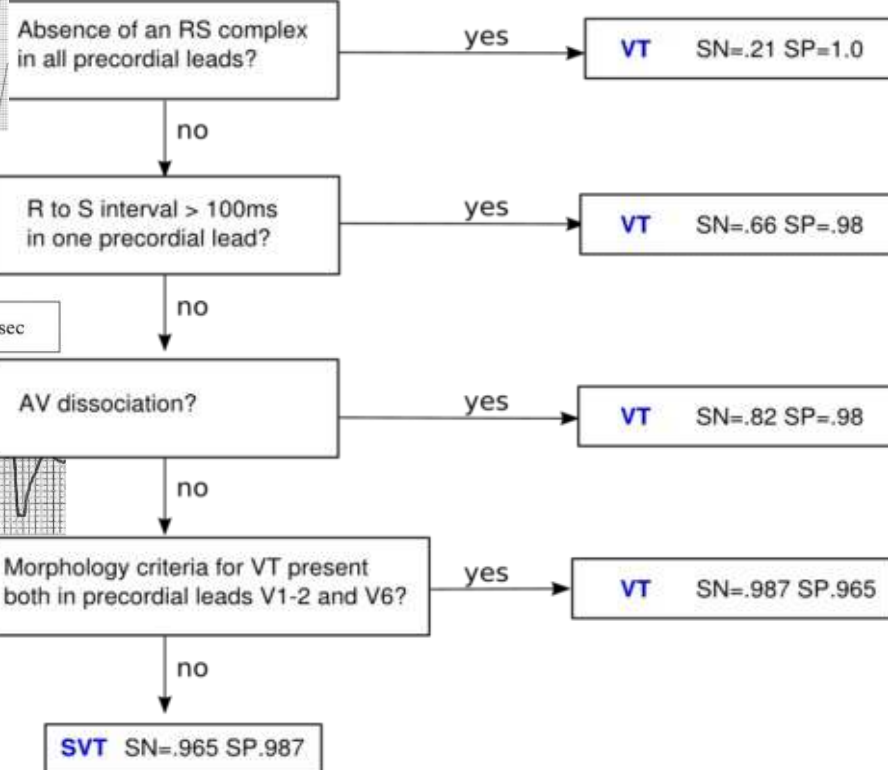
BONUS

Read on own

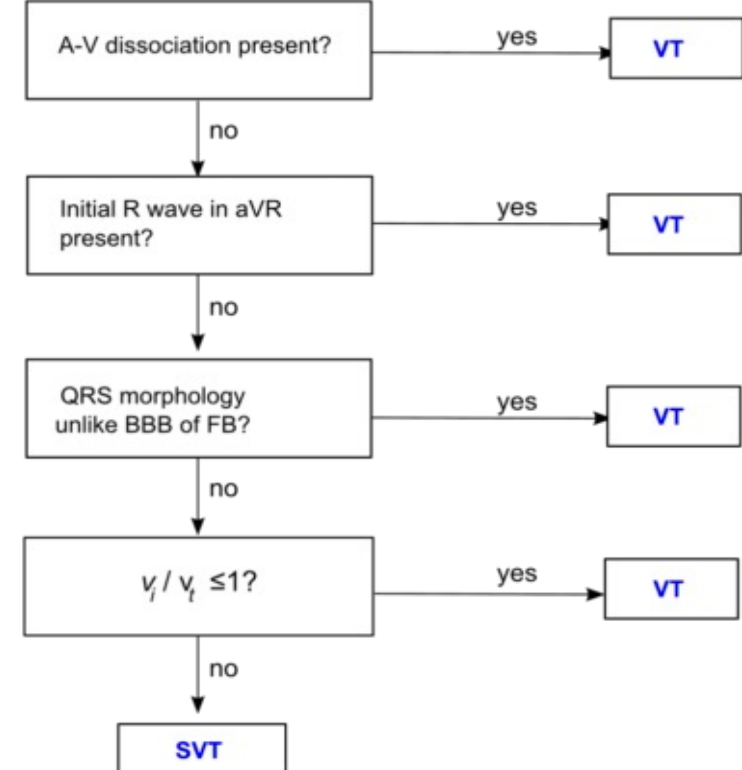
VT vs. SVT with aberrant conduction



Brugada Algorithm



Vereckei Algorithm



Arrhythmia	Private Car	Commercial Driver (Truck, Bus)
Sinus node dysfunction	No impaired LOC: OK to drive Impaired LOC/symptomatic pauses: NO DRIVING until PPM	
First degree, Wenckebach, isolated LBBB, LAFB, LPFB, RBBB, bifasicular block	OK to drive unless history of impaired LOC	
Second degree type II, alternating BBB, 3 rd degree AVB	NO DRIVING until PPM Exception: congenital 3 rd AVB (only restricted if impaired LOC)	
SVT/AF/AFL	No impaired LOC: OK to drive Impaired LOC: NO DRIVING until treated	
VF (no reversible causes)	3 months	NO DRIVING
VT/VF due to reversible cause (MI, drug, etc.)	NO DRIVING until underlying cause treated	
Hemodynamically unstable VT or impaired LOC	3 months	NO DRIVING
VT with structural heart disease and no impaired LOC, no ICD	3 months	NO DRIVING
VT with structurally normal heart (idiopathic VT)	1 week if well controlled	NO DRIVING



CCS 2023 Fitness to Drive Guidelines - Arrhythmias

Major updates from previous guidelines:

- **For ventricular arrhythmias, restriction for private driving has been reduced to 3 months** (previously 6 months). This is the same as an ICD placed for secondary prophylaxis with impaired LOC.
- N.B. ICD shock with impaired LOC will have the same restriction as hemodynamically unstable VT



CCS 2023 Fitness to Drive Guidelines - Pacemakers

Category		Private Car	Commercial Driver (Truck, Bus)
Pacemaker	Impaired LOC or high grade AV block	1 week post implant	
	No impairment in LOC/high grade AV block	OK to drive	
	Generator change (also applies for ICD private driving)	OK to drive	
	Lead upgrade/revision (also applies for ICD private driving)	1 week post implant if history of impaired LOC/high grade AV block Otherwise OK to drive	

Major updates from previous guidelines:

- Private and commercial driving have the **same recommendations**
- Pacemaker implantation in the **absence of impaired LOC/high grade AV block has no restriction**
- **Commercial driving post PPM restriction reduced to 1 week** (previously 1 month)

Category		Private Car	Commercial Driver (Truck, Bus)
ICD	Primary prophylaxis	1 week post implant	NO DRIVING
	Secondary prophylaxis, impaired LOC	3 month	
	Secondary prophylaxis, NO impaired LOC	1 week	
ICD therapy	ICD shock or therapy, impaired LOC or disabling	3 months	
	ICD shock or therapy, NO impaired LOC or disabling	1 week	
	Inappropriate ICD therapies	No restriction	
	Electrical storm (3 or more VT/VF events in 24 hours)	3-6 months after event (expert re-evaluation)	



CCS 2023 Fitness to Drive Guidelines - ICD

Major updates from previous guidelines:

- N.B. if you have an **ICD** you **cannot drive commercially** (on the basis of underlying condition that needs ICD)
- **For primary prophylaxis ICD**, restriction is now **1 week** private driving (previously 1 month)
- **For secondary prophylaxis ICD, if LOC was impaired**, restriction is now **3 months** (previously 6 months).
- Recommendations for inappropriate therapies and electrical storm are now made.

MCQ 2026

You are called to see a patient in the ER who had presented with an acute myocardial infarction. They have now developed rapid heart rate at 150 bpm. ECG is shown. BP 70/40. Next step in management?

- a) Verapamil IV
- b) Metoprolol IV
- c) **Electrical cardioversion**
- d) Lidocaine
- e) Procainamide



Answer: D. This patient is hemodynamically unstable. The rhythm is a narrow complex tachyarrhythmia at the rate of up to 300 bpm which is not compatible with organ perfusion. Regardless of anticoagulation status, this patient should be cardioverted as per ACLS guidelines. Post cardioversion, they require minimum of 4 weeks of OAC regardless of the CHADS.

Syncope

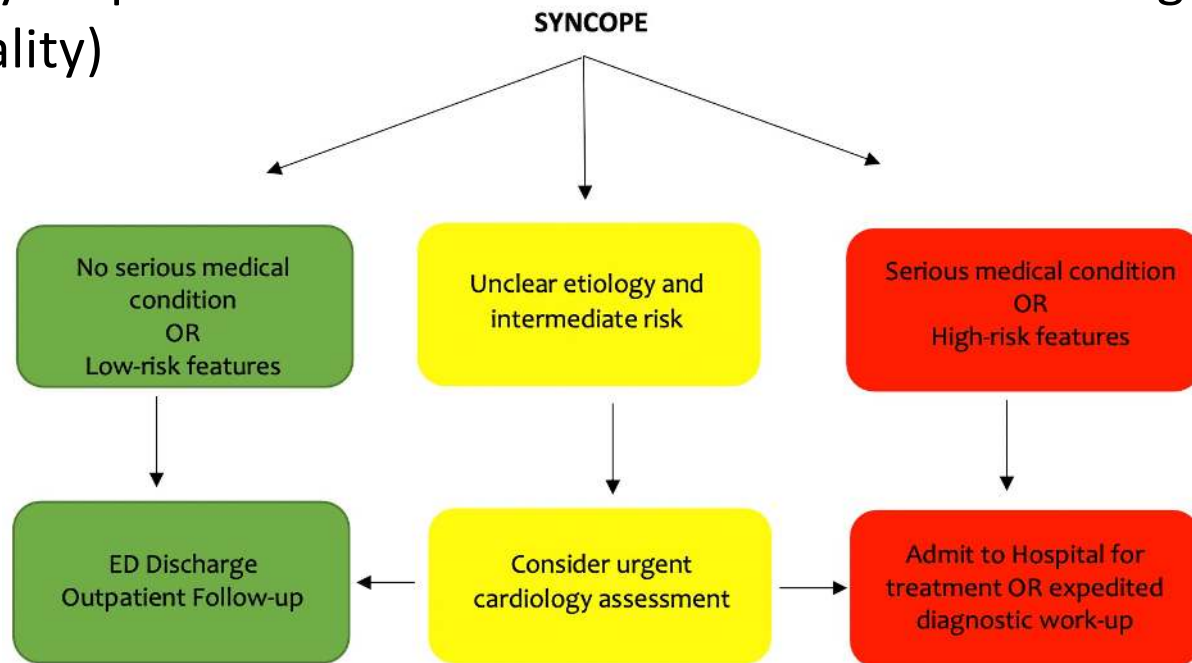
Key Resources:

- CCS 2020 Syncope Update
- JAMA RCE “Does this patient have cardiac syncope” – summarized in handout on internalmedicinereview.ca website

Syncope Practical Tips



- Syncope is all about history & risk stratification
- In the ED or from GIM POV → trying to find a **needle in the haystack** (i.e. Syncope related to a condition associated with high morbidity & mortality)



Syncope Practical Tips

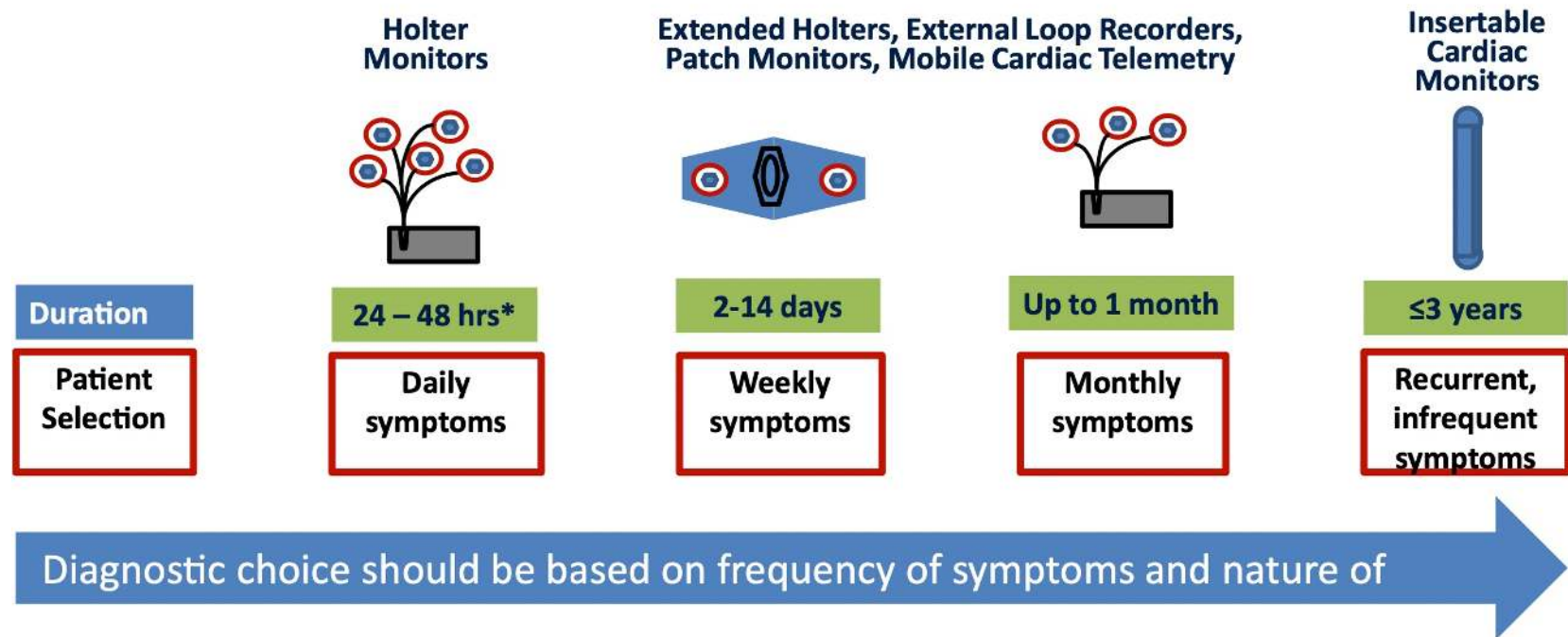
- “There are no single variables which can accurately diagnose syncope or seizure. Clinical prediction rules alone cannot be used to rule out cardiac syncope.” (JAMA RCE, 2019)
- The use of broad-based investigations in the workup of syncope is ineffective and costly
- Know what constitutes low vs. high risk on
 - History [Fam Hx Sudden cardiac death, symptoms sounded ischemic, syncope while exercising or lying supine or without prodrome, trauma due to syncope, known CVD]
 - Exam [abnormal vitals or cardiac exam]
 - Labs [elevated cardiac biomarkers – troponin or BNP]
 - ECG [any brady or tachy arrhythmia or conductive abnormality]
- Cardiac testing (i.e. echocardiogram or stress testing) should be based on high clinical suspicion of ischemic, structural, or valvular heart disease
- **Stress testing should be performed for patients who present with syncope that occurs before, during, or after exertion**

Syncope Practical Tips

- Brain imaging should be performed only for patients for whom intracranial disease is highly suspected as a possible contributing cause to syncope, or if there has been a suspicion of head trauma
- Carotid artery imaging in the absence of focal neurological findings **should not** be performed
- A tilt-table test should be considered only when there is diagnostic uncertainty; usually this includes atypical presentations, older patients with few clues in the history, distinguishing convulsive syncope from epilepsy, and nonhemodynamic collapses.

Cardiac Monitoring

Tailoring Cardiac Monitor Selection to Symptom Frequency



Select Management Strategies in Syncope

Vasovagal Syncope (VVS)

- Education / reassurance
- avoid triggers
- Increased H₂O and Na⁺
- Counterpressure maneuvers

Drugs for recurrent and refractory VVS:

1st line Fludrocortisone or

Midodrine

2nd line Beta blocker (age >42)

Combo therapy sometimes needed for refractory cases

Cardiac Pacing for VVS

- Patients ≥ 40 with highly symptomatic VVS
- Documented symptomatic asystole > 3s
- Documented asymptomatic asystole > 6s
- Tilt-table induced asystole > 3s or HR < 40 bpm for > 10s

Orthostatic Hypotension

- Education
- Avoid triggers
- increased H₂O and Na⁺
- Medication review
- Compression garments
- Head-up tilt sleeping

Failing non-pharm management:

- Fludrocortisone
- Midodrine



CCS 2023 Fitness to Drive Guidelines - Syncope

Category		Private Car	Commercial Driver (Truck, Bus)
Syncope	Single/recurrent vasovagal syncope	OK to drive	
	Reversible cause (orthostatic, dehydration) or avoidable trigger (micturition)	1 week	1 month
	Single unexplained syncope	1 week	12 months
	Recurrent unexplained syncope	3 months	12 months
	Tachy/brady/device syncope	Refer to respective recommendation	

Major updates from previous guidelines:

- **Restriction for vasovagal syncope in the seated position has been removed, and recurrent vasovagal syncope commented on.** Authors note that risk of vasovagal syncope while driving is extremely low, even in the setting of recurrent vasovagal syncope.
- **Reversible causes for syncope now have the same recommendations as avoidable triggers** (previously were recommended “treatment of underlying cause”). **Commercial restriction for these causes is now 1 month** (previously 1 week).

Peripheral Arterial Disease

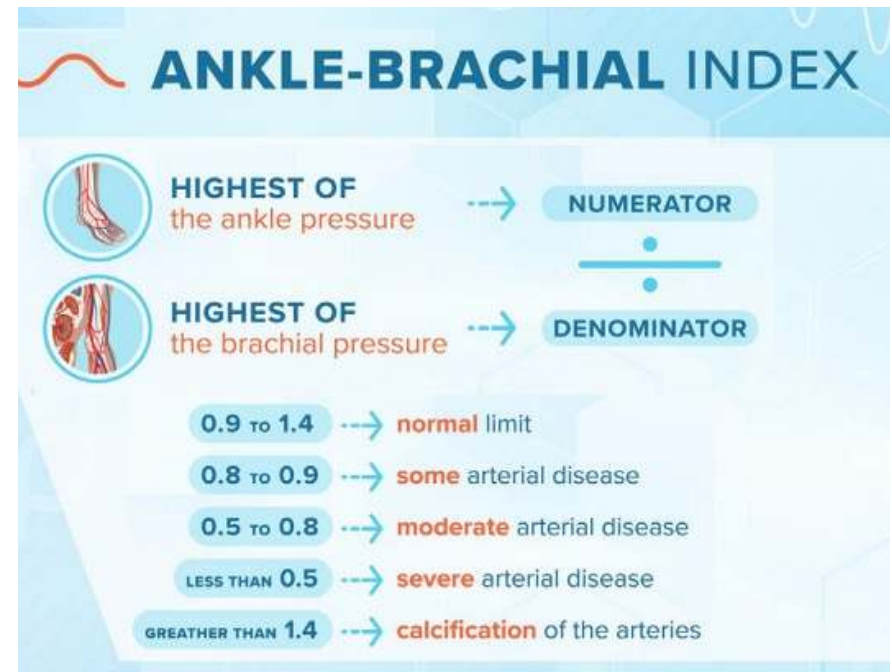
Key Resources:

- CCS 2022 Peripheral Arterial Disease Guidelines

MALE = Major Adverse Limb Events

Peripheral Arterial Disease

- PAD can manifest as intermittent claudication, chronic limb threatening ischemia or acute limb ischemia
- **Diagnosis:** ABI or TBI test of choice to confirm diagnosis of PAD in patients with symptoms
- **Screening:** adults age>50 with risk factors (smoking, diabetes), even if asymptomatic
 - No role for routine testing in asymptomatic individuals without risk factors
 - No role in also routinely screening for coronary artery disease/carotid artery disease unless they have symptoms of these conditions
 - *Guideline doesn't specify frequency of screening*



How to perform an ABI. N.B. requires continuous wave doppler and blood pressure cuff:

<https://youtu.be/5ux13-XjzgQ>

Peripheral Arterial Disease Management

- **Management:**

- **Smoking cessation** (including avoiding second hand smoke): NRT, bupropion, varenicline, CBT
- **Exercise** program, walking therapy, foot care, wound therapy
- **Diabetes management** tight control may reduce MALE, including SGLT2i (↓ MACE, no increased amputation risk)
- **Statins indicated** → add on ezetimibe/PSCK9 inhibitor to target
- **Hypertension** per Hypertension Canada, <140/90, ACEi/ARB first line therapy
- **Antithrombotic:** indicated ONLY if symptomatic (i.e. **not indicated if asymptomatic**)
 - First line: ASA + low dose rivaroxaban 2.5 bid if high risk of ischemic events
 - high-risk comorbidities such as polyvascular disease, diabetes, history of heart failure, or renal insufficiency) **and low bleeding risk**
 - Otherwise SAPT (clopidogrel > aspirin)
 - Recommend against full dose anticoagulation + antiplatelet in stable chronic PAD
 - GLP-1 agonists have shown significant symptom improvement in patients with T2DM and PAD (STRIDE trial, 2025)

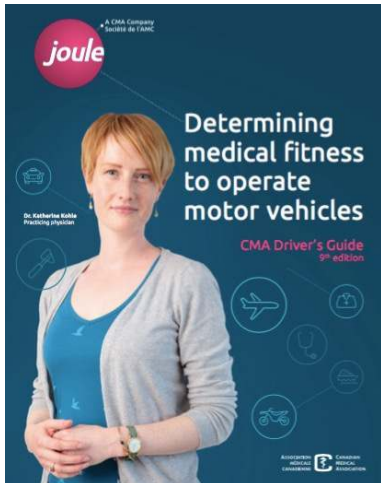
Peripheral Arterial Disease Management

- **Revascularization indicated if symptoms affecting QoL and good targets to revascularize:**
 - Method of repair depends on the location/anatomy/expertise available, requires team approach, hybrid methods may be necessary
 - AVOID endovascular repair in common femoral/femoral profundus
 - **Antithrombotic post revascularization:**
 - Elective endovascular repair: ASA + low dose rivaroxaban 2.5 bid first line
 - DAPT (ASA + clopidogrel) if cannot receive low dose rivaroxaban
 - Elective open repair: ASA + low dose rivaroxaban 2.5 bid
 - If cannot tolerate then single antiplatelet OR VKA
 - Emergent revascularization: (postop / at discharge)*:
 - full dose anticoagulation [they don't specify with what agent] + SAPT **OR**
 - ASA + low dose rivaroxaban OR
 - DAPT
- *THESE ARE weak recommendation, no high quality studies here
surgeon usually directs management in real life

Summary

- Cardiac symptoms are often the presenting symptom of a non-cardiac disease
 - Always look for an etiology
- Know how to manage acute cardiac symptoms but always remember to manage other contributing chronic risk factors (Cardiac and non-cardiac)
- If conflicting guidelines, follow recent Canadian (CCS) guidelines and do the safest thing
- Driving will always be an issue – know the restrictions

Fitness to Drive for non-Cardiac Conditions



- Available for free now without CMA membership
- Updated Version 10.0 – see <https://driversguide.ca>
 - Know with regards to Neuro Conditions (seizure, stroke), dementia + driving
- **2023 CCS guidelines** summarized in tables on the following slides are identical to CMA 10th edition drivers guide.
- **Duty to report**
 - Varies province to province including duty to report
 - Alberta, Nova Scotia, Quebec = reporting discretionary
 - In Ontario (Royal College is in Ottawa so default to Ontario rules if unsure) – MDs **must** fill out medical condition report and cannot be legally challenged for doing so:
 - “Medical Condition Report Ontario”
 - Ontario Ministry of Transportation suggests the CMA & CCS guide be used by physicians in determining requirements to report



*See main section for nuances about specific valve lesions

CCS 2023 Fitness to Drive Common Scenarios

Category		Private Car	Commercial Driver (Truck, Bus)	Category		Private Car	Commercial Driver (Truck, Bus)
Coronary Artery Disease	STEMI/NSTEMI with LVEF >40%	2 weeks post d/c	1 months post d/c	Heart Failure	NYHA I-II	OK to Drive	LVEF $\geq$ 30% OK
	STEMI/NSTEMI with LVEF $\leq$ 40% or no PCI performed	1 month post d/c	3 months post d/c		NYHA III	OK to Drive	NO DRIVING
	UA (ACS without MI)	48h w/ PCI 7 d w/o PCI	7 d w/ PCI 1 month w/o PCI		NYHA IV, inotropes	NO DRIVING	
	PCI in non-ACS	48h	48h		LVAD	2 months after implant if NYHA 1-2	
	CABG	1 month post d/c	3 months post d/c	Valvular Heart Disease	Medically treated AS, AI, MS, MR, TR	AS: OK if NYHA 1-2 MS, MR, AI, TR: OK if NYHA 1-3	Only OK if NYHA 1, no syncope, LVEF $\geq$ 50%*
PPM	Impaired LOC or high grade AV block	1 week post implant		Syncope	Vasovagal syncope	OK to drive	
ICD	Secondary proph w/ impaired LOC	3 months post implant	NO DRIVING		Reversible cause/trigger	1 week	1 month
Arrhythmia	VF, unstable VT or impaired LOC	3 months	NO DRIVING		Unexplained syncope	Single: 1 week Recurrent: 3 mo	Single/ recurrent:12mo



CCS 2023 Fitness to Drive Guidelines

Category		Private Car	Commercial Driver (Truck, Bus)
Coronary Artery Disease*	STEMI/NSTEMI with LVEF >40%	2 weeks post d/c	1 months post d/c
	STEMI/NSTEMI with LVEF ≤ 40%	1 month post d/c	3 months post d/c
	STEMI/NSTEMI with no PCI performed	1 month post d/c	3 months post d/c
	UA (ACS without MI)	48h w/ PCI 7 d w/o PCI	7 d w/ PCI 1 month w/o PCI
	PCI in non-ACS context	48h	48h
	Asymptomatic CAD, stable angina	OK to drive	OK to drive
	CABG	1 month post d/c	3 months post d/c

Category		Private Car	Commercial Driver (Truck, Bus)
Heart Failure**	NYHA I	OK to drive	LVEF ≥ 30% OK
	NYHA II	OK to drive	LVEF ≥ 30% OK
	NYHA III	OK to drive	NO DRIVING
	NYHA IV	NO DRIVING	NO DRIVING
	Home inotropes	NO DRIVING	NO DRIVING
	LVAD	2 months after implant if NYHA 1-2	NO DRIVING
	Heart transplant	6 weeks after discharge + NYHA 1-2 + stable immunosuppression + followed annually	6 months post discharge + NYHA 1 + LVEF ≥ 50% + followed annually

*Updates focus on LVEF post MI, regardless of mechanism

**Updates provide new recommendations for LVAD/TxP



CCS 2023 Fitness to Drive Guidelines

Category		Private Car	Commercial Driver (Truck, Bus)
Valvular Heart Disease	Medically treated AS, AI, MS, MR, TR	AS: OK if NYHA 1-2 MS, MR, AI, TR: OK if NYHA 1-3	Only OK if in the lowest risk group: NYHA 1, no syncope, LVEF $\geq$ 50% AS, AI: as above MR: also no pHTN or systemic embolism MS: no LVEF requirement TR: no RV dysfunction, symptomatic RV failure, RV arrhythmias
	Post TAVI or SAVR for AS, AR	1 month if stable QRS duration, no high grade AV block (2 nd type II or 3 rd) if no PPM and NYHA I-III AR requires as above + LVEF $\geq$ 50%	3 month if stable QRS duration, no high grade AV block (2 nd type II or 3 rd) if no PPM and NYHA I AR requires as above + LVEF $\geq$ 50%
	Post TEER for MR or TR	48h post d/c if NYHA I-III	1 month post d/c if NYHA I and LVEF $\geq$ 50%
	Post balloon valvuloplasty for MS	48h post d/c if NYHA I-III	1 month post d/c if NYHA I
	Post TMVR, TTVR or surgical valve replacement	1 month post d/c if NYHA I-III	3 months post d/c if NYHA I and LVEF $\geq$ 50%

Category		Private Car	Commercial Driver (Truck, Bus)	Arrhythmia	Private Car	Commercial Driver (Truck, Bus)
Pace-maker	Impaired LOC or high grade AV block	1 week post implant		Sinus node dysfunction	No impaired LOC: OK to drive Impaired LOC/symptomatic pauses: NO DRIVING until PPM	
	No impairment in LOC/high grade AV block	OK to drive		First degree, Wenckebach, isolated LBBB, LAFB, LPFB, RBBB, bifasicular block	OK to drive unless history of impaired LOC	
	Generator change (also applies for ICD private driving)	OK to drive		Second degree type II, alternating BBB, 3 rd degree AVB	NO DRIVING until PPM Exception: congenital 3 rd AVB (only restricted if impaired LOC)	
	Lead upgrade/revision (also applies for ICD private driving)	1 week post implant if history of impaired LOC/high grade AV block Otherwise OK to drive		SVT/AF/AFL	No impaired LOC: OK to drive Impaired LOC: NO DRIVING until treated	
				VF (no reversible causes)	3 months	NO DRIVING
ICD	Primary prophylaxis	1 week post implant	NO DRIVING	VT/VF due to reversible cause (MI, drug, etc.)	NO DRIVING until underlying cause treated	
	Secondary prophylaxis, impaired LOC	3 month		Hemodynamically unstable VT or impaired LOC	3 months	NO DRIVING
				VT with structural heart disease and no impaired LOC, no ICD	3 months	
	Secondary prophylaxis, no impaired LOC	1 week		VT with structurally normal heart (idiopathic VT)	1 week if well controlled	



CCS 2023 Fitness to Drive Guidelines

Category		Private Car	Commercial Driver (Truck, Bus)
ICD therapy	ICD shock or therapy, impaired LOC or disabling	3 months	NO DRIVING
	ICD shock or therapy, NO impaired LOC or disabling	1 week	
	Inappropriate ICD therapies	No restriction	
	Electrical storm (3 or more VT/VF events in 24h)	3-6 months after event (expert re-evaluation)	
Hyper-trophic cardio-myopathy	No high risk features	OK to drive	
	High risk features**	Asymptomatic: OK to drive	NO DRIVING
		Syncope: 3 months	
	Prior sustained ventricular arrhythmia	3 months	

Category		Private Car	Commercial Driver (Truck, Bus)
Syncope	Single/recurrent vasovagal syncope	OK to drive	
	Reversible cause (orthostatic, dehydration) or avoidable trigger (micturition)	1 week	1 month
	Single unexplained syncope	1 week	12 months
	Recurrent unexplained syncope	3 months	12 months
	Tachy/brady/device syncope	Refer to respective arrhythmia or device recommendation	

**High risk features: think ICD recommendations i.e. wall thickness >30mm, unexplained syncope, apical aneurysm, LVEF <50%),

Questions?



You've got this!

BONUS Material

- Choosing Wisely Canada Cardiology
- COVID and the heart
- Warning: Other CCS Guidelines / Position Statements
 - Adult Congenital Heart Disease → beyond the scope of IM exam, not reviewed
 - Dyslipidemia → see ENDO lecture for 2021 CCS Lipids
 - Cardiovascular disease in pregnancy → See OB MED Lecture
 - POTS Syndrome
 - Screening Competitive Athletes for Cardiac Disease
- High Yield Cardio Physical Exam Facts 
- BONUS MCQs – updated with latest guidelines

Applied exam... may have to “describe” things like how to measure a pulsus. Written exam: Stem may describe classic cardio finding

Choosing Wisely Canada

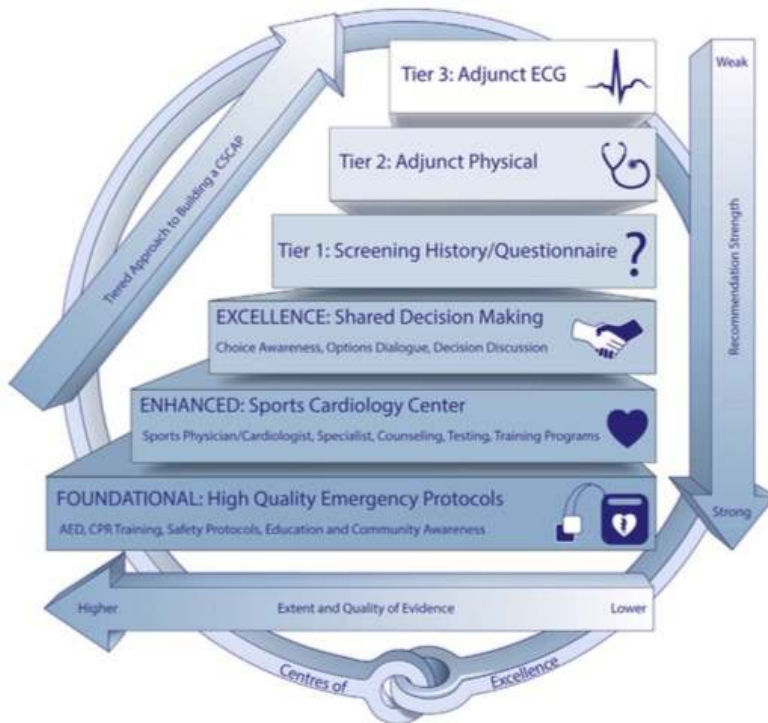
- Don't perform stress cardiac imaging or advanced non-invasive imaging in the **initial** evaluation of patients **without cardiac symptoms** unless high-risk markers are present.
- Don't perform annual stress cardiac imaging or advanced non-invasive imaging as part of routine follow-up in **asymptomatic** patients.
- Don't perform stress cardiac imaging or advanced non-invasive imaging as a pre-operative assessment in patients scheduled to undergo low risk non-cardiac surgery
- Don't perform echocardiography as routine followup for mild, asymptomatic native valve disease in adult patients with no change in signs or symptoms
- Don't order annual electrocardiograms (ECGs) for low-risk patients **without symptoms**.
- **Don't offer therapies on the basis of survival benefit without establishing your patient's prognosis, preferences, and goals of care**

CCS 2020: POTS Guideline

- **Postural Orthostatic Tachycardia Syndrome** (orthostatic tachycardia without orthostatic hypotension)
 - Orthostatic intolerance (sustained $\uparrow$ HR >30 bpm supine within 10 mins of standing with no drop in BP 20/10)
 - Rule out 2o causes (hypovolemia, thyroid etc.)
 - Rule out systemic disease e.g. Ehlers Danlos, Autoimmune neuropathy...
 - Symptoms associated: palpitations, chest discomfort, lightheadedness, chronic fatigue, chronic pain, nausea
- Workup – if diagnosis clear - do not need specialized cardiac testing. Recommend basic labs (CBC, electrolytes, Cr, TSH, am cortisol)
- Non-Pharm management: exercise training, Na 10g/d, H₂O 3-4 L /d, compression stockings waist high
- Pharm management: Midodrine, fludrocortisone
 - Other weak recommendations: Ivabradine, Methyldopa, Clonidine

CCS 2018

Society Position Statement Canadian Cardiovascular Society/Canadian Heart Rhythm Society Joint Position Statement on the Cardiovascular Screening of Competitive Athletes



- **Consensus:**

- Have emergency protocols in place at sports venues
- Get consent for screening
- Tiered approach:
 - Start with history.
 - Consider physical.
 - **Recommend against routine screening ECG.** (only do ECG if Hx & PE concerning)
- Sport restriction if patient has condition putting them at risk of SCD (long list – ARVC, HOCM w/ VT or multiple RFs, aortopathy...)

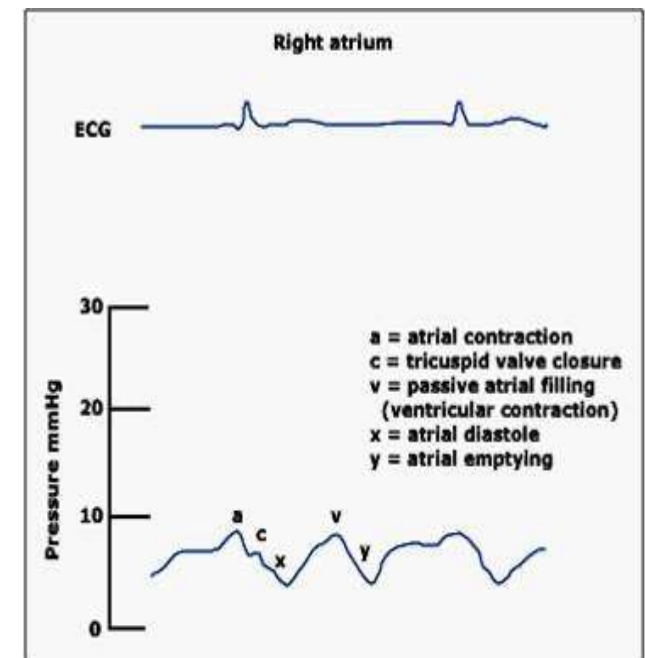
Physical Exam

Ask for the vitals first - may provide a clue

- General inspection
- **JVP** - describe it and commit to 'a' or 'v' wave dominant. 'a' is normally the larger component
- Feel the carotid or brachial pulses - time the peak of the pulse with the largest wave. If they coincide - 'v' is higher than 'a'
- Carotid pulse - describe # pulses, volume, timing; listen for bruit or radiating murmur
- Precordium - inspect and palpate - comment on findings
 - For palpation - feel for thrills over each valve area, a palpable P2 (pHTN), RV heave and then describe the location, quality and size of the apex

Auscultate

- Each valve area and carotids - S3/S4 most commonly heard at LLSB or apex if heard
- Time systolic or diastolic by palpating the carotid or brachial pulse
- S3 just follows S2; S4 just precedes S1 - again use the carotid to time
- Know the signs in AI - all reflect high volume arterial pulsation

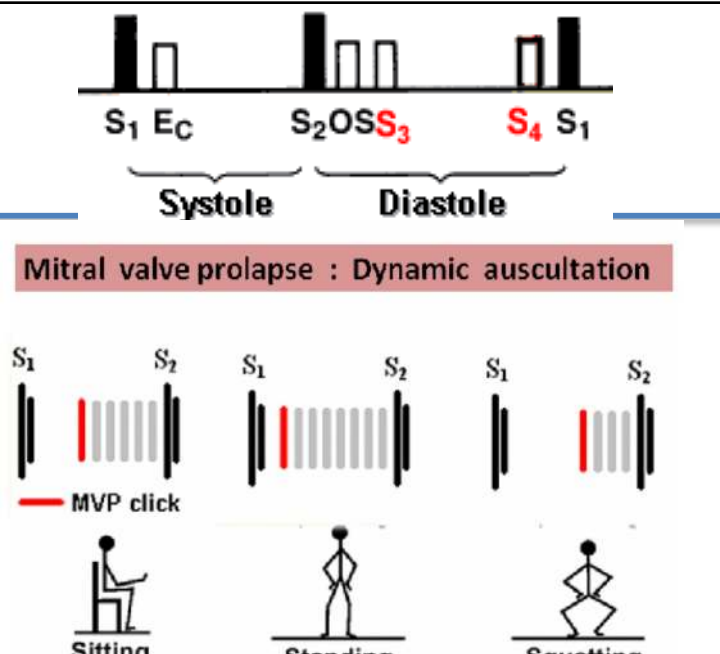


Tips for Tackling Murmurs



1. Systolic versus diastolic?
 - MR. AS married MS. ARD
2. Right sided murmurs increase with inspiration
3. Left sided murmurs increase with expiration
4. Save your energy, and bundle your murmurs!
 - Increase venous return → leg raise/squat; decrease → standing/Valsalva
 - Increase afterload → hand grip; decrease → amyl nitrate
 - AS/MS: louder with increased venous return & with decreased afterload
 - AR/MR: louder with increased venous return & **increased** afterload
 - HOCM/MVP (the opposite of AR/MR): louder with decreased venous return & decreased afterload

Systolic and regurgitant (holosystolic)	- "Blowing" quality - main ddx MR vs TR - TR (holosystolic) - respiratory variation and large cV wave - If JVP was not elevated - might be chronic MR - Chronic MR (holosystolic) should demonstrate an enlarged LV - displaced apex - Acute MR - JVP elevated, no resp changes with murmur, LV is normal size and patient in heart failure - Listen for 'click' if MR - could be MVP - VSD (holosystolic) radiates from LSB to RSB	Diastolic - Low pitched rumble	- Probably mitral stenosis - listen for opening snap after S2. Usually best heard at the apex - If no OS and AI heard at RUSB - think functional mitral stenosis (Austin-flint murmur)
Systolic ejection murmur/mid or late-peaking	- "Crescendo" - peaks mid or late and does not obscure S2 - Main DDx - aortic sclerosis, stenosis or HCM obstruction - Aortic stenosis - radiates to carotid, diminished carotid pulse, maybe apical-carotid delay - louder with increased venous return & with decreased afterload - HCM - double carotid pulse, 'triple ripple' apical beat, non-displaced apex, MR murmur may be present (SAM), Use maneuvers - louder with decreased venous return & decreased afterload - Aortic sclerosis – systolic murmur only with none of the features of AS or HCM	Diastolic - Higher pitched/may be very short	- Heard best at the left sternal border, probably AI - Look for signs of pHTN or dilated LV to determine acuity (acute is short murmur with normal LV size) - If you hear AI - check for Austin-Flint murmur at the apex (diastolic rumble mimicking MS, because the AI jet makes it difficult for the MV to open) - Other clues: Widened pulse pressure may be a clue to AI if given - Murmur becomes shorter (a.k.a. ends quicker) as severity worsens

No murmur - extra sounds only	- S3 – Early diastole (just after S2), due to blood distending a baggy ventricle (may be tensing of the chords). This most often indicates LV systolic dysfunction. Look for other signs of LV enlargement and CHF. - S4 – Late diastole (just before S1), due to blood distending a stiffened ventricle, most often associated with diastolic heart failure (HFpEF). - Scratching sound - more than two components - pericardial rub. Often heard best LLSB. If high JVP, think tamponade from effusion in pericarditis. Check for pulsus paradoxus.	 The diagram illustrates heart sounds and dynamic auscultation for mitral valve prolapse (MVP). At the top, a waveform shows the sequence of heart sounds: S₁, E_c (early closure), S₂, O_s (opening snap), S₃ (in red), and S₄ (in red) followed by S₁. Brackets below the waveform label the 'Systole' (S₁ to S₂) and 'Diastole' (S₂ to S₁). Below this, a red box contains the text 'Mitral valve prolapse : Dynamic auscultation'. Underneath, three scenarios are shown: 'Sitting', 'Standing', and 'Squatting'. Each scenario features a stick figure and a waveform. A red vertical line in the waveform indicates the 'MVP click'. In the 'Sitting' position, the click occurs mid-systole. In the 'Standing' position, the click occurs earlier in systole. In the 'Squatting' position, the click occurs later in systole. The waveforms also show S₁ and S₂ sounds.	
Continuous Murmur - through systole and diastole	- PDA - Coarctation - Ruptured sinus of valsalva to atrium - Aortopulmonary window - Internal mammary artery (mammary souffle) - Venous hum in jugular (stops with pressure) - Arterial stenosis (i.e subclavian) - Any arterial to venous fistula (Pulmonary AVMs)	Mid systolic click	- When you hear a mid systolic click, think mitral valve prolapse (MVP). The click comes from distension of the chords as the mitral valve leaflets prolapse into the LA - "LV volume dependent" – if the LV is full (squatting = more preload/afterload) the click will be later and MR shorter (leaflets don't "flop" as much because the chords are tense LV is more full) If the LV is less full (standing = less preload/afterload), the click will occur earlier (leaflets "flop more" because the chords are less tense because the LV is less full)

Heart Sounds

Feature	Scenario
Loud S1	<i>Short PR</i> <i>Exercise</i> <i>Increased transvalvular flow (ASD, PDA)</i> <i>AV Valve Obstruction – MS, TS</i>
Soft S1	<i>Calcific MS</i> <i>MR</i> <i>Valves close early (long PR, AI, LBBB)</i>
Variable S1	<i>Afib</i> <i>AV Dissociation</i> <i>Severe tamponade</i>
Wide Split S1	<i>RBBB, ASD, Ebstein's anomaly</i>

Feature		Scenario
↑intensity	Loud A2	<i>Severe HTN</i> <i>CoA, Aneurysm</i>
	Loud P2	<i>Pulm HTN</i> <i>ASD</i>
↓intensity	Soft A2	<i>AI or severe calcific AS</i>
	Soft P2	<i>Low pulm artery pressure, PS</i>
Wide Split S2	<i>RBBB, LV PPM</i>	
Wide Fixed S2	<i>ASD, RV Failure</i>	
Paradox Split S2	<i>LBBB, WPW, Fixed LVOT, AS, HOCM</i>	

Aortic Stenosis

“Does this patient have an abnormal systolic murmur?”

Etchells E et al. JAMA. 1997;277(7):564-71

Rule in AS

- slow rate of rise of carotid pulse (+LR 2.8-130)
- mid to late peak murmur (+LR 8-101)
- soft S2 (+LR 3.1-50)

Rule out AS

- Absence of radiation to right carotid (LR 0.05 – 0.10)

HCM Risk

Q. Why does it matter?

A. Risk of Sudden Cardiac Death with HOCM

2003 AHA/ACC Consensus statement on HOCM (Maron et al JACC 2003):

Risk Factors for Sudden Death in HCM:

N.B. Cardiac arrest, VF or sustained VT are class I indications for ICD

MAJOR (class IIa recommendations for ICD)

Family history of SCD

Spontaneous sustained VT

UNEXPLAINED syncope

Apical aneurysm

LVEF 50% or less

LV thickness $\geq 30\text{mm}$ (*positive predictive value low – most who die $<30\text{mm}$ thickness)

MINOR risk factors (class IIb recommendations for ICD)

NSVT (on Holter)

Extensive LGE on Holter

We suggest
you memorize
these.

AS vs HCM (see multiple online IMR Videos)

Problem	Carotid pulse	Pre-cordial Exam	Heart Sounds	Murmur	Decrease PreLoad*	Increase Afterload **
Aortic stenosis	<i>Parvus et tardus</i>	<i>PMI can be sustained</i>	<i>Dec intensity of S2</i>	<i>Cresc-decresc Loc'n: 2nd ICS, RUSB Radiates: Clavicle, Carotid</i>	<i>Softer</i>	<i>Softer (useful to ddx AS vs MR which is louder w grip)</i>
HOCM	Brisk initially Bisferiens	Diffuse/sustained apical impulse or "double tap"	+ S4 (>50%) Paradoxical split S2 with severe obstruction (10%)	Midsystolic cresc-decresc Location: apex, left lateral sternal border Radiates: axilla, base (not usu neck)	Louder - underfilled ventricle increases (LR+14, LR-0.3)	Softer (LR+3.6, LR-0.1)

*DECREASE PRELOAD = Valsalva X 20 seconds, or have patient move from SQUAT → STAND

** INCREASE AFTERLOAD = Isometric hand grip (listen after 1 min)

Septal Defects

ASD

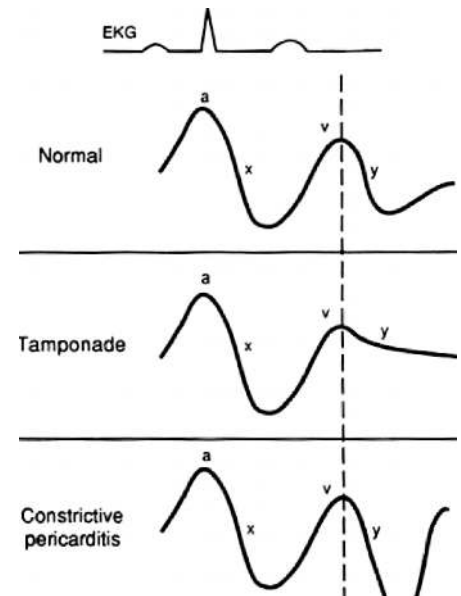
- Wide, fixed split S2
- Don't actually hear murmur from flow through defect. What you hear is a diastolic rumble due to increased flow over tricuspid valve
 - Pulmonary ejection murmur (systolic flow murmur) – over 2nd L ICS

VSD

- Holosystolic murmur 3-4th ICS with thrill
- Smaller defects generate *higher pitched* and *louder murmurs* (due to a larger pressure gradient through a small orifice).
- Larger defects have *quieter, lower pitched murmurs* as the gradients are not as large, however there is often greater flow through these defects, resulting in LV enlargement (L to R shunt, but since the shunting occurs in systole, the extra volume from the shunt goes to the pulmonary artery and LV, therefore an enlarged and displaced apical impulse may be present).

The Constriction/Tamponade Exam

Diagnosis	Pulsus > 10 mmHg	X and Y Descent	Kussmaul's
<i>Restrictive Cardiomyopathy</i>	<i>Rarely</i>	<i>Deep Y descent</i> <i>Other audible S3</i>	✓
Tamponade	✓ SN 98% SP 70%	Absent Y descent	✗
<i>Constrictive pericarditis</i>	<i>Rarely</i>	<i>Prominent X and Y</i>	✓



- JAMA RCE For Cardiac Tamponade (Roy CL et al, JAMA; 2007. 297:1810.)
 - Did not look at X, Y descents. Kussmaul also not included in data.
 - RULE IN:
 - Presence of Pulsus > 12mmHg (LR+ 5.9)
 - RULE OUT:
 - Absence of Pulsus >10-12 (LR- 0.03 for both cutoffs)
 - Also sensitive: tachycardia (SN 77%), elevated JVP (SN 76%)
 - Not helpful : hypotension (SN 26%), diminished heart sounds (SN 28%)

Pulsus Paradoxus (*SEE MULTIPLE ONLINE VIDEOS*)

Method

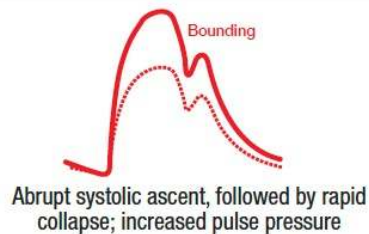
- Deflate until korotkoff sounds heard during quiet **expiration** (POINT 1)
- Continue until heard throughout respiratory cycle (POINT 2)
- Difference in BP (Point 1- Point 2) = pulsus

What is a significant pulsus? (>10-12 mmHg)

- In what settings might one hear a pulsus paradoxus
 - Cardiac Tamponade – sens 98%
 - Severe asthma (or exaggerated inspiratory efforts)
 - Has also been described in RV infarction, PE, severe pectus excavatum, severe asthma/COPD (McGee 2nd ed)
- Predicting Airflow Obstruction (JAMA RCE) – SN 45%, SP 88%, LR+ 3.7, LR- 0.62 using cutoff of 15mmHg

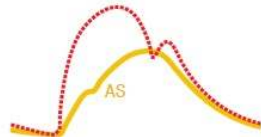
Carotid Pulse Examination

CORRIGAN'S PULSE (BOUNDING)



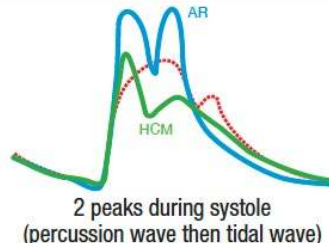
Aortic regurgitation, PDA, high output states (hyperthyroidism, AV fistula)

PULSUS PARVUS ET TARDUS (ANACROTIC)



Aortic stenosis

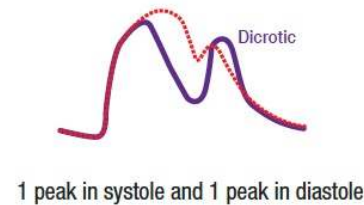
PULSUS BISFERIENS (BIFID)



"Spike and dome" HCM

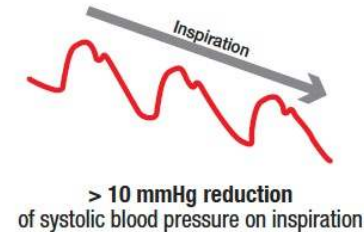
Also seen in aortic regurgitation

DICROTIC PULSE



Inaortic balloon pump, severe HF, tamponade, hypovolemia, sepsis

PULSUS PARADOXUS



Tamponade!

Also constriction, COPD, severe PE with RV failure, severe pneumothorax

PULSUS ALTERNANS



Severe heart failure, severe aortic regurgitation, tamponade, bigeminy

Peripheral Vascular Disease (*SEE ONLINE VIDEOS*)

Does the clinical examination predict lower extremity PAD?

(Khan et al, JAMA 2006)

Bottom line = clinical exam alone not sufficient to rule in or rule out.

Symptomatic leg (pain consistent with claudication)

Tests that help RULE IN PAD:

- Any pulse abnormality in symptomatic leg (LR+4.7)
- Presence of bruit in symptomatic leg (LR+5.6)
- Symptomatic leg cooler to touch (LR+5.9)
- Wounds or sores (LR+5.9)
- Discoloration (LR+2.8)

Tests that help RULE OUT PAD:

- ABSENCE of any pulse abnormality (LR-0.38)

Screening for PAD

Tests that help RULE IN PAD

- Any pulse abnormality (LR+3.0)
- Presence of femoral bruit (LR+4.8)

Tests that help RULE OUT PAD in screening setting

- No tests were particularly robust – absence of pulse abnormality LR- 0.47.

2023/4:

This is likely not to be testable in view of new PAD guidelines which say to do ABIs!

BONUS MCQ 1 – updated 2024

A 60F with diabetes and hypertension undergoes a coronary angiogram for progressive angina symptoms, and is found to have 80% mid LAD stenosis, 80% mid LCx stenosis, and 70% proximal RCA stenosis. Her LVEF is 56%. Which of these statements is **true**?

- A) Peri-procedural stroke rates are higher with PCI than CABG
- B) Repeat revascularization rates are higher with CABG
- C) PCI would provide a mortality benefit over optimal medical therapy
- D) This patient will derive a mortality benefit from undergoing CABG over PCI

Bonus MCQ #2 2025

A 35-year-old man with a family history of HCM is seen in evaluation for exertional dyspnea, chest pain, and syncope. Echocardiography reveals a hyperdynamic left ventricle (LVEF >65%) with asymmetric septal hypertrophy, septum up to 22 mm, with systolic anterior motion (SAM) and significant left ventricular outflow tract (LVOT) obstruction. Which of the following medical therapies is generally NOT indicated as first-line therapy for LVOT obstruction in HCM?

- A) Beta-blockers
- B) Non-dihydropyridine calcium channel blockers
- C) ACE inhibitors
- D) Maintaining hydration and stopping diuretics

BONUS MCQ 3

A 72 year old female with a history of htn (on HCTZ) and DM2 (diet therapy) is found to have a holosystolic murmur at the apex on routine exam. She has no exertional dyspnea, angina, or heart failure symptoms. Echocardiogram shows severe mitral regurgitation. Her EF is 66% with LV End-systolic dimension is 44 mm. What do you recommend?

- A) Start medical therapy and repeat echo in 6 months
- B) Refer to cardiovascular surgery for potential valve repair/replacement
- C) Start a beta-blocker and repeat echo q6months
- D) Refer to interventional cardiology for balloon valvuloplasty

BONUS MCQ 4

A 65M with dilated cardiomyopathy (LVEF 32%) presents in clinic with NYHA class II dyspnea. He has been on Bisoprolol 10 mg OD, Ramipril 10 mg OD, and Spironolactone 25 mg for 4 months. He has never had documented VT/VF. His ECG shows sinus rhythm with a HR of 65 bpm and QRS duration of 125 ms in RBBB morphology. Which of the following would you recommend for medical therapy:

- A) Change Ramipril to Sacubitril/Valsartan and add empagliflozin
- B) Change Ramipril to Sacubitril/Valsartan
- C) Add Ivabradine to current therapies
- D) Continue current medical therapy

BONUS MCQ 5

A 78 F with hx of rheumatic mitral stenosis, PMR, and hospitalization 8 months ago for UGIB is diagnosed with atrial fibrillation 3 months after requiring elective PCI to her LCx with a Drug-Eluting stent. She feels well. She has been on ASA, Clopidogrel, Prednisone, and Pantoprazole. What would you recommend with regards to her antiplatelets/anticoagulation?

- 1) Continue ASA/Clopidogrel and add Rivaroxaban for 9 additional months, then Rivaroxaban and Clopidogrel alone
- 2) Stop ASA. Start Warfarin in addition to Clopidogrel for 3 additional months, then Warfarin alone
- 3) Stop ASA and Clopidogrel and start Warfarin alone
- 4) Stop Clopidogrel and add Apixaban with ASA

BONUS MCQ 6

A 63 F with no prior medical history presents to the ER with a 2 hour history of palpitations. She measured her HR at the gym 4 hours previously at 74 bpm. She is found to be in rapid atrial fibrillation with a heart rate of 170 bpm. She has no other medical history and her CHADS65 score is 0. Her labs are unremarkable. A TEE guided cardioversion is completed in the ER and she is started on a beta-blocker. Which of the following is the best next step:

- 1) Start anticoagulation with a DOAC for at least 4 weeks.
- 2) Start anticoagulation with warfarin for at least 4 weeks.
- 3) Start ASA alone for at least 4 weeks
- 4) Given a CHADS65 score of 0, the patient does not need anticoagulation and can be discharged without further anticoagulation/antiplatelet therapy

BONUS MCQ 7

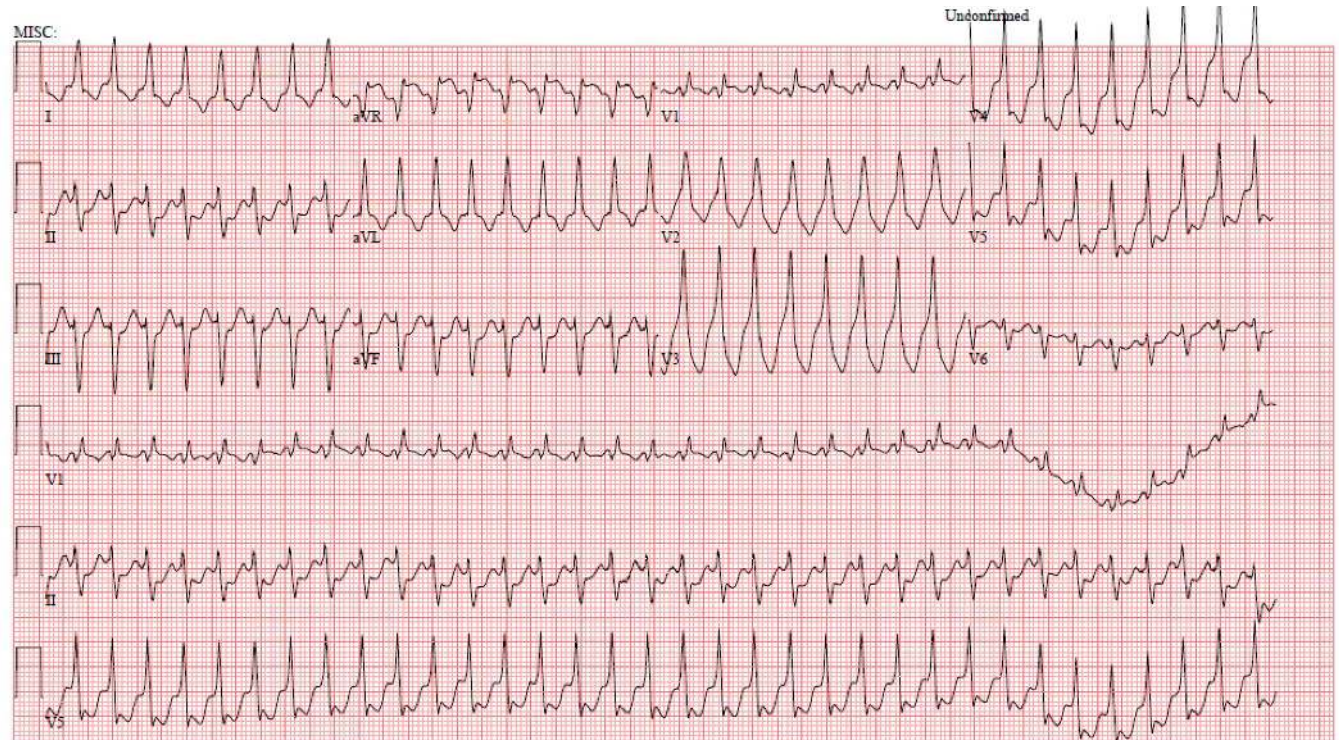
A 64 year old female with a history of palpitations is found to have a diastolic murmur at the apex on routine exam. She has had increasing exertional dyspnea but no chest discomfort or heart failure symptoms. Echocardiogram shows mitral stenosis with an estimated valve area of 1.3 cm² and moderate mitral regurgitation. Her EF is 67%. What do you recommend as the next step?

- A) Start medical therapy and repeat echo in 6 months
- B) Refer to interventional cardiology for percutaneous mitral balloon valvuloplasty
- C) Refer to cardiac rehab for exercise training program
- D) Refer to cardiovascular surgery for potential valve repair/replacement

BONUS MCQ 8

Patient presents with palpitations and chest tightness (ECG shown). Blood pressure is 90/60. What is your next step in management?

- A) DC cardioversion
- B) Adenosine 6 mg IV push
- C) Amiodarone 150 mg IV
- D) Metoprolol 5 mg IV
- E) Procainamide 100mg IV



BONUS MCQ 9

A 22 year old woman presents with syncope. She has no past medical history. Exam shows a blood pressure of 100/50. Heart rate is 86 bpm. There is a fixed split S2 and left parasternal heave. ECG shows incomplete RBBB. What is your working diagnosis?

- A) VSD
- B) ASD
- C) PE
- D) Severe aortic stenosis

Bonus MCQ 10 2026

A patient presents with new onset of shortness of breath. Exam reveals BP 190/100, JVP elevated, pansystolic murmur, no pleural rub. Echo shows mild MR and EF 60%. EKG suggests mild LVH. His heart failure is most likely due to?

- a) Myocarditis
- b) Mitral regurgitation
- c) Diastolic dysfunction
- d) Systolic dysfunction
- e) Pericarditis

Bonus MCQ #11 2025

A 62-year-old woman presents with chest pain, found to have inferior ST depressions and elevated troponin. CTPA is negative for pulmonary embolism in the emergency department. She is admitted and managed as an NSTEMI. Coronary angiography is performed, is carefully reviewed, and reveals no-obstructive coronary arteries. Before a diagnosis of MINOCA can be made, which of the following conditions MUST be excluded?

- A) Supply-demand mismatch
- B) Takotsubo cardiomyopathy
- C) Myocarditis
- D) All of the above

BONUS MCQ 12

You are asked to evaluate a 45 year old male with no known medical history who was incidentally found to have murmur. He works as a dog walker and walks at least 5km/day without any symptoms. He takes no medications, and has no allergies. His sister unfortunately passed away when she was 40 years old unexpectedly. O/E: BP 120/80, HR 75. Auscultation of the heart reveals a 3/6 systolic murmur at the base of the heart. ECG shows sinus rhythm with LVH by voltage. You are debating b/w AS and HOCM as a cause of his murmur. What maneuvers can you do to differentiate?

- A) Squat – HOCM murmur will get louder
- B) Valsalva – HOCM murmur will get quieter
- C) Leg raise – HOCM murmur will get quieter
- D) Hand grip – HOCM murmur will get louder

BONUS MCQ 13 – updated 2024

A 66-year-old woman undergoes mechanical mitral valve replacement for severe symptomatic mitral regurgitation. Her other risk factors include HTN. A preoperative coronary angiogram showed no atherosclerosis. What long-term anticoagulation should she be on?

- A. Apixaban 5mg BID
- B. Warfarin with goal INR 2-3
- C. Warfarin with goal INR 2.5-3.5 and aspirin
- D. Warfarin with goal INR 2.5-3.5

BONUS MCQ 14

A college basketball player presents for routine physical examination. Blood pressure 120/50, HR 60bpm. Precordial exam reveals a systolic crescendo–decrescendo murmur at the LLSB radiating to the base. S2 is paradoxically split. Which of the following is not a cause of paradoxically split S2?

- a) HOCM
- b) LBBB
- c) WPW
- d) ASD
- e) Aortic Stenosis

Bonus MCQ #15 2025

A 22-year-old asymptomatic woman presents for a routine physical exam. She has no known medical history. There is a family history of sudden cardiac death in a cousin, which she does not know the details of. Auscultation reveals a 3/6 midsystolic click followed by a late systolic murmur, best heard at the apex. The click becomes later and the duration shorter with squatting. Which of the following murmurs is MOST likely?

- A) Aortic stenosis
- B) Mitral regurgitation
- C) Hypertrophic cardiomyopathy
- D) Mitral valve prolapse

BONUS MCQ 16

A 60F with hypertension undergoes a coronary CT angiogram for chronic stable angina symptoms, and is found to have 70% mid LAD stenosis, 80% mid LCx stenosis, and 70% Left Main stenosis. Her LVEF is 56%. Which of these statements is **true**?

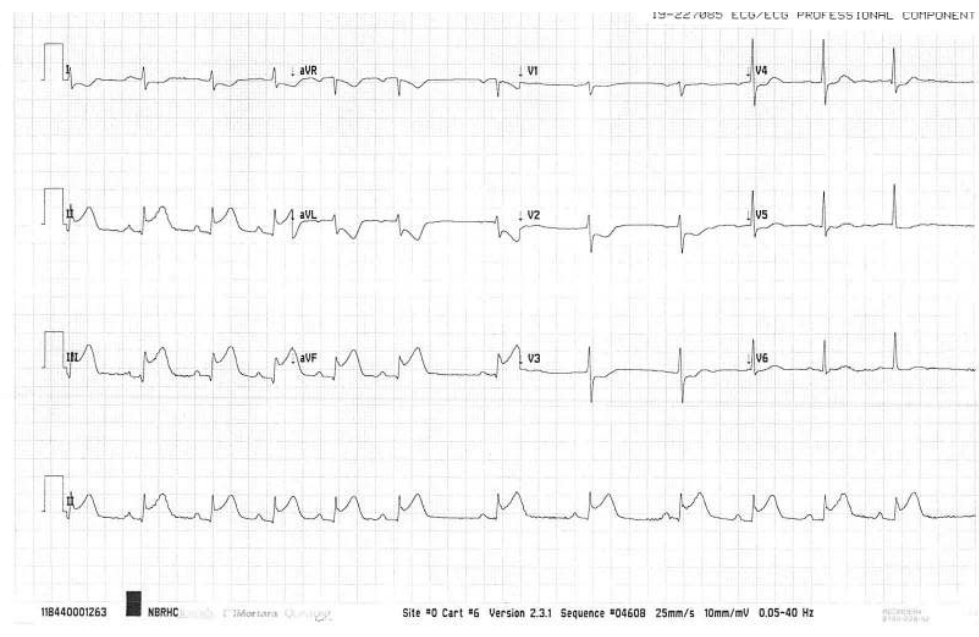
- A) Peri-procedural stroke rates are higher with PCI than CABG
- B) Repeat revascularization rates are lower with CABG
- C) The safety and efficacy of left main PCI has never been studied.
- D) This patient will not derive a mortality benefit from undergoing CABG

BONUS MCQ 17

A 78y woman presents to a PCI capable centre with sudden onset severe nausea and diaphoresis prior to her granddaughter's wedding citing a panic attack. ECG is shown. She has a PMH of anxiety, HTN, **Ischemic Stroke (R MCA 2014)**. She is on Clopidogrel 75mg, Perindopril, Crestor. CBC Hgb 120 Plt 148. Cr, INR normal.

Which of the following is the most appropriate antiplatelet choices?

- a) ASA 160 mg chew + Clopidogrel 300 mg po
- b) ASA 325 mg chew + Clopidogrel 600 mg po
- c) ASA 160 mg chew + Prasugrel 60mg
- d) ASA 160 mg chew + Ticagrelor 180mg
- e) ASA 650 QID + Colchicine 0.6 BID



BONUS MCQ 18

A 65M with dilated cardiomyopathy (LVEF 32%) and history of familial angioedema presents in clinic with NYHA class II dyspnea after a recent hospitalization for CHF. He has been on Bisoprolol 10 mg OD, Candesartan 32 mg OD, and Spironolactone 50 mg OD, apixaban 5mg bid for 4 months. He has never had documented VT/VF.

Labs show Normal CBC. K 4.2, Cr 102 (CrCl 55). A1C 5.5%, LDL 1.5. His ECG shows atrial fibrillation with a HR of 74 bpm and QRS duration of 125 ms in RBBB morphology.

Which of the following would you recommend next to reduce risk of cardiovascular mortality?

- A) Change Candesartan to Sacubitril/Valsartan
- B) Add Ivabradine to current therapies
- C) Add Dapagliflozin to current therapies
- D) Add amiodarone to current therapies
- E) Continue current medical therapy, refer for CRT

BONUS MCQ 19

A 48 M with a history of asthma develops progressive exertional dyspnea. Examination shows BP 85/50, HR 105, elevated JVP which rises on inspiration, rapid X and Y descents, and an audible sound after S2. He is found to have a pulsus paradoxus of 11 mmHg. Which of the following is the LEAST likely etiology?

- A) Cardiac tamponade
- B) Constrictive pericarditis
- C) Restrictive cardiomyopathy

MCQ 20 2026

You are called to see a patient in the ER who had presented with an acute myocardial infarction. They have now developed rapid heart rate at 150 bpm. ECG is shown. BP 70/40. Next step in management?

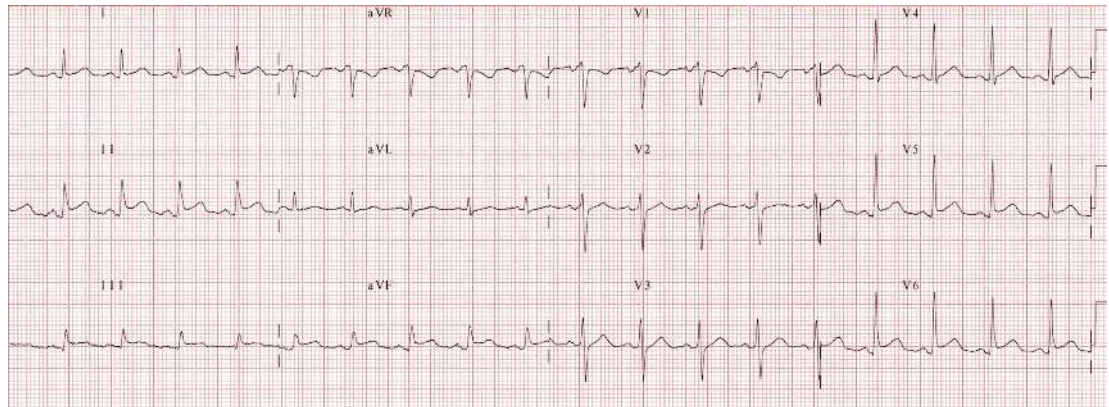
- a) Verapamil IV
- b) Metoprolol IV
- c) Electrical cardioversion
- d) Lidocaine
- e) Procainamide



BONUS MCQ 21

A 53-year-old male with a history of inferior STEMI treated with PCI to RCA 12 weeks ago returns to the ED with chest pain. Home medications include Clopidogrel, rosuvastatin, Losartan, indapamide, bisoprolol, hydralazine. He has a history of severe asthma and nasal polyps. He describes sharp, stabbing pain, easily localizable, worse with respiration and specific positions. His Tn is mildly elevated, BNP is normal. His ECG is shown below. Focused cardiac ultrasound is performed in the ED which reveals preserved LV function with no effusion, no regional wall motion abnormalities. He has a fever of 38.2C and feels unwell. BP 168/92. He is euvolemic on exam. What is the *next* best management strategy for this patient?

- A) High dose ASA and colchicine
- B) High dose colchicine
- C) Stop Hydralazine
- D) Admit to hospital for primary PCI for inferior STEMI, likely related to in-stent thrombosis
- E) Prednisone



BONUS MCQ 22

A 62-year-old female with palpitations, fatigue, and chest pain that started approximately 3 days ago. This corresponded with an increased heart rate detected by her Apple Watch. Her past medical history significant for stable CAD, breast cancer treated with lumpectomy, migraines, rheumatoid arthritis. She takes ASA, rosuvastatin, methotrexate, folic acid. Her vitals are stable and her physical exam is unremarkable. ECG reveals atrial fibrillation with rapid ventricular response. Focused cardiac ultrasound reveals preserved EF and no significant valvular abnormalities. BNP is mildly elevated and Tn is negative. Which of the following are appropriate management steps?

- A)** TEE, cardioversion (chemical or electrical), 4 weeks anticoagulation, then stop DOAC and change to ASA indefinitely
- B)** Start unfractionated heparin (with bolus), cardioversion (chemical or electrical), start and continue DOAC for 4 weeks post-cardioversion, then stop DOAC
- C)** Stop ASA, start DOAC and treat for ≥ 3 weeks before planned cardioversion (chemical or electrical), continue DOAC indefinitely
- D)** TEE, cardioversion (chemical or electrical), 3 weeks anticoagulation, then stop DOAC and change to ASA indefinitely
- E)** None of the above

BONUS MCQ 23

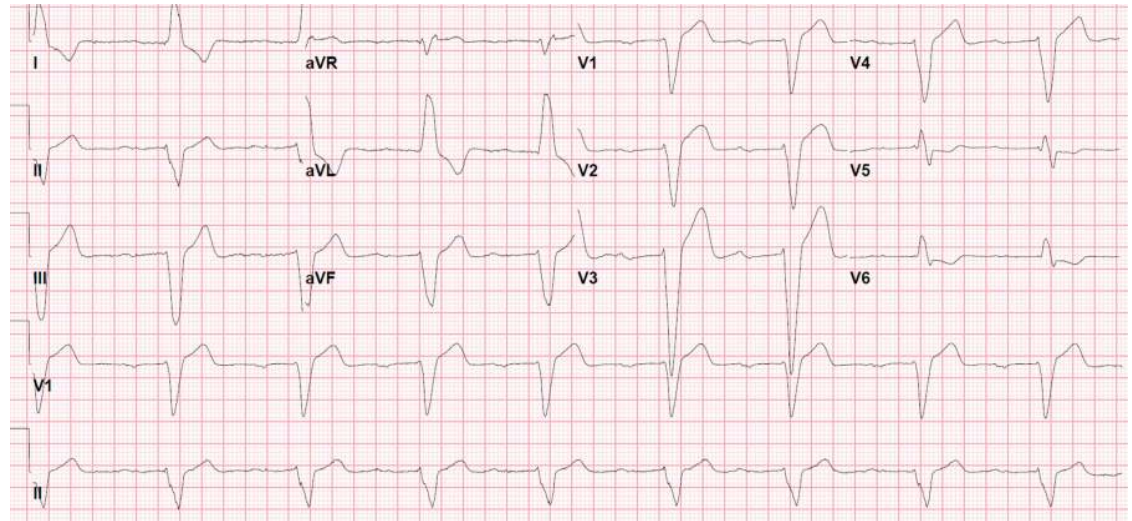
A 37-year-old male with a history of HTN and recent cocaine use presents to the ED with "tearing" chest pain radiating to the back. ECG reveals sinus tachycardia, SBP 160mmHg. He is in significant discomfort and agitated. A CT scan of the chest reveals aortic dissection involving the descending thoracic aorta. Which of the follow is the best *next* management strategy?

- A) STAT cardiac surgery consultation for urgent aortic surgery
- B) Admit patient to monitored unit; target HR first to < 60-80 bpm with IV labetalol; target SBP next with IV nitroprusside to < 120mmHg
- C) Admit patient to monitored unit; target HR first to < 60-80 bpm with IV calcium channel blocker; target SBP next with IV nitroprusside to < 120mmHg
- D) Admit patient to monitored unit; target SBP first with IV alpha blocker, then use nitroprusside as 2nd line agent as required

BONUS MCQ 24

A 42-year-old male presents to your clinic with exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea and dry cough for 4 months. He eventually developed peripheral and scrotal edema, which prompted him to seek evaluation. His ECG is shown below. His BNP is elevated. Tn is within normal range. He does not take any medications. His TTE reveals LVEF 22%, with severe RV dysfunction. In addition to coronary evaluation and lifestyle measures, which of the following best describes the *next* best therapy steps based on Canadian Guidelines?

- A)** Beta blocker, ACEI, SGLT2i, MRA, furosemide as needed; consider device therapy after 3 months of optimal medical therapy
- B)** ARNI, SGLT2i, MRA, furosemide as needed; consider ICD after 3 months of optimal medical therapy
- C)** Beta blocker, ARNI, SGLT2i, MRA, ivabradine, furosemide as needed; consider device therapy after 3 months of optimal medical therapy
- D)** ARNI, SGLT2i, MRA, furosemide as needed; refer to EP for ICD now



BONUS MCQ 25 – new 2024

A 55 year old man with a history of diabetes and hypertension has a lateral MI with PCI to his left circumflex artery. His transthoracic echocardiogram prior to discharge shows an LVEF of 52% with a lateral wall motion abnormality. He does not have any symptoms of heart failure. He asks about interventions he can do to reduce his cardiovascular risk. Based on the best available evidence all of the following interventions will reduce his risk of cardiovascular death or hospitalization EXCEPT:

- a. Add SGLT2
- b. Flu shot
- c. Routine stress testing at 1 year
- d. Polypill of aspirin/statin/ACE inhibitor

BONUS MCQ 26 – new 2024

A 45yM is admitted with NSTEMI. He underwent an exercise myocardial perfusion scan which showed an anterior wall perfusion defect with stress, and subsequently underwent coronary angiography which showed a severe lesion of the LAD which was treated with 1 drug eluting stent. There is no other significant residual disease. His transthoracic echocardiogram shows normal LV function and wall motion. He feels well. No heart failure or arrhythmias.

He works as a bus driver. He inquires if there are any restrictions to his ability to drive. You tell him prior to discharge:

- A) No restrictions
- B) 48 hours post PCI for private driving, 1 week for commercial driving
- C) 2 weeks post PCI for private driving, 1 month for commercial driving
- D) 1 month post PCI for private driving, 3 months for commercial driving

BONUS MCQ 27 – new 2024

65yM with ischemic cardiomyopathy, previous anterior MI 2021 with primary PCI to the LAD. At the time he had residual moderate lesions in the circumflex and right coronary arteries. He is admitted to your care for decompensated heart failure and was found to have a significant troponin elevation which lead to a coronary angiogram which shows severe lesions of the proximal LAD, circumflex and RCA. A cardiac MRI was performed which shows a transmural infarction in the distal LAD territory, however there is otherwise good viability in all other territories. His LVEF is 30%.

Home Meds: aspirin, valsartan, bisoprolol, spironolactone, atorvastatin.

Which of the following interventions will NOT reduce his risk of cardiovascular death and hospitalization for heart failure?

- a. Coronary artery bypass grafting
- b. Percutaneous coronary intervention
- c. Change valsartan to combination sacubitril-valsartan
- d. SGLT-2 inhibition

BONUS MCQ 28 – new 2024

A 66 y F is seen in routine follow-up. PMH: HTN, T2DM, inferior MI 3 months ago where she received one drug eluting stent to the RCA. Her left ventricular systolic function is normal.

She has been feeling well since leaving hospital except for experiencing intermittent palpitations over the last 2 weeks. She had called your office and a 48 hour Holter monitor was arranged prior to this visit. You review the Holter monitor with her which shows paroxysmal atrial fibrillation with the longest episode lasting 24 hours.

Her current medications include aspirin 81mg daily, ticagrelor 90mg BID, ramipril 2.5mg daily, bisoprolol 5mg daily, metformin 500mg BID, dapagliflozin 5mg daily, rosuvastatin 10 mg daily.

Which of the following modifications do you make to her medications?

- A) Discontinue ticagrelor, start rivaroxaban 20 mg daily
- B) Discontinue aspirin, start rivaroxaban 15mg daily
- C) Discontinue aspirin and ticagrelor, start rivaroxaban 15mg daily
- D) Discontinue aspirin and ticagrelor, start clopidogrel 75mg daily and rivaroxaban 15mg daily

BONUS MCQ 29 – new 2024

50yM with a history of HTN, DLD, T2DM, CAD and HFmrEF. PCI to the RCA 3 years ago after an inferior MI with no recurrence of anginal symptoms. He has baseline NYHA II symptoms and his LVEF was 45% on TTE last month. Recently to ED with acute onset palpitations, found to be in atrial fibrillation and was electrically cardioverted. His aspirin was discontinued and he was prescribed for apixaban 5mg BID.

He has since had 2 more episodes spontaneously converting to sinus. He finds these episodes bothersome and is interested in adjusting his medications to prevent these episodes. He is worried about starting amiodarone due to long term side effects, and is asking if there are other options.

His current medications include apixaban 5mg BID, ramipril 2.5mg daily, bisoprolol 5mg daily, spironolactone 25mg daily, dapagliflozin 10mg daily, furosemide 40 mg daily.

In addition to investigating etiology of his paroxysms of atrial fibrillation, which would be the most appropriate change to his medications?

- A) Discontinue bisoprolol, start sotalol
- B) Continue bisoprolol, start flecainide
- C) Continue bisoprolol, start dronedarone
- D) Discontinue bisoprolol, start propafenone

BONUS MCQ 30 new 2024

A 75-year-old man presents to clinic regarding right leg pain that occurs with exertion present over the last 2 months. The symptoms are consistently relieved by rest. He notes decreased hair on his right leg. His right DP pulse is weaker than the left. You suspect peripheral arterial disease and intermittent claudication as a cause for symptoms. His comorbidities include diabetes and a 50 pack year smoking history for which he continues to smoke. He does not have any known cardiac disease.

You discuss your thoughts including next steps to confirm your diagnosis. All of the following statements are true, EXCEPT:

- a) If he were asymptomatic, performing an ABI would still be indicated in this patient to screen for PAD
- b) An ABI requires a continuous wave doppler and blood pressure cuff to be performed and is the diagnostic test of choice to diagnose PAD
- c) An ABI less than 0.9 indicates the presence of some degree of PAD
- d) If diagnosed with PAD, he should also be screened for CAD/carotid artery stenosis

MCQ #31 2026

A 65F with DM2 and CCS3 angina presents with LAD and RCA disease on angiogram. Her surgeon advises that she go for CABG but she wants stents (PTCA). With regards to future risk of revascularization, What do you advise:

- a) CABG
- b) PTCA
- c) Medical treatment
- d) No difference

BONUS MCQ #32 new 2024

A 45 year old man with diabetes and a strong family history of CAD presents with 1 month exertional chest pain. He undergoes a stress echocardiogram which shows wall motion abnormalities in the LAD and RCA territory at peak stress, concerning for multivessel coronary artery disease. He is admitted that day to undergo elective but expedited coronary angiography which shows severe lesions in the LAD, circumflex and RCA. He is recommended continued admission for planning for coronary artery bypass grafting (CABG) assessment. Based on the new CCS 2023 Antiplatelet Update, which of the following options is INCORRECT regarding management of this patient:

- A. A second antiplatelet like ticagrelor should not routinely be administered prior to elective coronary angiography
- B. Ticagrelor should be held 2-3 days prior to CABG
- C. Aspirin should be held 2-3 days prior to CABG
- D. DAPT, with aspirin and ticagrelor, should be started post-operatively when safe to do so from a bleeding perspective

MCQ #33 new 2024

A 56 year old man with a history of hypertension, dyslipidemia and smoking is admitted to hospital for an inferior STEMI. He underwent PCI to the RCA 3 days ago and has recovered well, with no further episodes of chest pain, clinical heart failure or arrhythmias. His transthoracic echocardiogram was performed with showed low normal ejection fraction, LVEF 51%, with inferior hypokinesis otherwise normal RV and valvular function. He is being discharged on appropriate medical therapy. He works as a truck driver. Based on the new CCS 2023 Fitness to Drive Guidelines, what recommendations will you provide him regarding when to return to driving?

- A. May return to private driving after 1 month, commercial driving after 3 months
- B. May return to private driving after 2 weeks, commercial driving after 1 month
- C. May return to private driving after 48 hours, commercial driving after 7 days
- D. No restrictions

Bonus MCQ #34 2025

Which of the following patients would be LEAST likely to benefit from an early rhythm control strategy for their atrial fibrillation?

- A) A 55-year-old woman with newly diagnosed paroxysmal atrial fibrillation and a severely reduced ejection fraction (LVEF 30%)
- B) A 68-year-old man with persistent atrial fibrillation, significant symptoms (NYHA Class III), and a history of multiple admissions for heart failure
- C) A 42-year-old man with recent-onset (within 1 year) paroxysmal atrial fibrillation and few comorbidities
- D) A 72-year-old woman with persistent atrial fibrillation over 5 years, well-controlled symptoms (NYHA Class I) on rate control, and a normal ejection fraction (LVEF)

MCQ #35 new 2024

A 45 year old man with a history of hypertension and tobacco use undergoes a transthoracic echocardiogram for a murmur heard incidentally on routine physical examination. This demonstrates an ascending aorta aneurysm of 4.6 cm, with a trileaflet aortic valve with mild-moderate aortic regurgitation. Of the following recommendations, which would not be appropriate at this time?

- A. Refer to an Aortic Team for further evaluation and surveillance.
- B. Obtain a CT angiogram or MRI to characterize of the thoracic aorta.
- C. Obtain a full family history and refer for genetic testing. Genetic testing and/or screening imaging of family members will need to be performed.
- D. Start him on a calcium channel blocker to manage his blood pressure if greater than 130/80mmHg.

MCQ #36 new 2024

A 68yF is investigated for palpitations. She is a retired nurse and noticed that for the last two years her HR is irregularly irregular. 48h Holter reveals she is in Afib throughout the study, 95% of the time rates controlled <110bpm. She has a history of hypertension on perindopril 8mg/indapamide 2.5mg, otherwise healthy, hates taking medications. Labs are normal including TSH, LFTs, ECHO shows normal LV function with mild LVH and enlarged LA, LAVI, mild MR. The sensation of irregular heart beat bothers her and she would like treatment.

She is started on a DOAC for stroke prevention. What are the next best steps for this patients symptoms?

- a) Amiodarone oral load then 200 mg daily
- b) Propafenone 150 mg po TID
- c) Diltiazem CD 120 mg daily, and if still symptomatic book for elective cardioversion
- d) Metoprolol 25 mg po BID, if symptoms persist start Dronedarone

MCQ #37 new 2024

Based on the new CCS 2023 Fitness to Drive Guidelines, which of the following scenario and private driving recommendation pairs is incorrect (assuming the patient was well, medically optimized at the time of discharge and otherwise fit to drive)?

- A. A 62 year old man discharged after receiving a dual chamber pacemaker for syncope and complete heart block. Recommended to not drive for 1 week after implant.
- B. A 69 year old woman with an elective dual chamber pacemaker for sinus node dysfunction for exertional fatigue with no history of syncope or high grade AV block. Recommended not to drive for 1 week after implant.
- C. A 56 year old man with a dilated cardiomyopathy with an LVEF of 31% and NYHA II symptoms, received a single chamber ICD for primary prophylaxis. Recommended not to drive for 1 week after implant.
- D. A 45 year old woman with a dilated cardiomyopathy and who had an ICD shock associated with syncope. Device interrogation showed monomorphic VT. Her medical therapy was optimized in the device clinic. Recommended not to drive for 3 months.

MCQ #38 new 2025

A 58-year-old woman with a history of migraine headaches and hypertension presents to the emergency department with sudden onset chest pain radiating to her jaw. Her ECG shows T wave inversions in the inferior leads and elevated troponin. She is admitted to hospital and treated as an NSTEMI. Given ongoing chest pain, she is brought to the cath lab urgently, however coronary angiography reveals no significant coronary artery stenosis. Which of the following tests would be LEAST helpful in establishing a diagnosis of MINOCA?

- A) Transthoracic echocardiography
- B) Cardiac magnetic resonance imaging (CMRI) with late gadolinium enhancement (LGE)
- C) Exercise ECG testing
- D) Careful review of the coronary angiogram with consideration of intracoronary imaging

MCQ #39 new 2025

In a patient with atrial fibrillation and a CHADS-65 score of 0, all of the following comorbidities/situations would warrant anticoagulation for stroke prophylaxis EXCEPT?

- A) Cardiac amyloidosis
- B) New atrial fibrillation discovered after non-cardiac surgery
- C) Hypertrophic cardiomyopathy
- D) Thyrotoxicosis until euthyroid

MCQ #40 new 2025

A 60-year-old man with a history of hypertension presents to the clinic complaining of exertional dyspnea. On physical exam, you palpate a brisk carotid upstroke followed by a palpable dicrotic notch. On auscultation, a diastolic murmur is heard at the left lower sternal border. Which of the following conditions is MOST consistent with this physical exam?

- A) Aortic stenosis
- B) Mitral regurgitation
- C) Hypertrophic cardiomyopathy
- D) Aortic regurgitation

MCQ #41 2026

33 y/o M who presents with a murmur. Brisk carotid upstroke. Systolic murmur heard best at the LLSB, that increases with valsalva, quieter with passive leg raise. What is the patient not at risk for in the next 5 years?

- a) Sudden cardiac death
- b) Stroke
- c) Aortic dissection
- d) CHF

BONUS MCQ ANSWERS

- MCQ 1: Answer D multivessel disease + diabetes would be associated with a mortality benefit with CABG
- MCQ 2: **Answer: C.** ACE inhibitors are generally avoided in HCM with LVOT obstruction, as they reduce afterload which can worsen outflow tract obstruction. Beta blockers and non-dihydropyridine CCBs reduce heart rate, which improves LV filling, and reduces inotropy, reducing contractility, both which reduce LVOT obstruction. Maintaining hydration and stopping diuretics will improve LV filling and reduce LVOT obstruction by increasing preload.
- MCQ 3: Answer B - VALVE Question: refer to CV Surg LVESD is >40
- MCQ 4: Answer A change ramipril to sacubitril/valsartan, add empagliflozin – ok to do concurrently
- MCQ 5: Answer C (3): Stop ASA and Clopidogrel and start warfarin alone.
- MCQ 6: Answer 1 – start anticoagulation with a NOAC for at least 4 weeks.
- MCQ 7: Answer D) refer to CV surg for potential valve repair/replacement.
- MCQ 8: Answer (A) DC cardioversion
- MCQ 9: Answer – (B) ASD
- MCQ 10: Answer C - **Diastolic dysfunction**
- MCQ 11: Answer D - **Answer: D.** All the options represent alternative diagnoses that need to be ruled out before confirming a diagnosis of MINOCA. Note that SCAD is also an exclusion criteria but often challenging to diagnose on primary angiogram. Where available Optical Coherence Tomography (special vessel imaging used angiography) suggested to delineate if SCAD present (or embolism, can also show plaque composition etc).
- MCQ 12: Answer - C - Leg raise – HOCM murmur will get quieter. Recall: maneuvers that make HCM/MVP louder: Standing/Valsalva (decreased venous return) and Amyl Nitrate (decreased afterload)

BONUS MCQ ANSWERS

MCQ 13: d Warfarin with goal INR 2.5-3.5 per new ACC/AHA Valve dz guideline

For patients with a mechanical mitral valves, warfarin alone is not necessarily adequate.

- The new ACC/AHA Guidelines no longer recommend routine use of ASA with mechanical valves... “For patients with a mechanical SAVR or mitral valve replacement who are managed with a VKA and have an indication for antiplatelet therapy, addition of aspirin 75 to 100 mg daily may be considered when the risk of bleeding is low.” - so for this patient with HTN only we would not otherwise prescribe ASA!
- Patients with mitral valve prosthesis should be on warfarin with an with an international normalized ratio (INR) goal of 3 (2.5-3.5).
- Patients with a ‘current generation’ aortic valve prosthesis should be on warfarin with an with an international normalized ratio (INR) goal of 2.5 (2-3). Patients with older AVR (ball-cage) may benefit from higher target 3.0.
- Higher INR goals may be considered in patients who have thromboembolic events while anticoagulated within this INR range. This patient has not had a thromboembolic event therefore a higher INR goal would not be indicated.

MCQ 14: ASD (causes wide, fixed split S2)

MCQ 15: **Answer:** D. The presence of a midsystolic click and a murmur that changes duration with position (becoming earlier and softer with increased preload from squatting) strongly suggests mitral valve prolapse. Aortic stenosis with a bicuspid aortic valve could have a click but would be very early after the S1, described as an ejection click, but the click is not dynamic like with MVP. MR would become louder with squatting and has no click. HCM would become softer with quieter, but has no click.

BONUS MCQ ANSWERS

MCQ 16: Answer B. Similar to MCQ 1 but has LM disease. LM disease excluded from ISCHEMIA trial, so we don't really know about the PCI vs OMT question.

MCQ17: **Answer D.** Getting primary PCI. No prasugrel (hx stroke). E is Pericarditis dosing. This is not a pericarditis ECG. NB. There is an Emerging role for Colchicine post MI – 0.5mg/day reduces MACE compared to placebo [COLCOT Trial, NEJM Nov. 2019]

MCQ18: Answer C. Note FAMILIAL ANGIOEDEMA = synonymous with HEREDITARY ANGIOEDEMA DAPA is now in the guidelines for non diabetic w EF <40%. Pt is in afib so ivabradine not appropriate, and RBBB w QRS <150 w AFIB so CRT unlikely to confer mortality benefit.

MCQ19: Answer A tamponade

MCQ20: c – electrical Cardioversion

MCQ21 – C This is a classic RC style question where you see a couple of answers that you would “do both” and it's like mind-reading to guess what the “NEXT” thing they want you to do is. This pt has pericarditis 12 weeks post MI. They are on hydralazine which can cause pericarditis and likely have NSAID allergy (Sampter's triad). So you would admit, hold hydralazine, give colchicine. In terms of the NEXT THING to do ... well, you probably don't want them to have a dose of hydralazine in ED so pick C, chase with a dose of colchicine.

Bonus MCQ Answers

MCQ 22 - Answer: A. CHADS₂. Symptoms >48h so need to either do TEE then cardiovert OR anticoagulated 3 weeks then cardiovert. After cardioversion must remain on OAC 4 weeks.

MCQ 23 – C is best answer...Type B Dissection due to Cocaine = medical management → = control HR first, but avoid beta blocker (cocaine)

MCQ 24 – B. Purposefully tricky question. Young man w dilated cardiomyopathy, ECG w large 1o AVB and LBBB, consider differential ischemic and non ischemic causes (?sarcoid). Beta blocker may be risky up front. Ivabradine doesn't cause AV conduction delay but not indicated here w HR <70. = Get cardiac MRI NOW

MCQ #25: ANSWER C (routine stress testing post PCI does not show any benefit per the POST-PCI trial. SGLT2 inhibition would be indicated given CAD + diabetes and would improve outcomes. Flu shot per IAMI and polypill per the SECURE trial would also show benefit).

MCQ #26: Answer C – given this is a NSTEMI with LVEF >40%, he can drive privately from 2 weeks, commercially 1 mo, per new CCS 2023 Fitness to drive guidelines. If he had an LVEF 40% or less, then D would be correct. If he was managed medically without PCI done, then D would also be correct.

Bonus MCQ Answers

MCQ #27: Answer B – PCI will not reduce CV mortality or dec hosp; and the recent REVIVED-BCIS trial demonstrated the lack of benefit in multivessel PCI even if there was viability as in this patient. CABG based on the follow-up from the STICH trial, which showed CABG in a patient with an ischemic cardiomyopathy like this, had benefits on reducing all cause and CV mortality, hospitalization. SGLT2i addition and/or changing valsartan to sacubitril-valsartan would also carry benefit based on EMPEROR-Reduced, PARADIGM-HF trials etc.

MCQ #28: Answer D – this patient has an indication for anticoagulation with a CHADS65 score of 3. Ticagrelor with a DOAC is a combination generally avoided due to higher bleeding risk, therefore ticagrelor should be changed to clopidogrel (A, B wrong). In the setting of a DES placed 3 months ago for an MI, a dual pathway strategy would be preferred to complete 12 months of therapy with clopidogrel + rivaroxaban (C wrong, D right) after which rivaroxaban would be indicated alone back at the full dose of 20mg!

MCQ #29: Answer A - Given a history of CAD and HF with LVEF >40%, antiarrhythmic options are limited to sotalol or amiodarone. Sotalol would be the preferred option to try as he is young and amiodarone will have cumulative side effects. Stop bisoprolol if starting another beta blocker; propafenone, flecainide and dronedarone are not appropriate in a patient with heart failure. Amiodarone could be considered for this patient with CAD and HF but is not a great long term solution due to side effects).

MCQ #30: Answer D - PER NEW CCS 2022 PAD GUIDELINES, DO NOT SCREEN ASYMPTOMATIC PTS FOR CAD/CAS

MCQ #31: Answer: A. CABG – per Freedom trial

MCQ #32: Answer C. Aspirin should be continued to day of surgery, while ticagrelor should be held 2-3 days before, clopidogrel 2-7 days before. DAPT is now (weakly) routinely recommended post CABG. Loading on DAPT for elective coronary angiography is not routinely recommended as this could delay timing of a CABG due to bleeding risk.

Bonus MCQ Answers

MCQ #33 Answer: B. STEMI with LVEF more than 40% is now recommended 2 weeks private and 1 month commercial restriction (previously 1 and 3 months)

MCQ #34 Answer: D. This patient has long-standing persistent AFib, well-controlled symptoms, and normal ejection fraction and is doing well on a rate control strategy. Per ACC/AHA 2023 AF Update, early rhythm control strategy is recommended for patients with reduced LVEF, and reasonable for symptomatic AF to improve symptoms, recent diagnosis (<1 yr) to reduce progression, in combination of atrial fibrillation and HF.

MCQ #35 Answer: D. Given his history of aortopathy and hypertension, a beta blocker followed by ARB would be the preferred agents for BP control (may help to slow progression). Given his aortic disease, especially at a young age, referral to an Aortic Team is recommended for further evaluation and surveillance. An echocardiogram only captures portions of the aorta, so cross-sectional imaging with a CT or MRI (preferably MRI since age<50) would be important to see how large the aorta truly is. Likewise, family history and evaluation for genetic/familial aortopathies are important parts of his initial evaluation.

MCQ #36 Answer: C. given that the patient is in atrial fibrillation for likely over 2 years given the provided history, this would be persistent atrial fibrillation. Starting on a rate controlling agent, reassessing and booking cardioversion would be recommended per the CCS 2020 AF update algorithm. If they feel better after cardioversion in sinus rhythm, rhythm control agents could be trialed in this patient even though they have over 1 year of symptoms.

MCQ #37 Answer: B. No driving restriction is required if elective pacemaker for sinus node dysfunction without syncope or high grade AV block. Note the patient in scenario C would be eligible to drive commercially prior to the ICD due to having an LVEF more than 30%, but would now be ineligible to drive commercially for having an ICD! Note that the patient in scenario D has the same recommendation not to drive for 3 months for and ICD shock and hemodynamically significant VT. This used to be 6 months in the old guideline!

Bonus MCQ Answers

MCQ #38: C. Exercise ECG testing would be useful to risk stratify a patient with chest pain prior to angiography, usually in the ambulatory setting, but does not add value to this patient and can be harmful in the setting of a recent MI (can provoke arrhythmias). Echo will be useful to assess for MINOCA. Cardiac MRI is useful in the MINOCA workup to rule out other causes such as myocarditis. Careful review of the coronary angiogram and consideration of intracoronary imaging may yield other causes of MINOCA like SCAD that was initially missed, plaque disruption, etc.

MCQ #39: B. The benefit for anticoagulation for atrial fibrillation newly diagnosed after non-cardiac surgery is “uncertain” per most recent ACC/AHA Guidelines. Patients are often at high risk of occurrence so the decision will be made with shared decision making. Cardiac amyloidosis, HCM and thyrotoxicosis are all situations when patients are at a higher risk of thromboembolism and anticoagulation is indicated regardless of the CHADS-65 score.

MCQ #40: D. Pulsus bisferiens, characterized by two strong systolic peaks in the carotid pulse, with the presence of a diastolic murmur, is highly suggestive of severe aortic regurgitation. Hypertrophic cardiomyopathy with LVOT obstruction can also give a pulsus bisferiens due to initial outflow with systole, then a dip from LVOT obstruction, followed by a second peak due to continued flow through the LVOT (spike and dome). The murmur of LVOTO would be systolic. Aortic stenosis would be associated with a pulsus parvus et tardus (slow and delayed upstroke), and the murmur would be systolic. MR may have a hyperdynamic pulse (bounding) due to volume load on the LV, and the murmur would be systolic.

MCQ # 41: C- Aortic dissection . **this murmur = HOCM - systolic LLSB, louder with less preload and softer with increase preload, risks for sudden cardiac death, stroke, and CHF**